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(54) **ADJUSTMENT OF A SURGICAL DEVICE
FUNCTION BASED ON SITUATIONAL
AWARENESS**

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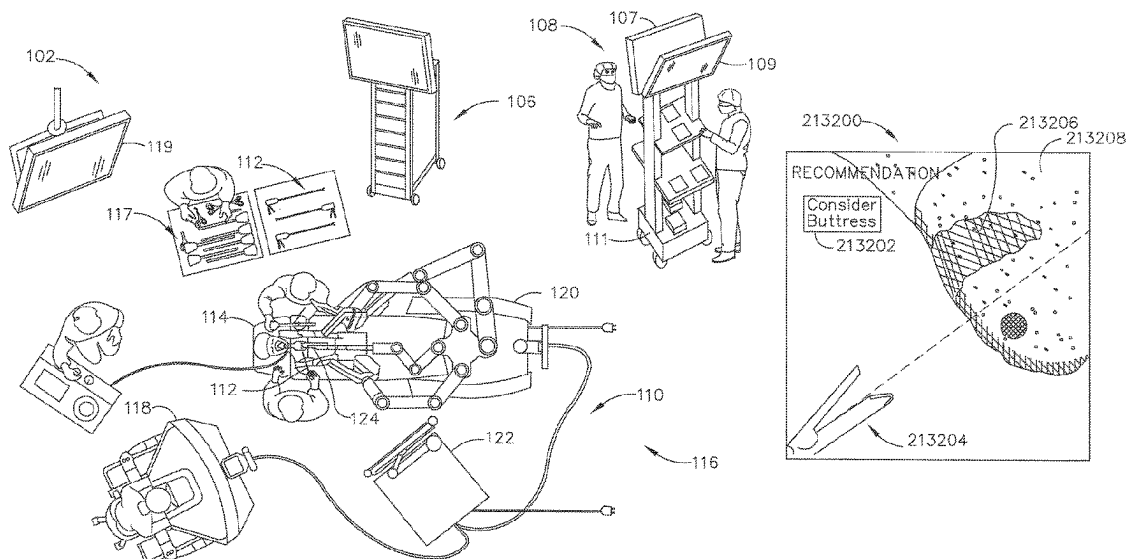
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(57) **ABSTRACT**

Surgical devices and surgical systems are disclosed. A
surgical system can include a display and a control circuit
configured to receive a signal from a situationally-aware
surgical hub indicative of a surgical state, display a recom-
mendation on the display based on the signal, and assign a
priority level of the recommendation.

20 Claims, 25 Drawing Sheets



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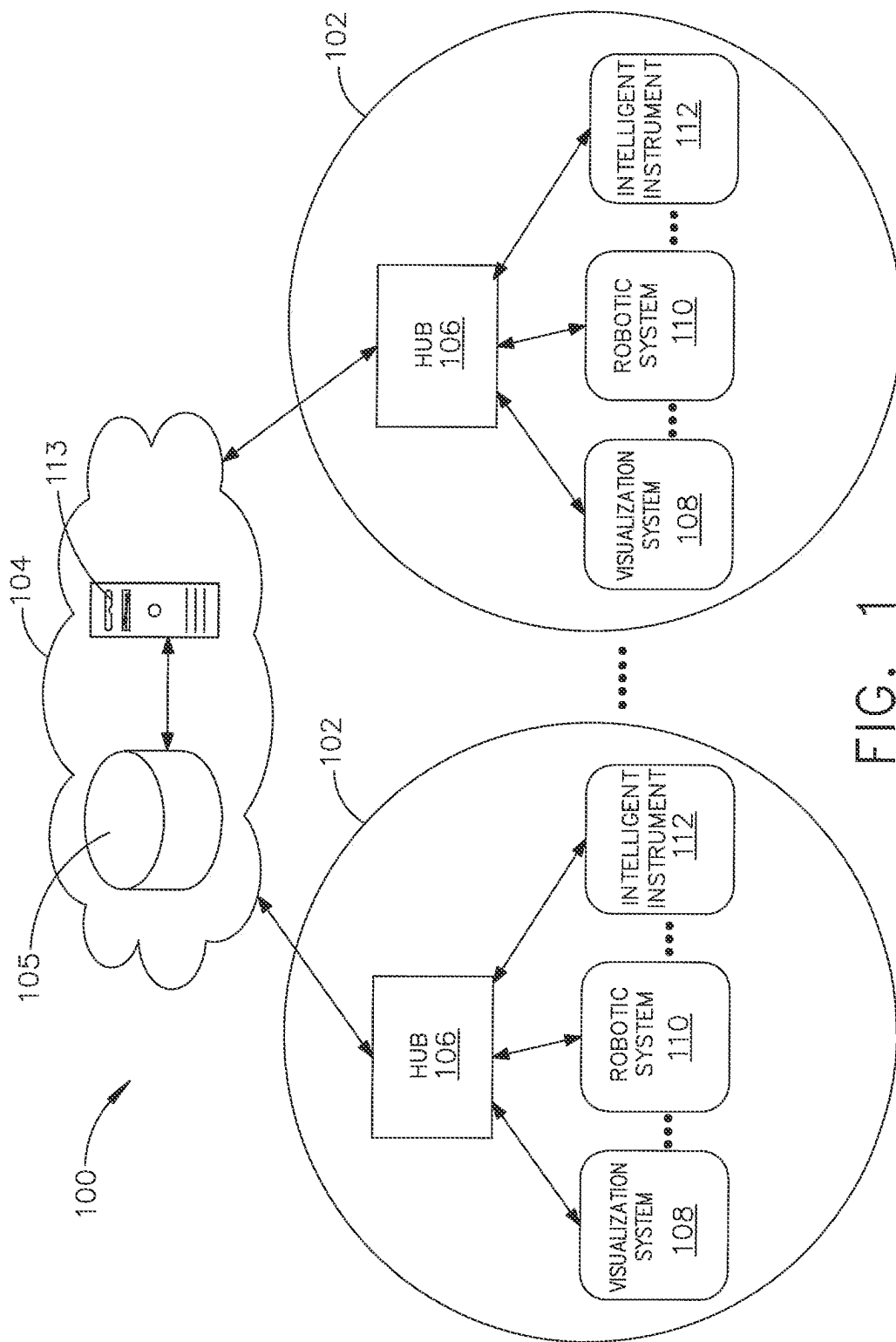
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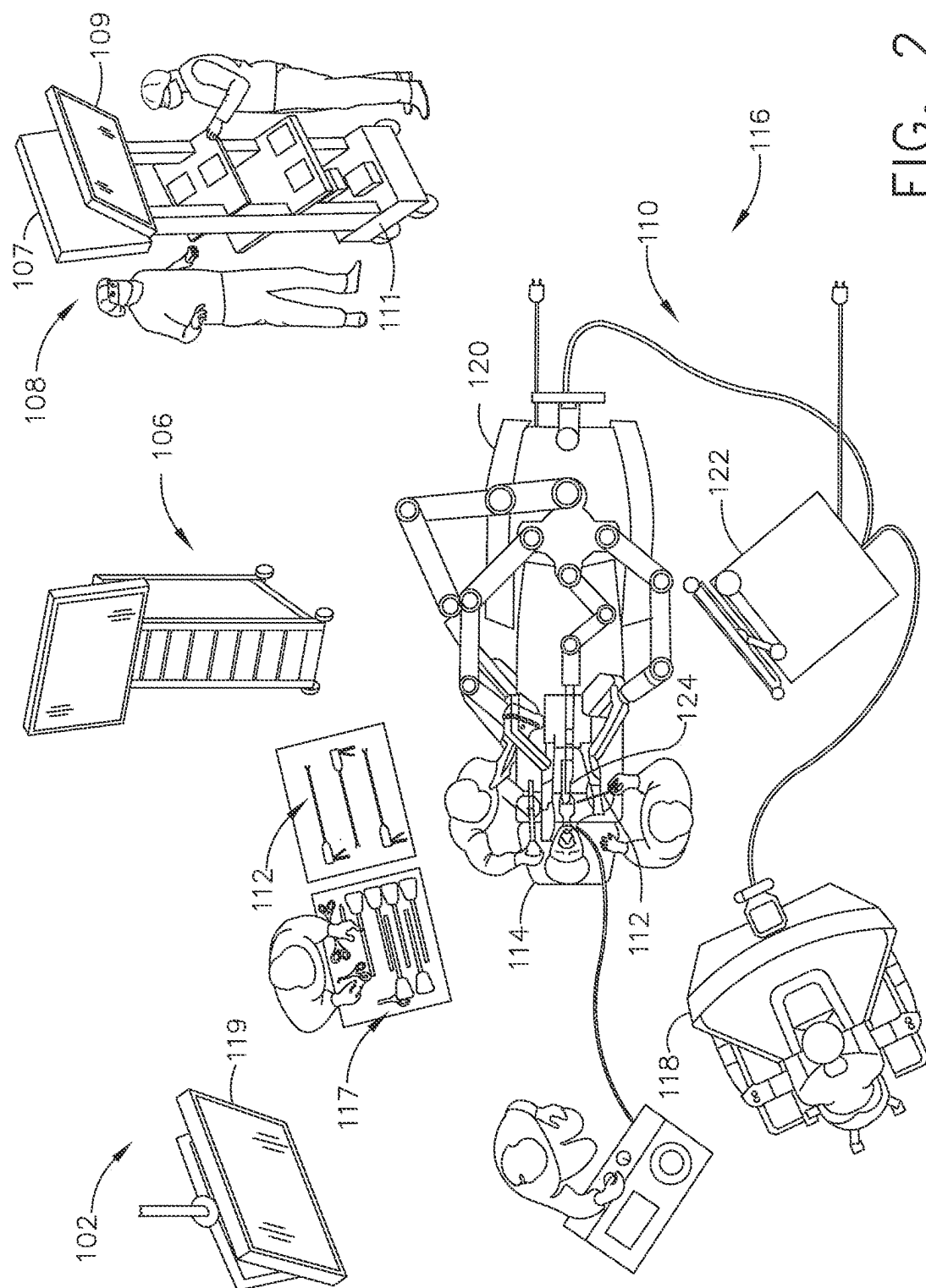


FIG. 2

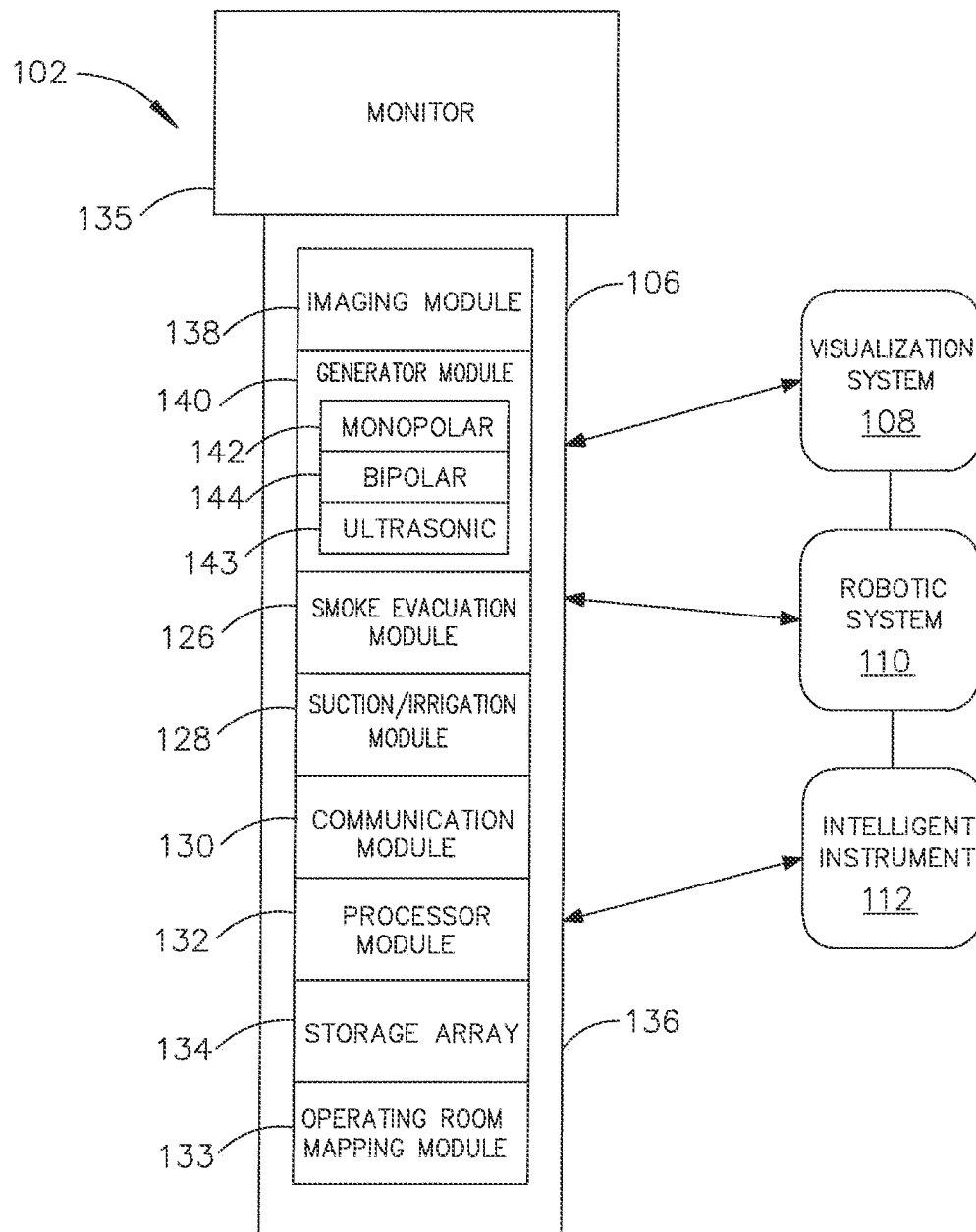


FIG. 3

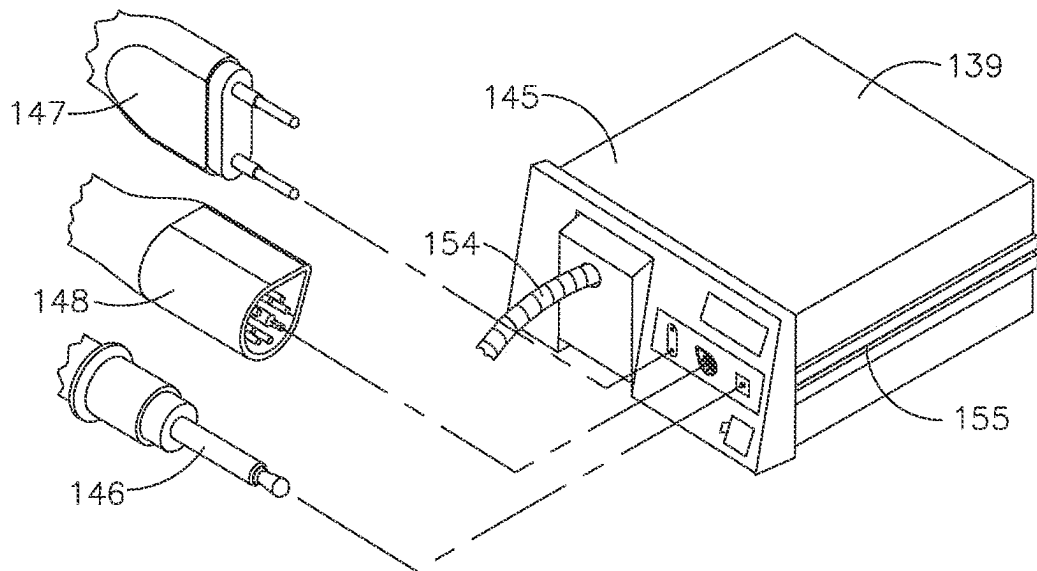


FIG. 5

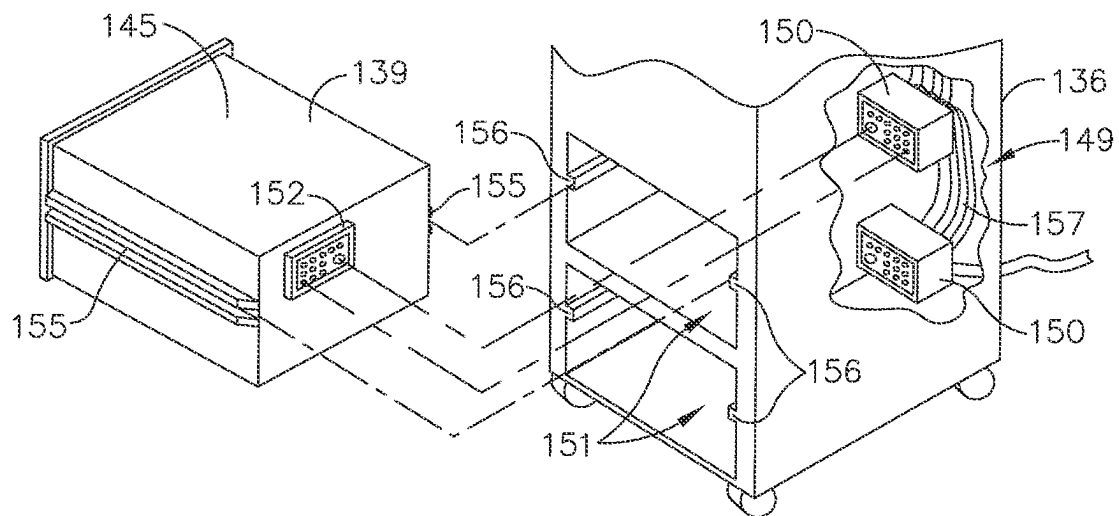


FIG. 4

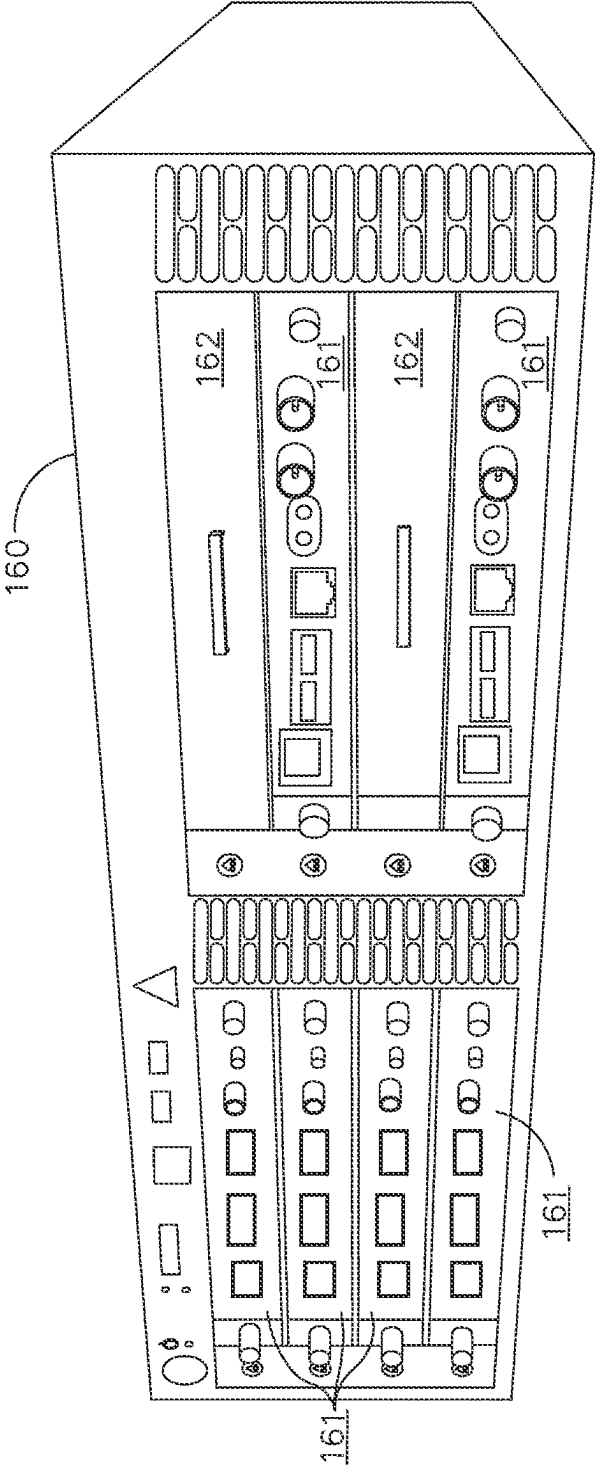


FIG. 6

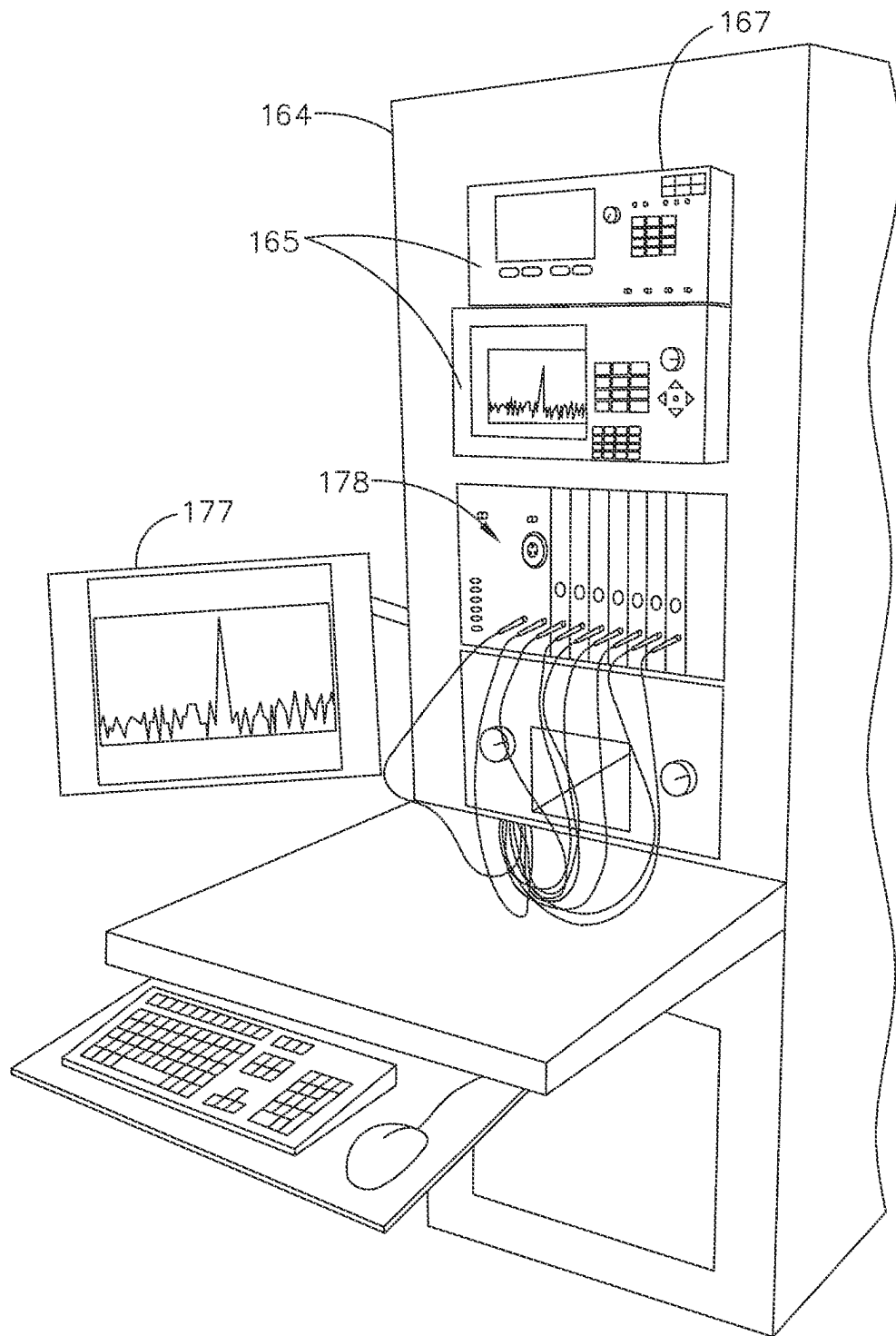


FIG. 7

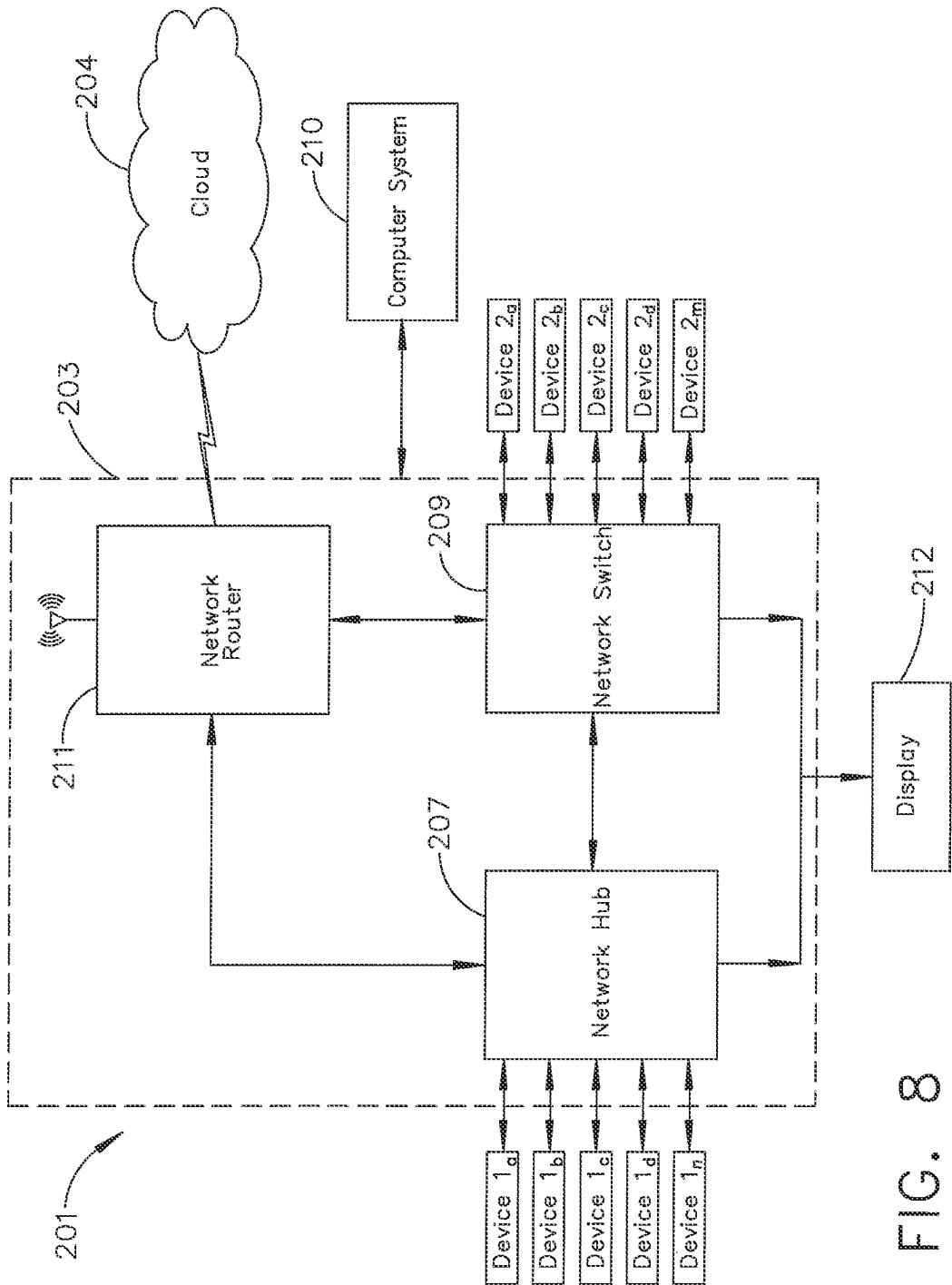


FIG. 8

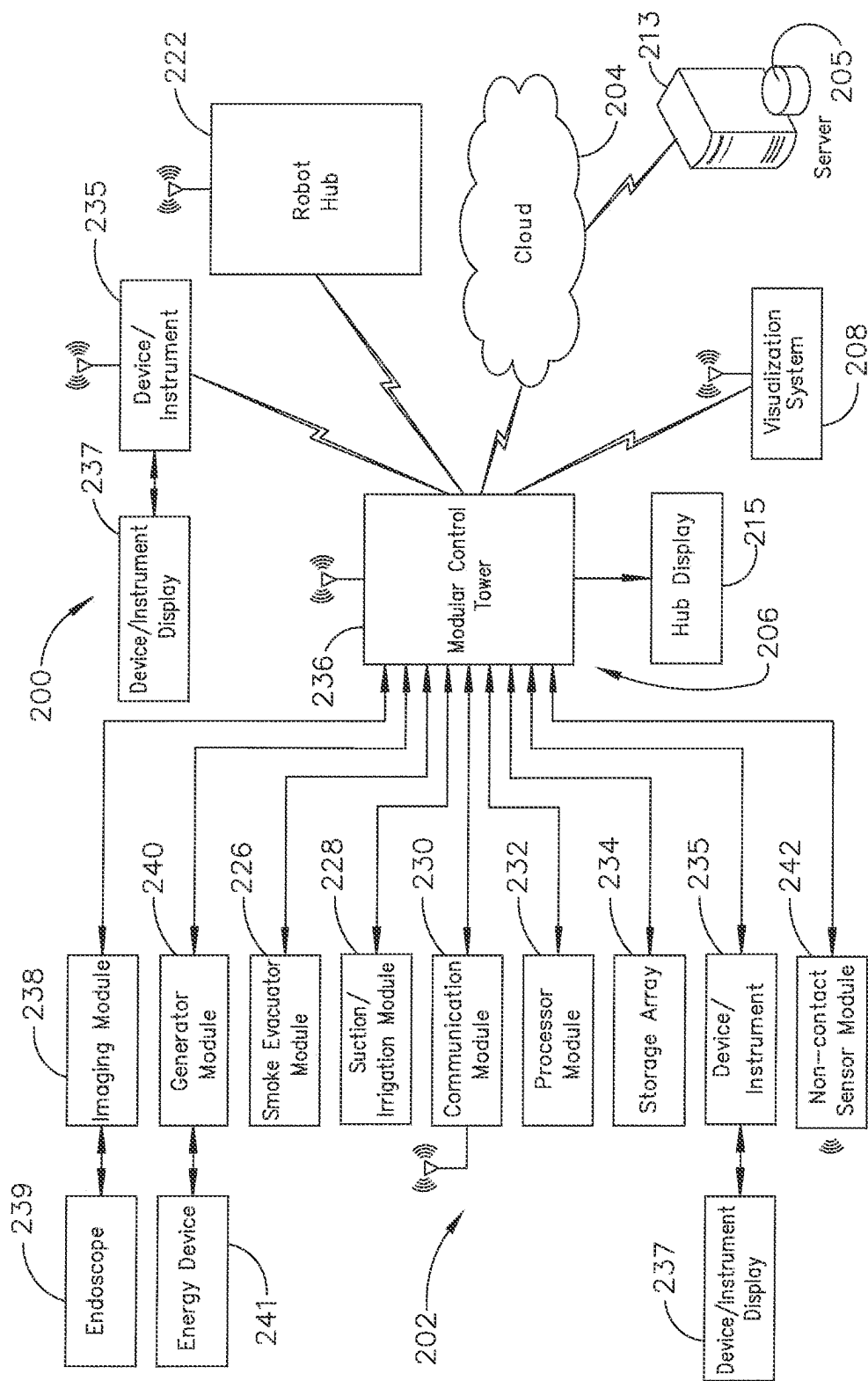


FIG. 9

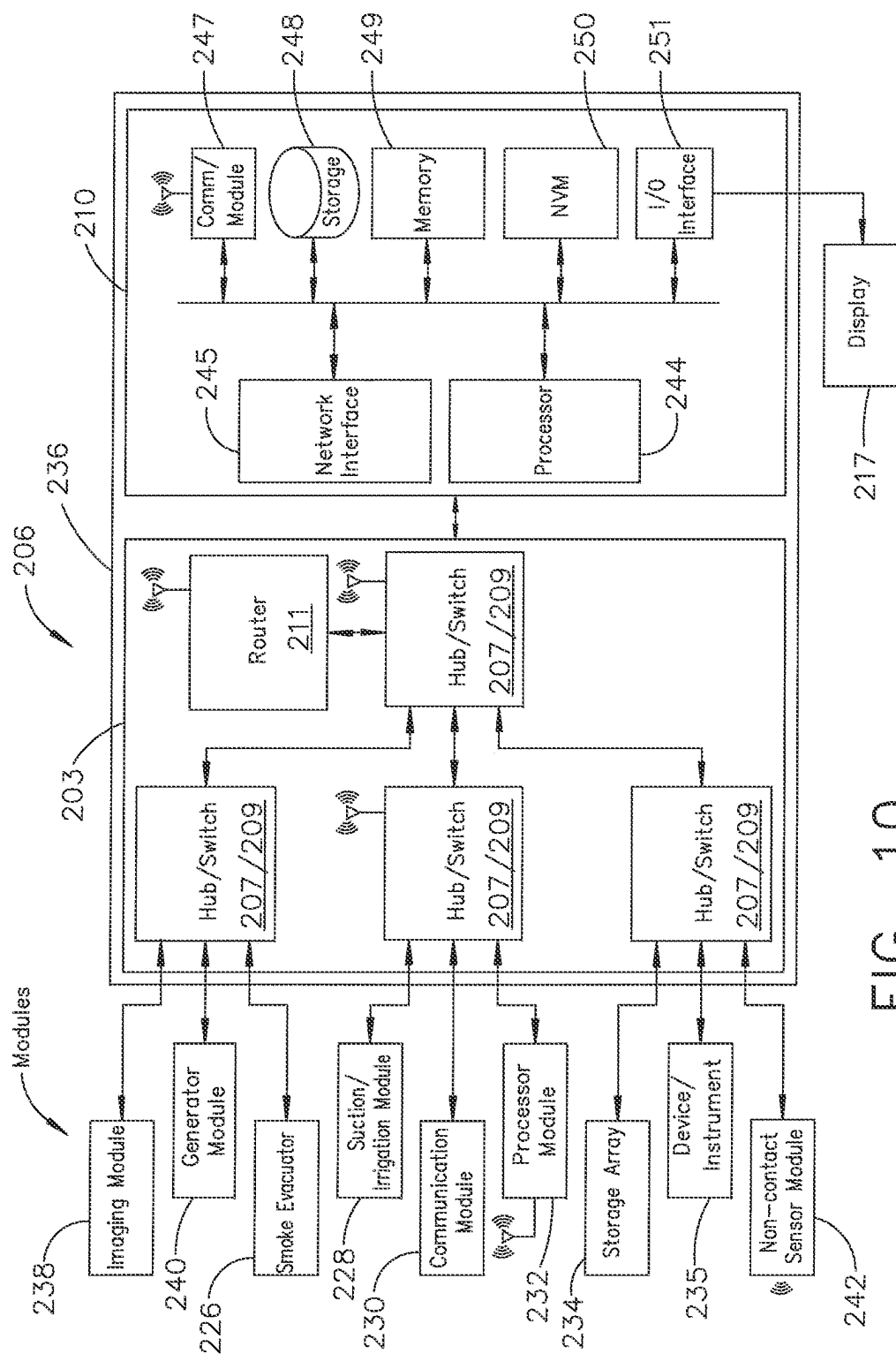


FIG. 10

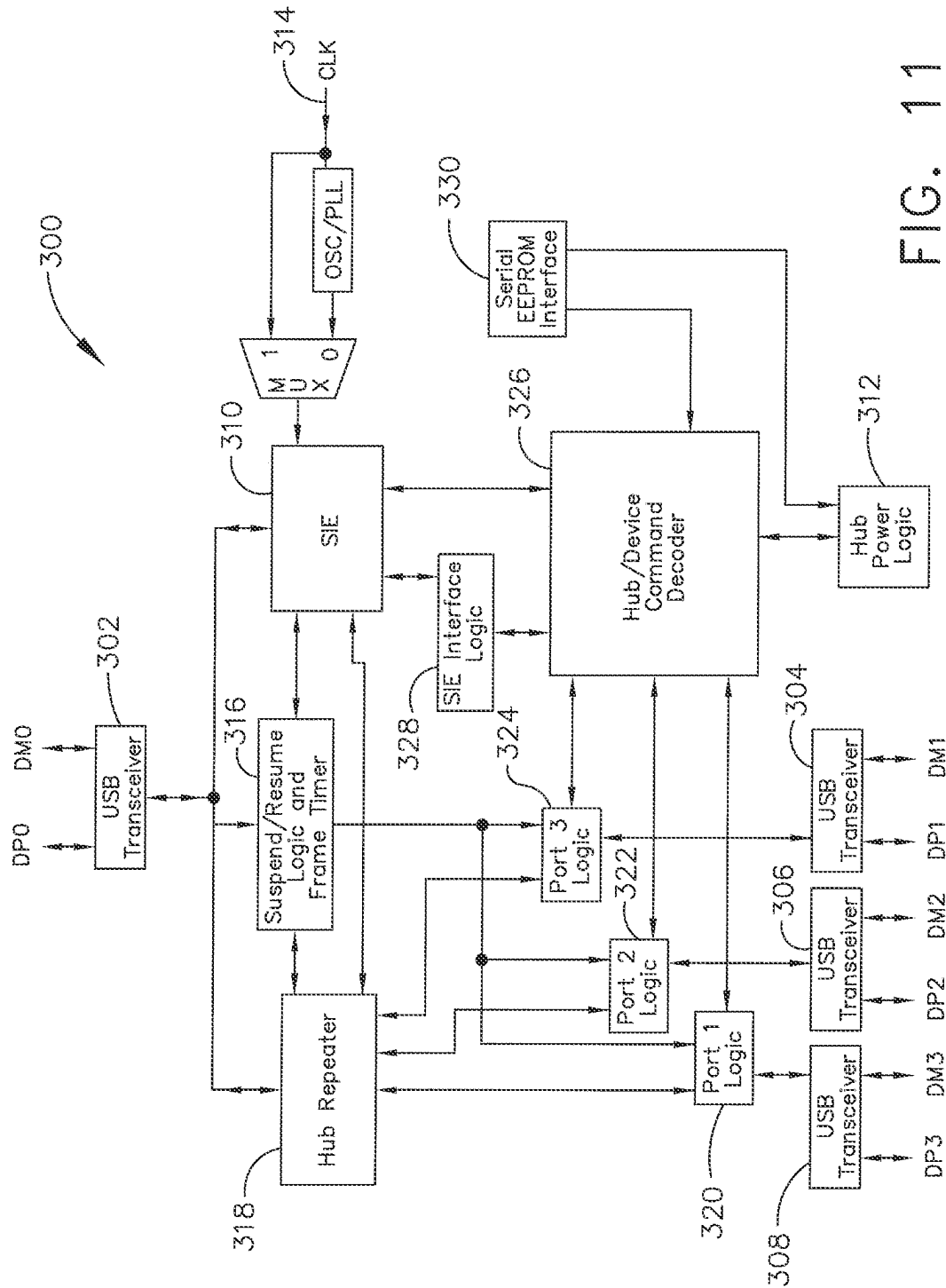


FIG. 11

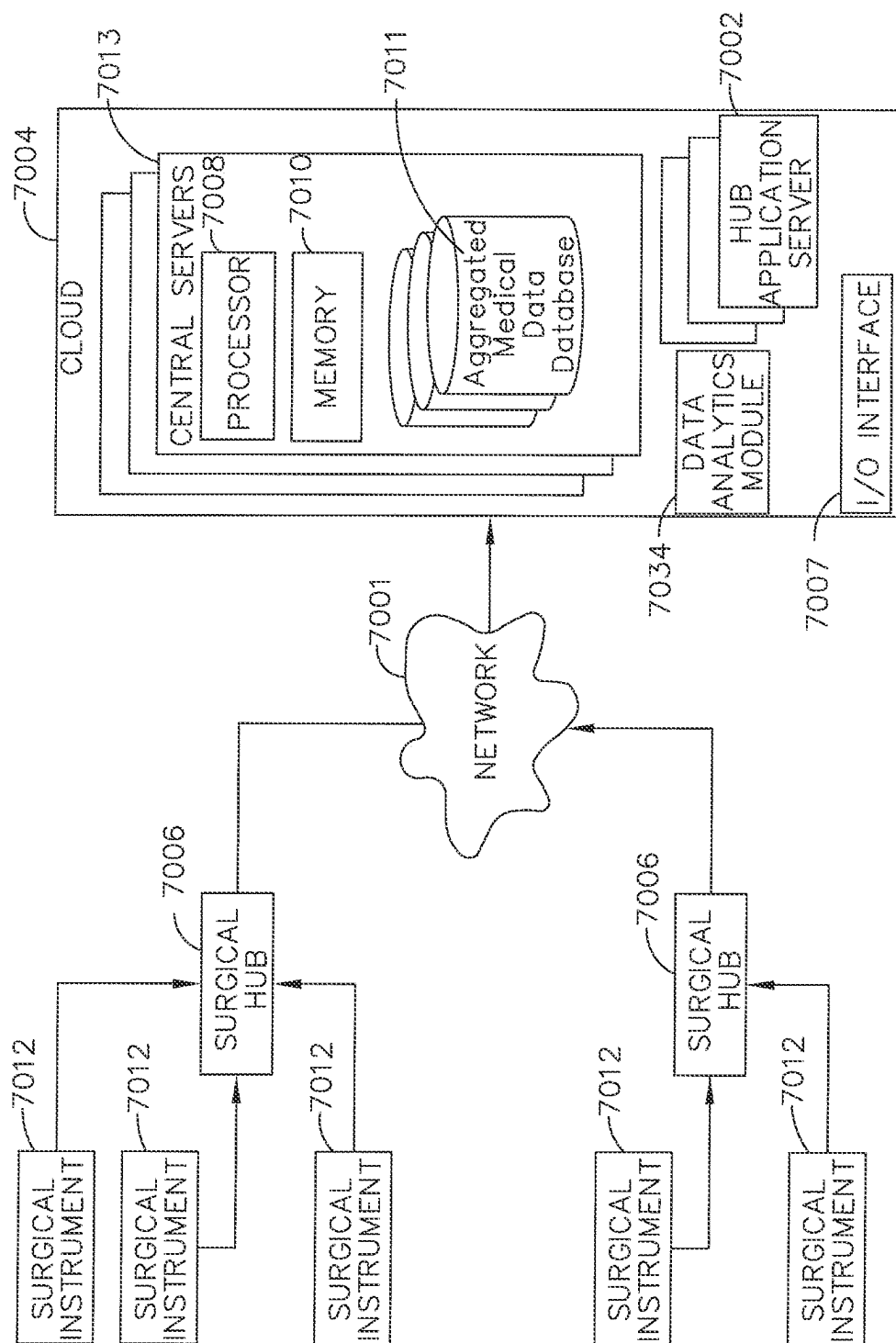


FIG. 12

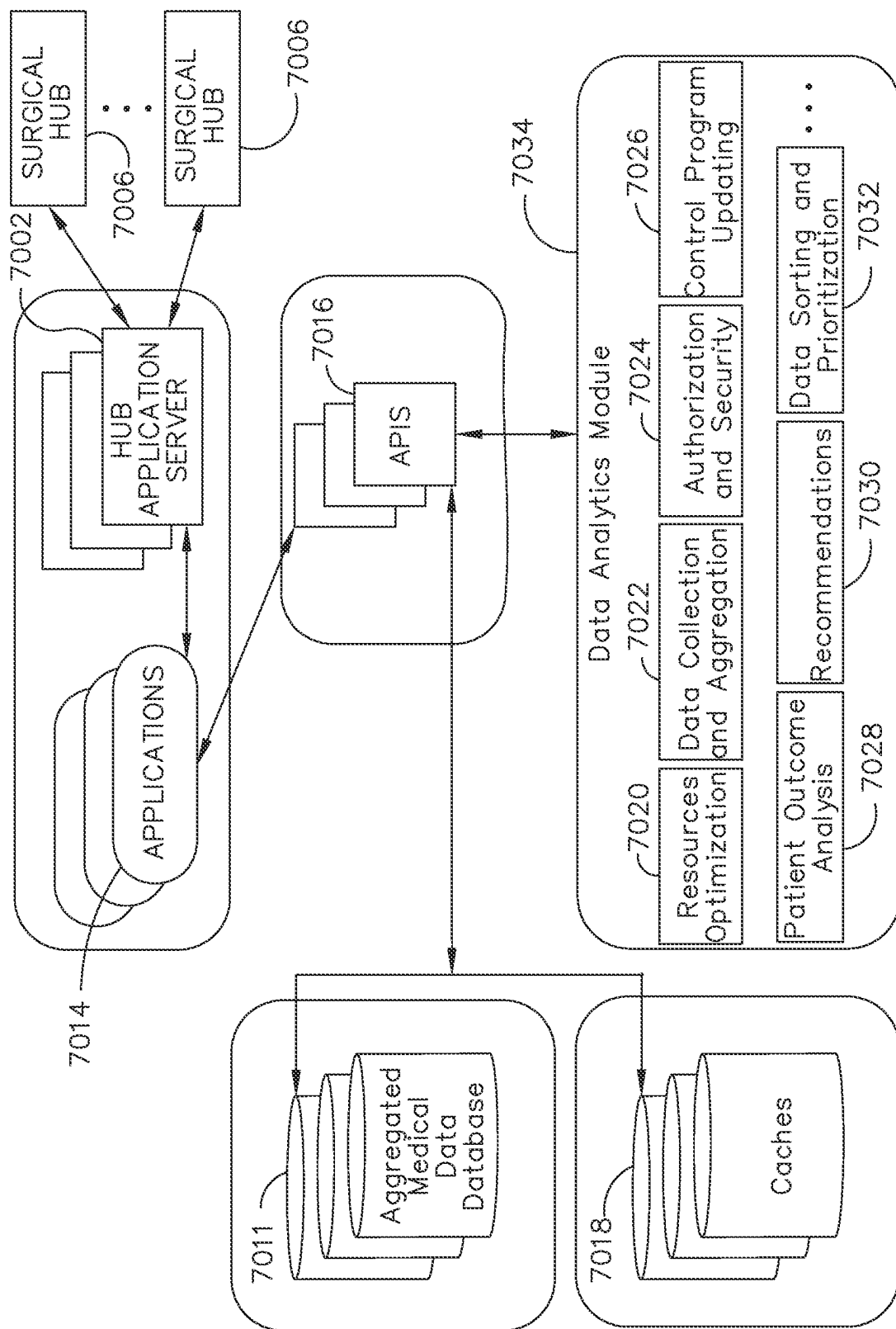


FIG. 13

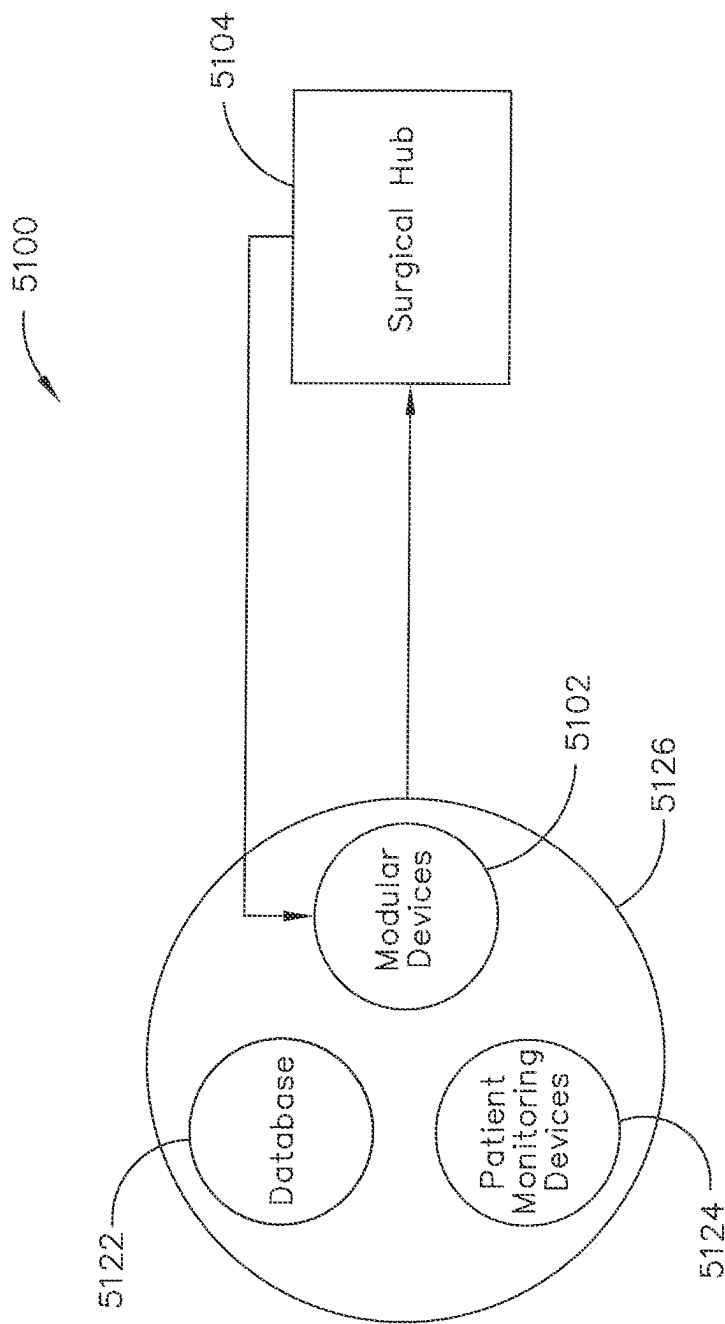


FIG. 14

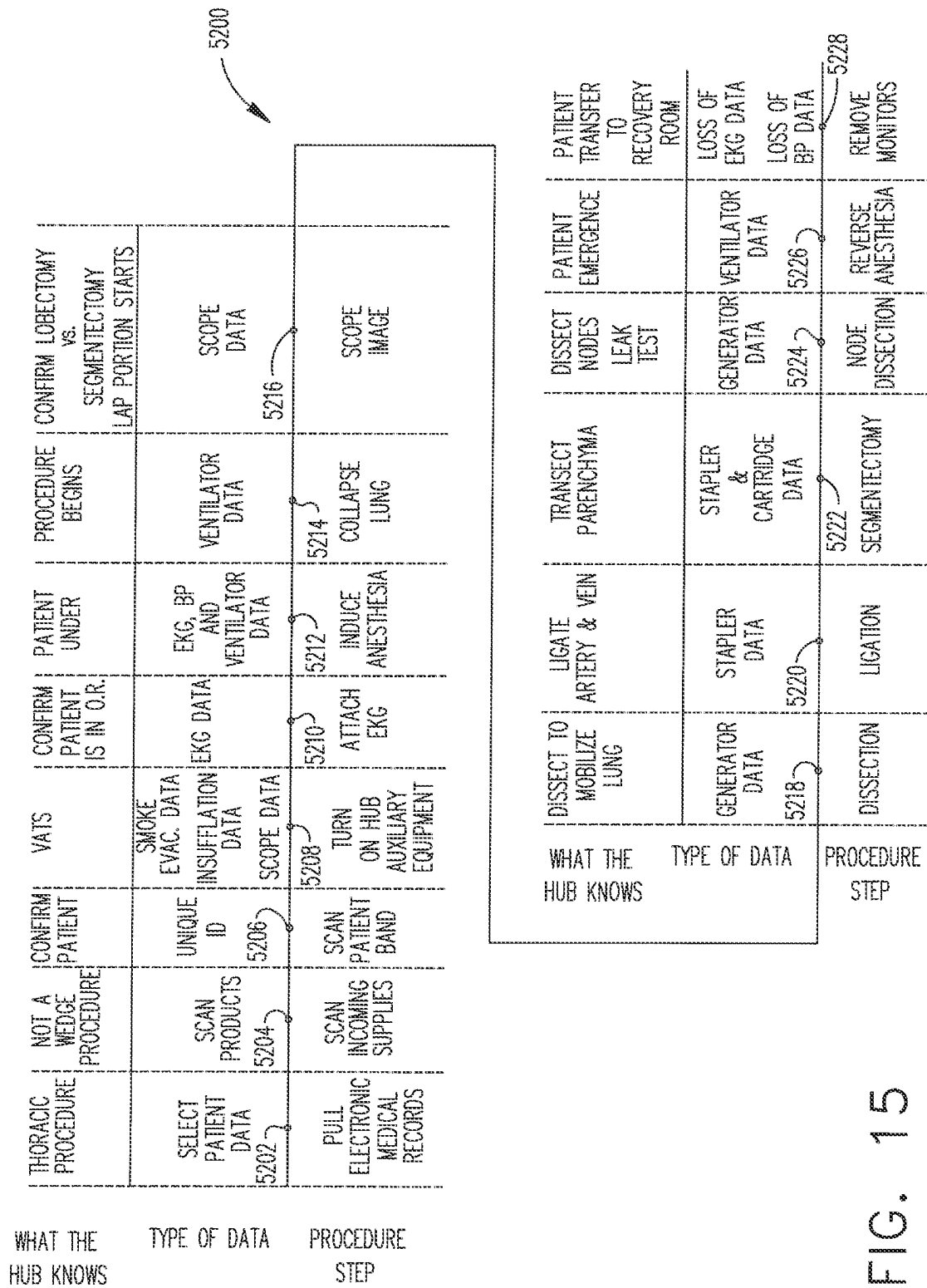


FIG. 15

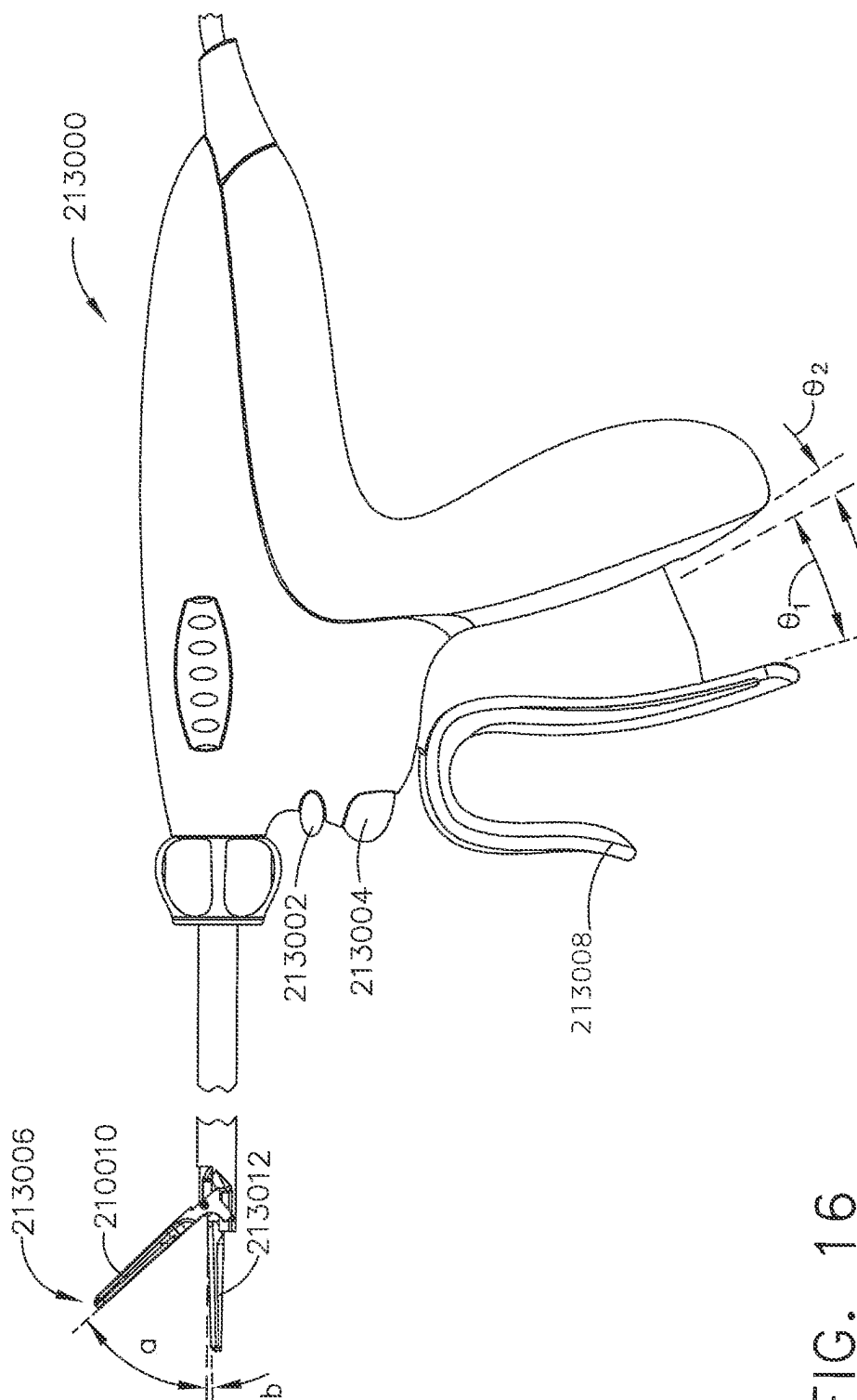


FIG. 16

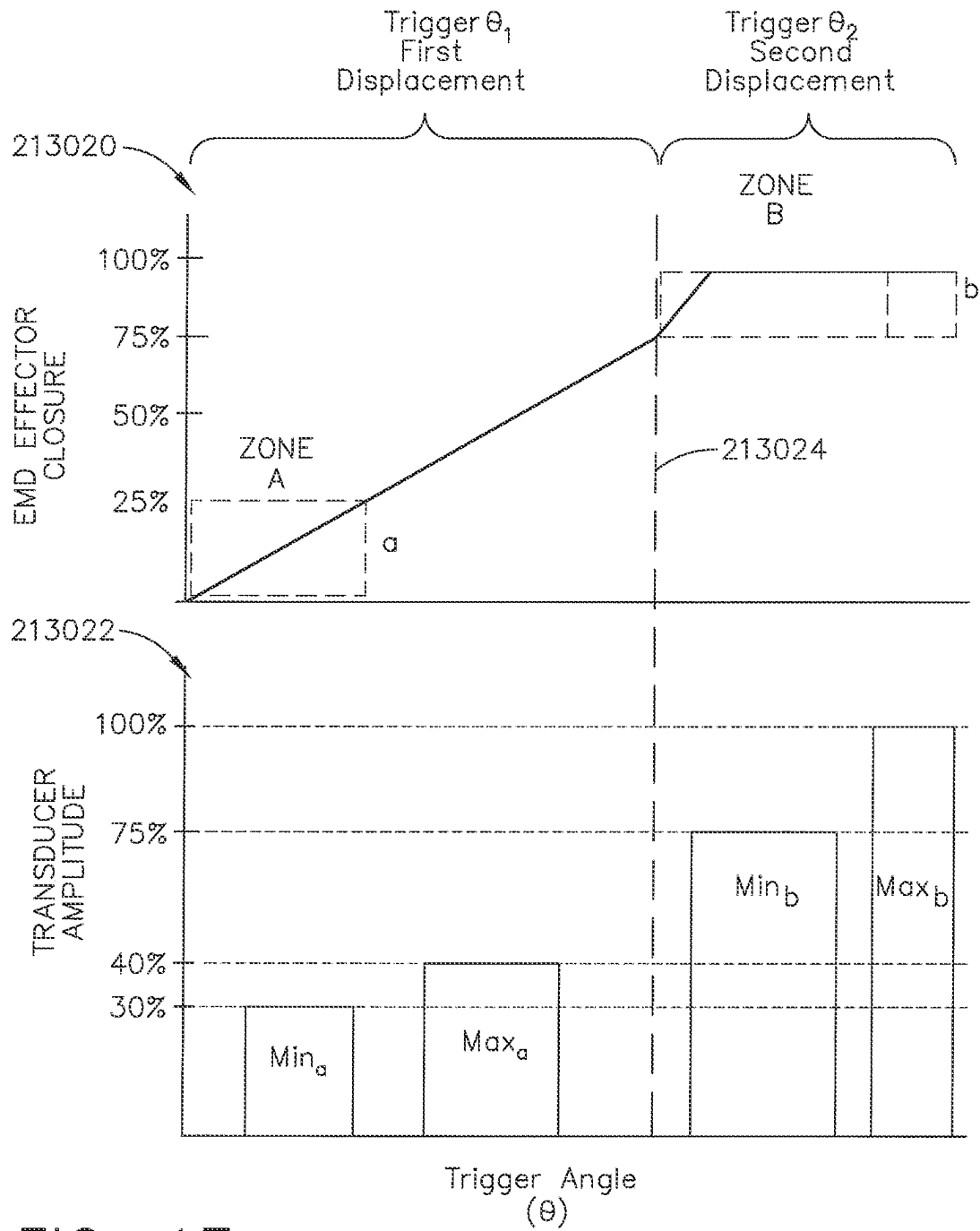
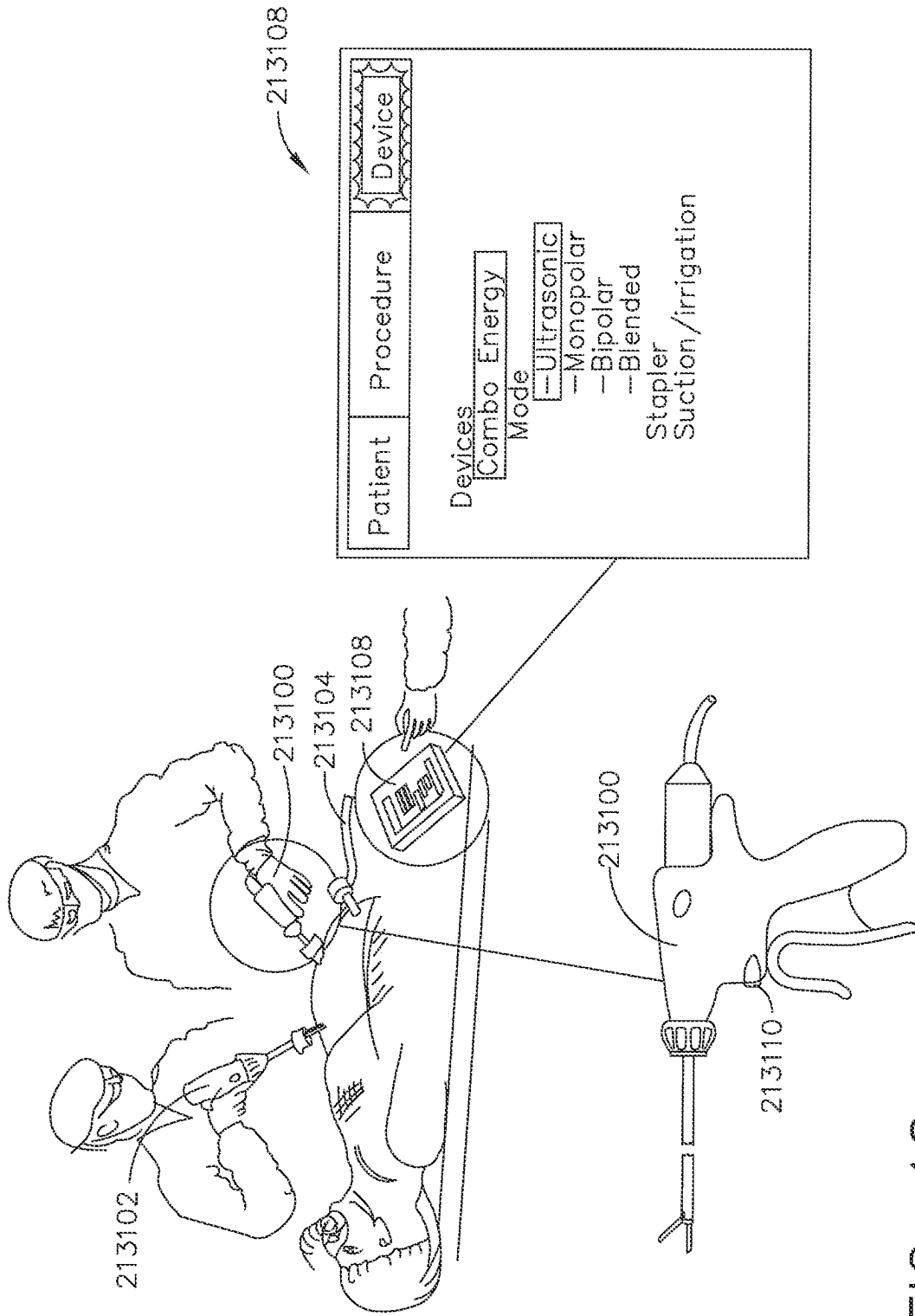


FIG. 17



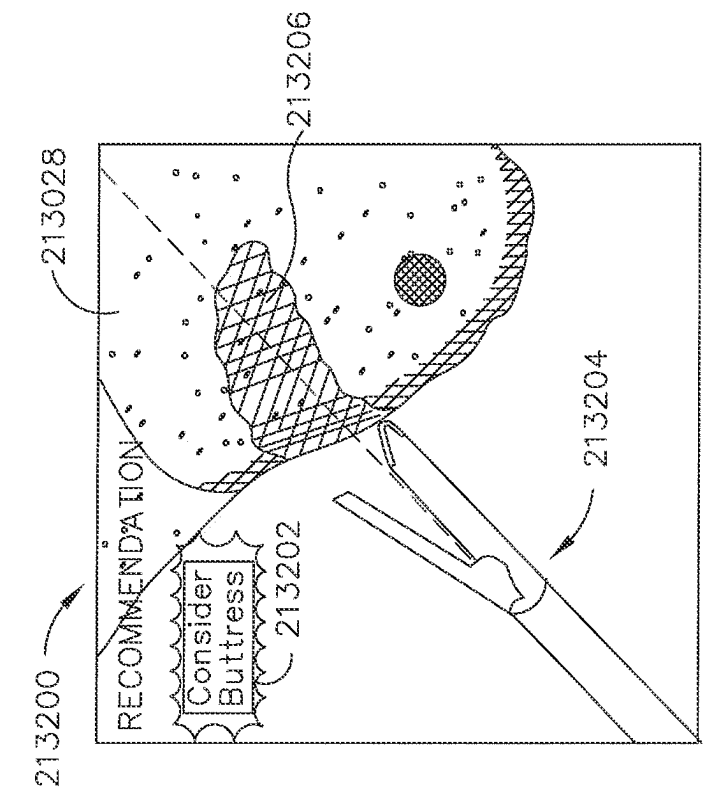


FIG. 19B

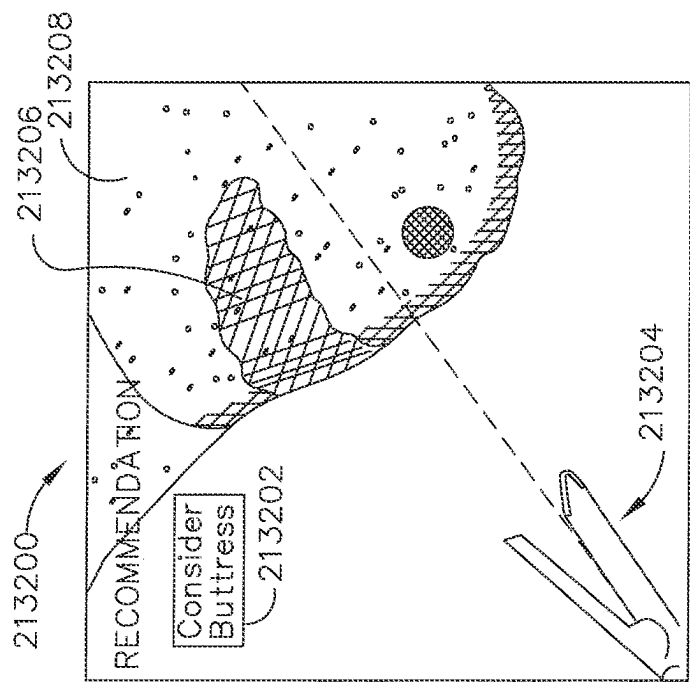


FIG. 19A

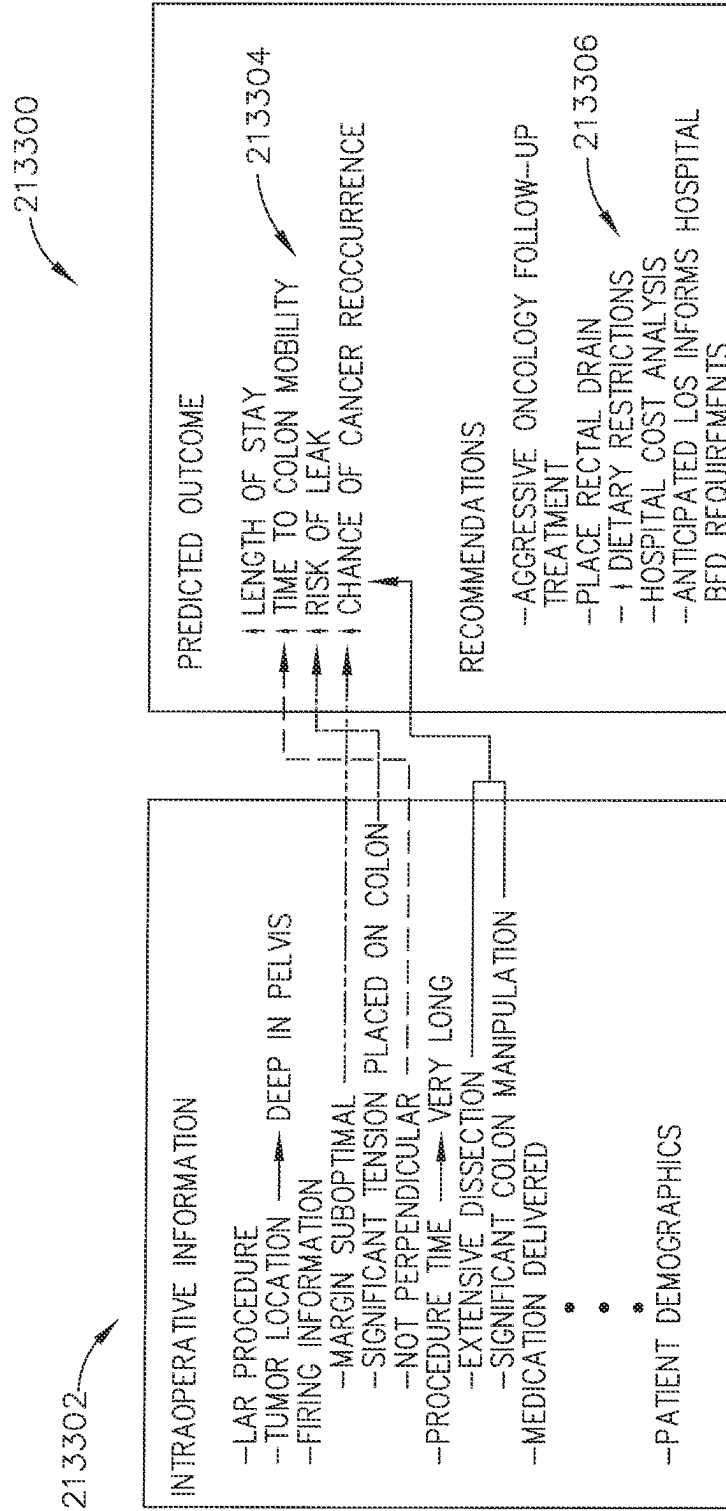


FIG. 20

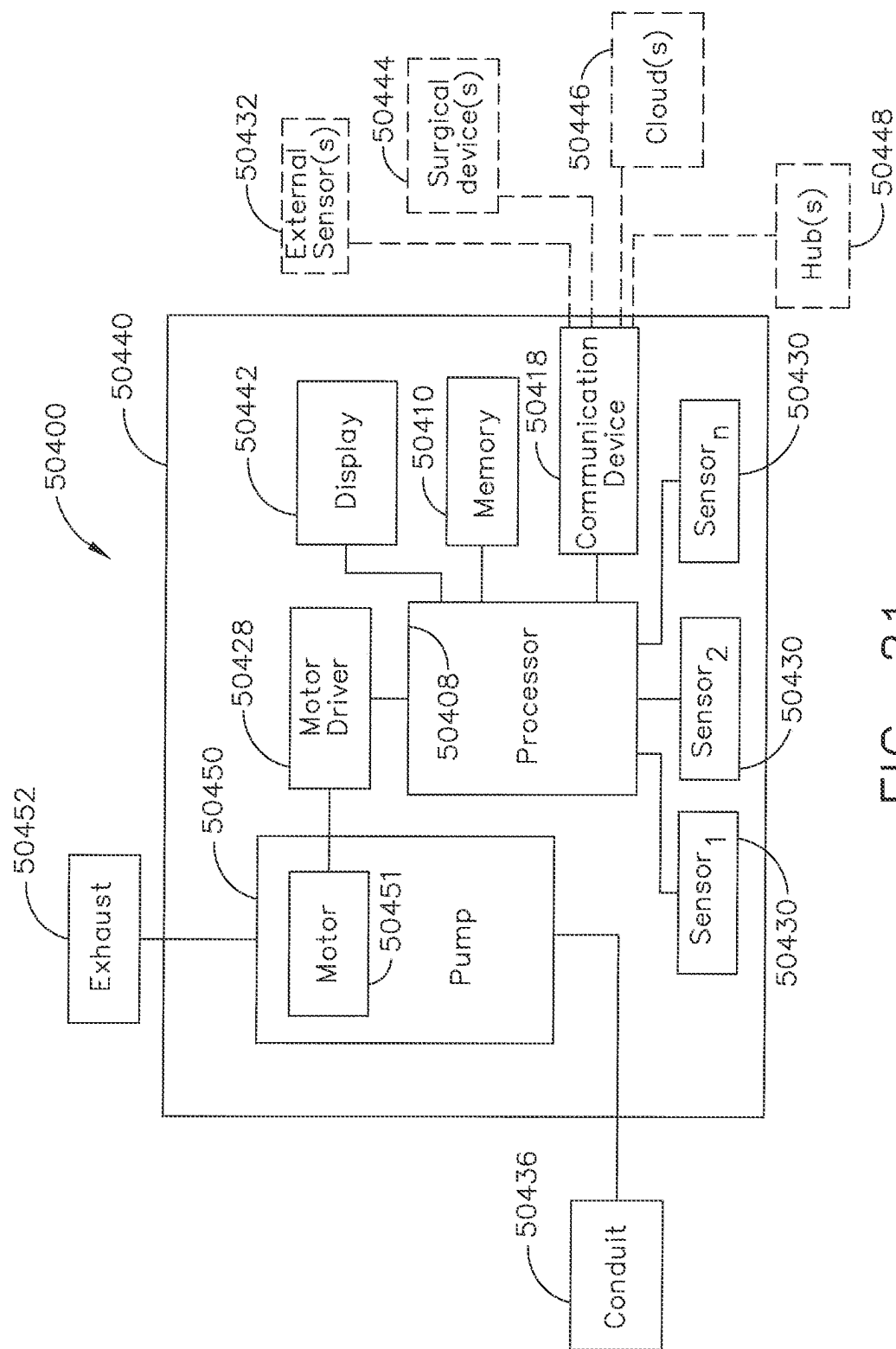


FIG. 21

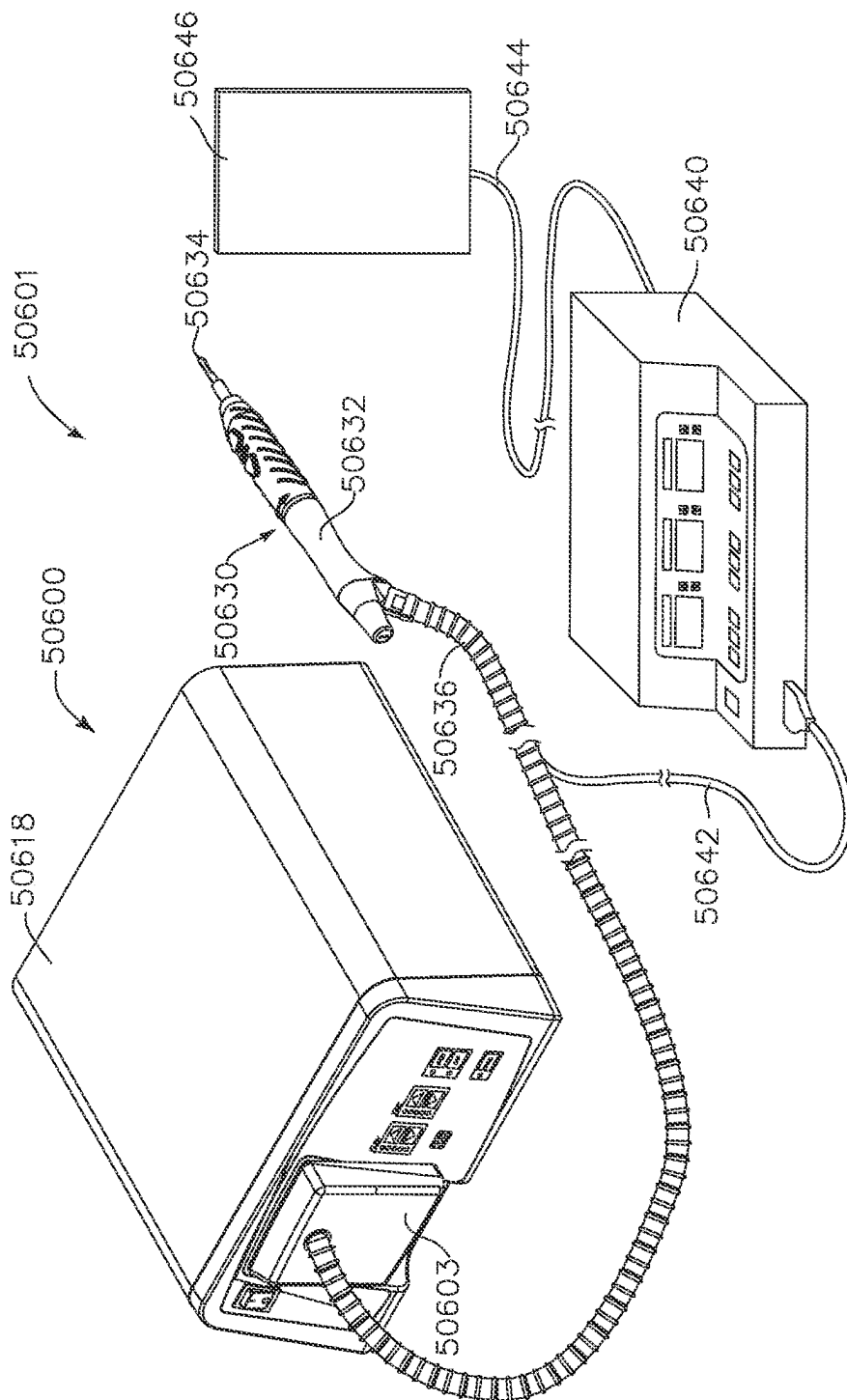


FIG. 22

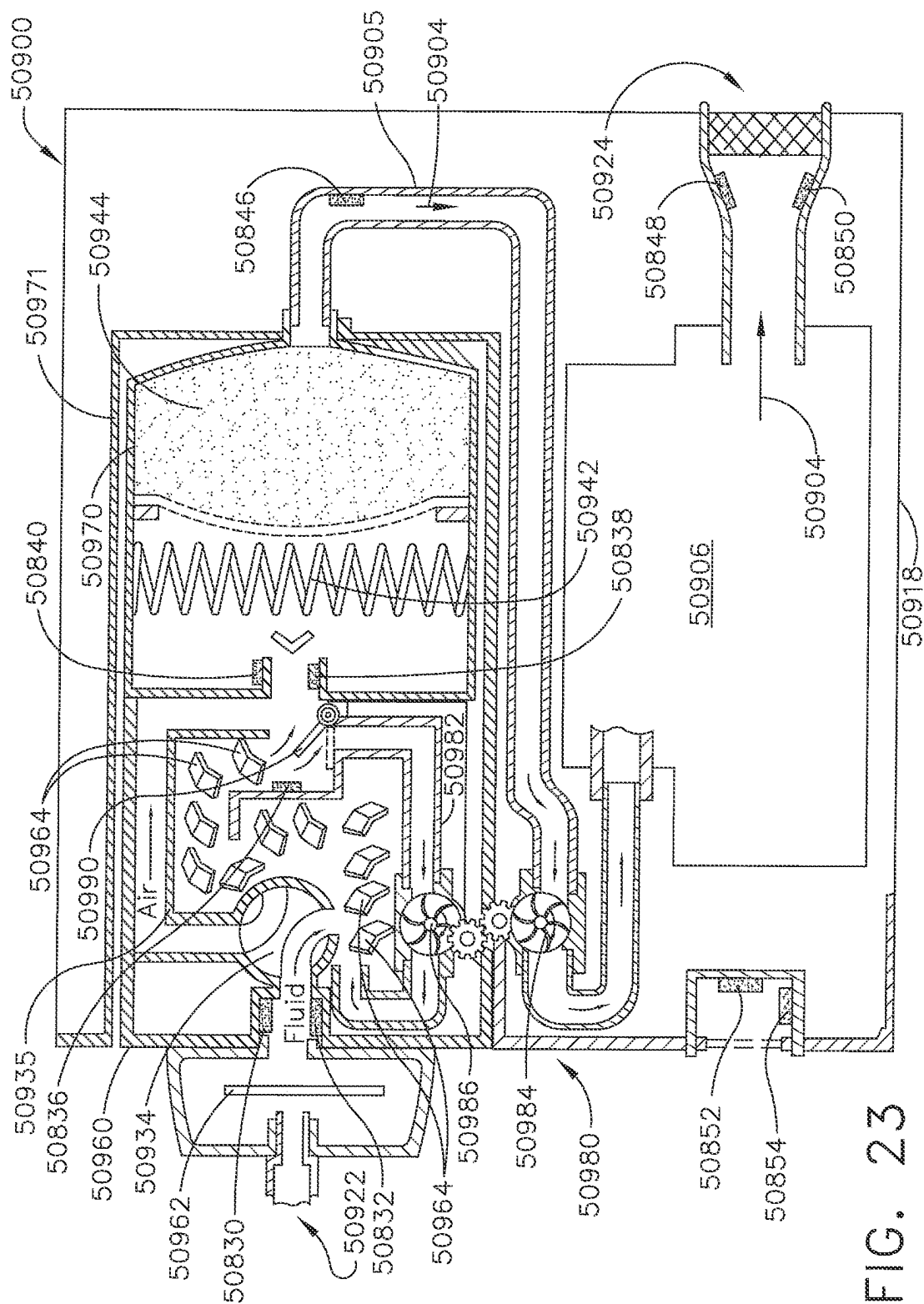


FIG. 23

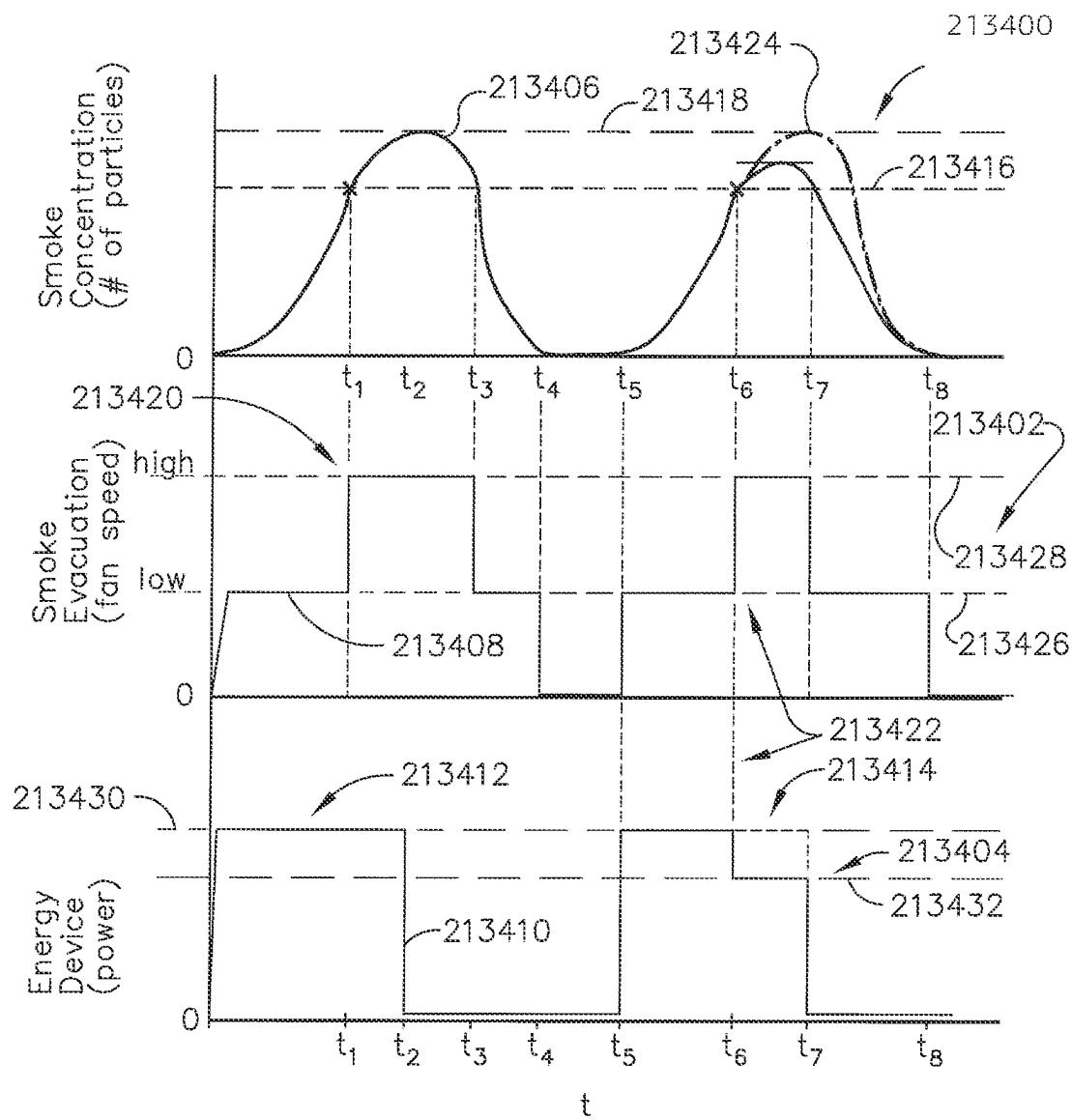


FIG. 24

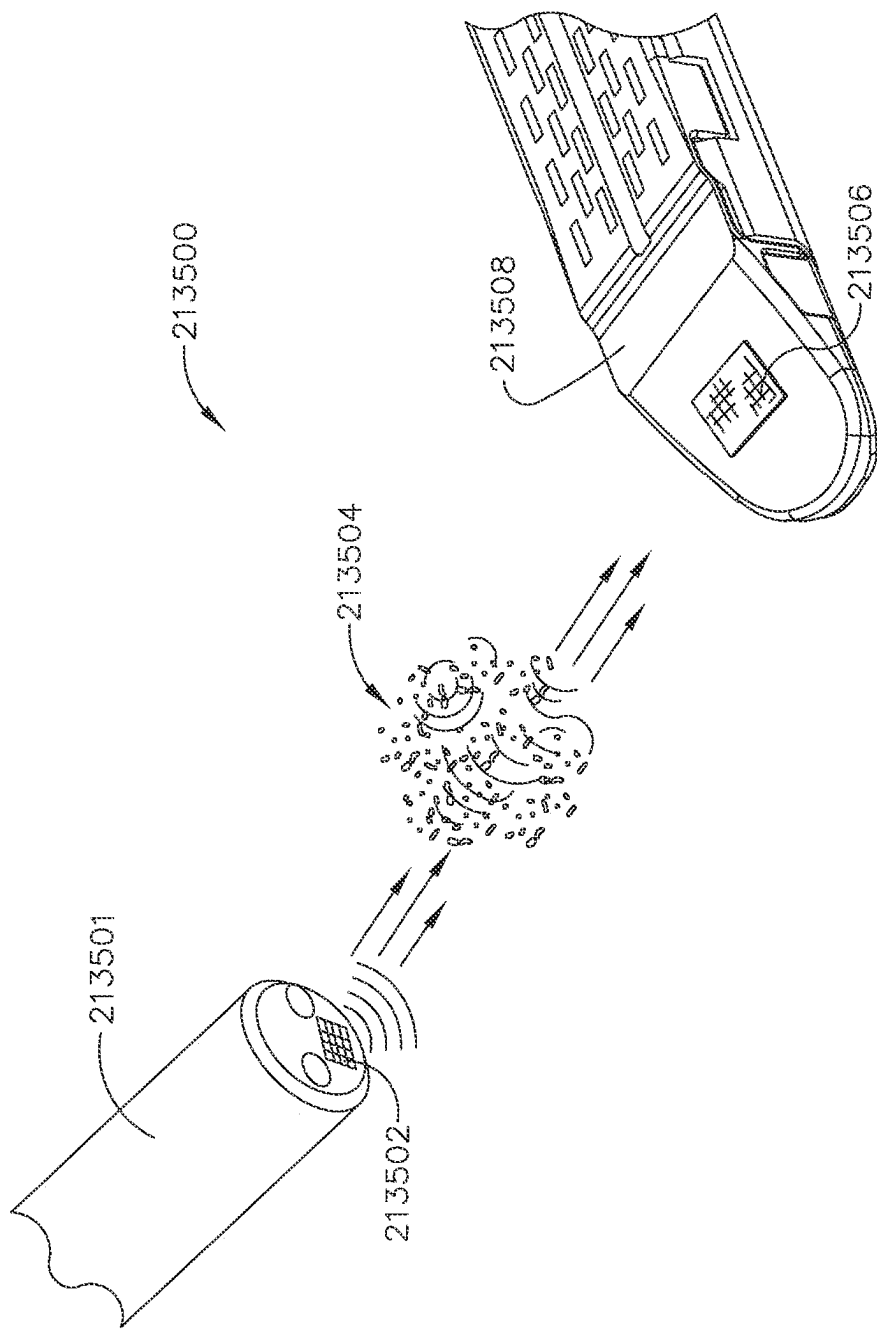
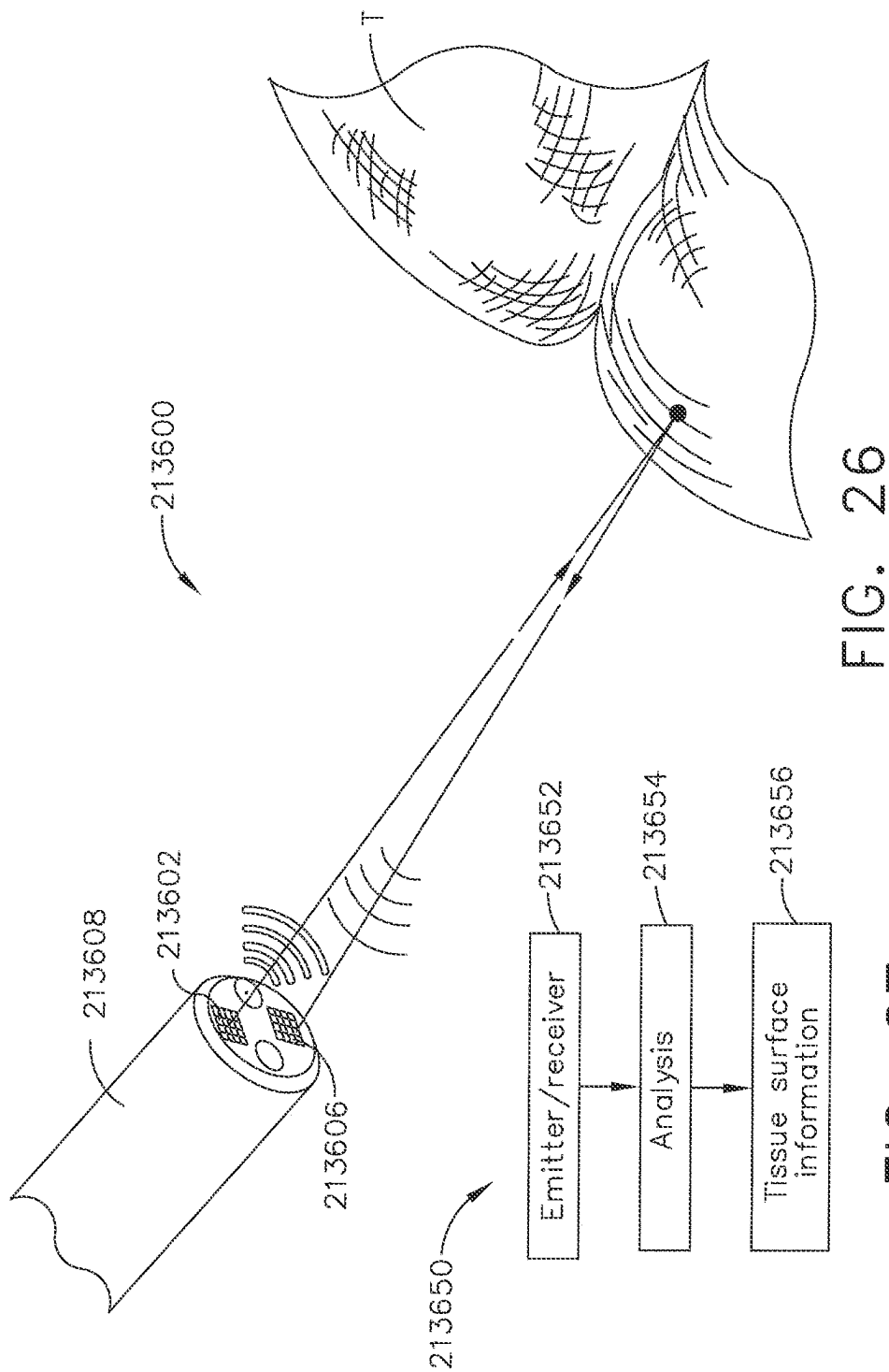


FIG. 25



ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional application claiming priority under 37 35 U.S.C. § 121 to U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS, filed on Nov. 6, 2018, now U.S. Patent Application Publication No. 2019/0201127, the disclosure of each of which is herein incorporated by reference in its entirety.

This application is a divisional application claiming priority under 37 35 U.S.C. § 121 to U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS, filed on Nov. 6, 2018, now U.S. Patent Application Publication No. 2019/0201127, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/729,183, titled A CONTROL FOR A SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE THAT ADJUSTS ITS FUNCTION BASED ON A SENSED SITUATION OR USAGE, filed on Sep. 10, 2018, the disclosure of each of which is herein incorporated by reference in its entirety.

This application is a divisional application claiming priority under 37 35 U.S.C. § 121 to U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS, filed on Nov. 6, 2018, now U.S. Patent Application Publication No. 2019/0201127, which also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/692,747, titled SMART ACTIVATION OF AN ENERGY DEVICE BY ANOTHER DEVICE, filed on Jun. 30, 2018, to U.S. Provisional Patent Application No. 62/692,748, titled SMART ENERGY ARCHITECTURE, filed on Jun. 30, 2018, and to U.S. Provisional Patent Application No. 62/692,768, titled SMART ENERGY DEVICES, filed on Jun. 30, 2018, the disclosure of each of which is herein incorporated by reference in its entirety.

This application is a divisional application claiming priority under 37 35 U.S.C. § 121 to U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS, filed on Nov. 6, 2018, now U.S. Patent Application Publication No. 2019/0201127, which also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed on Apr. 19, 2018, the disclosure of which is herein incorporated by reference in its entirety.

This application is a divisional application claiming priority under 37 35 U.S.C. § 121 to U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS, filed on Nov. 6, 2018, now U.S. Patent Application Publication No. 2019/0201127, which also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/650,898, filed on Mar. 30, 2018, titled CAPACITIVE COUPLED RETURN PATH PAD WITH SEPARABLE ARRAY ELEMENTS, to U.S. Provisional Patent Application Ser. No. 62/650,887, titled SURGICAL SYSTEMS WITH OPTIMIZED SENSING CAPA-

BILITIES, filed Mar. 30, 2018, to U.S. Provisional Patent Application Ser. No. 62/650,882, titled SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM, filed Mar. 30, 2018, and to U.S. Provisional Patent Application Ser. No. 62/650,877, titled SURGICAL SMOKE EVACUATION SENSING AND CONTROLS, filed Mar. 30, 2018, the disclosure of each of which is herein incorporated by reference in its entirety.

This application is a divisional application claiming priority under 37 35 U.S.C. § 121 to U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS, filed on Nov. 6, 2018, now U.S. Patent Application Publication No. 2019/0201127, which also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, filed Mar. 8, 2018, and to U.S. Provisional Patent Application Ser. No. 62/640,415, titled ESTIMATING STATE OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR, filed Mar. 8, 2018, the disclosure of each of which is herein incorporated by reference in its entirety.

This application is a divisional application claiming priority under 37 35 U.S.C. § 121 to U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS, filed on Nov. 6, 2018, now U.S. Patent Application Publication No. 2019/0201127, which also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, to U.S. Provisional Patent Application Ser. No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, and to U.S. Provisional Patent Application Ser. No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of each of which is herein incorporated by reference in its entirety.

BACKGROUND

The present disclosure relates to surgical systems and devices. Clinical information can be collected and/or stored in variety of different surgical devices preoperatively, intraoperatively, and/or postoperatively. Certain information may be helpful to a clinician in making one or more clinical decisions preoperatively, intraoperatively, and/or postoperatively.

FIGURES

The various aspects described herein, both as to organization and methods of operation, together with further objects and advantages thereof, may best be understood by reference to the following description, taken in conjunction with the accompanying drawings as follows.

FIG. 1 is a block diagram of a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

FIG. 2 is a surgical system being used to perform a surgical procedure in an operating room, in accordance with at least one aspect of the present disclosure.

FIG. 3 is a surgical hub paired with a visualization system, a robotic system, and an intelligent instrument, in accordance with at least one aspect of the present disclosure.

FIG. 4 is a partial perspective view of a surgical hub enclosure, and of a combo generator module slidably receiv-

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able in a drawer of the surgical hub enclosure, in accordance with at least one aspect of the present disclosure.

FIG. 5 is a perspective view of a combo generator module with bipolar, ultrasonic, and monopolar contacts and a smoke evacuation component, in accordance with at least one aspect of the present disclosure.

FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

FIG. 7 illustrates a vertical modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

FIG. 8 illustrates a surgical data network comprising a modular communication hub configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to the cloud, in accordance with at least one aspect of the present disclosure.

FIG. 9 illustrates a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

FIG. 10 illustrates a surgical hub comprising a plurality of modules coupled to the modular control tower, in accordance with at least one aspect of the present disclosure.

FIG. 11 illustrates one aspect of a Universal Serial Bus (USB) network hub device, in accordance with at least one aspect of the present disclosure.

FIG. 12 is a block diagram of a cloud computing system comprising a plurality of smart surgical instruments coupled to surgical hubs that may connect to the cloud component of the cloud computing system, in accordance with at least one aspect of the present disclosure.

FIG. 13 is a functional module architecture of a cloud computing system, in accordance with at least one aspect of the present disclosure.

FIG. 14 illustrates a diagram of a situationally aware surgical system, in accordance with at least one aspect of the present disclosure.

FIG. 15 is a timeline depicting situational awareness of a surgical hub, in accordance with at least one aspect of the present disclosure.

FIG. 16 is an elevation view of a surgical instrument including an end effector and a trigger, wherein the degree to which the end effector is closed is controlled by the degree to which the trigger is actuated, in accordance with at least one aspect of the present disclosure.

FIG. 17 is a pair of graphs depicting end effector closure and transducer amplitude relative to trigger angle for the surgical instrument of FIG. 16, in accordance with at least one aspect of the present disclosure.

FIG. 18 is a schematic of a surgical procedure involving a plurality of devices in the surgical field, including a combination energy instrument, a surgical stapler, and a suction/irrigation device, and further depicting a detail elevation view of the combination energy instrument and a detail plan view of a sterile field display that includes touch screen controls for adjusting the surgical functions of the plurality of devices in the sterile field, in accordance with at least one aspect of the present disclosure.

FIG. 19A depicts a display screen including a recommendation for a clinician based on input from a situationally-aware surgical hub, in accordance with at least one aspect of the present disclosure.

FIG. 19B depicts the display screen of FIG. 19A, where the recommendation is indicated with an elevated priority level based on an anticipated surgical act and the input from

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the situationally-aware surgical hub, in accordance with at least one aspect of the present disclosure.

FIG. 20 is a block diagram illustrating intraoperative data correlated with predicted outcomes, and further depicting corresponding recommendations based on the predicted outcomes, in accordance with at least one aspect of the present disclosure.

FIG. 21 is a schematic of a surgical evacuation system, in accordance with at least one aspect of the present disclosure.

FIG. 22 is a perspective view of a surgical system including a surgical evacuation system, in accordance with at least one aspect of the present disclosure.

FIG. 23 is a schematic of an evacuator housing of an evacuation system, in accordance with at least one aspect of the present disclosure.

FIG. 24 is a series of graphs depicting particulate concentration, fan speed, and energy device power relative to time, in accordance with at least one aspect of the present disclosure.

FIG. 25 is an ultrasonic pass-through particle detection system, in accordance with at least one aspect of the present disclosure.

FIG. 26 is an ultrasonic reflection particle detection system, in accordance with at least one aspect of the present disclosure.

FIG. 27 is a flowchart for detecting tissue surface information using the ultrasonic reflection particle detection system of FIG. 26, in accordance with at least one aspect of the present disclosure.

DESCRIPTION

Applicant of the present application owns the following U.S. patent applications, filed on Nov. 6, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 16/182,224, titled SURGICAL NETWORK, INSTRUMENT, AND CLOUD RESPONSES BASED ON VALIDATION OF RECEIVED DATASET AND AUTHENTICATION OF ITS SOURCE AND INTEGRITY, now U.S. Pat. No. 11,308,075;

U.S. patent application Ser. No. 16/182,230, titled SURGICAL SYSTEM FOR PRESENTING INFORMATION INTERPRETED FROM EXTERNAL DATA, now U.S. Patent Application Publication No. 2019/0200980;

U.S. patent application Ser. No. 16/182,233, titled SURGICAL SYSTEMS WITH AUTONOMOUSLY ADJUSTABLE CONTROL PROGRAMS, now U.S. Patent Application Publication No. 2019/0201123;

U.S. patent application Ser. No. 16/182,239, titled ADJUSTMENT OF DEVICE CONTROL PROGRAMS BASED ON STRATIFIED CONTEXTUAL DATA IN ADDITION TO THE DATA, now U.S. Pat. No. 11,423,007;

U.S. patent application Ser. No. 16/182,243, titled SURGICAL HUB AND MODULAR DEVICE RESPONSE ADJUSTMENT BASED ON SITUATIONAL AWARENESS, now U.S. Pat. No. 11,273,001;

U.S. patent application Ser. No. 16/182,248, titled DETECTION AND ESCALATION OF SECURITY RESPONSES OF SURGICAL INSTRUMENTS TO INCREASING SEVERITY THREATS, now U.S. Pat. No. 10,943,454;

U.S. patent application Ser. No. 16/182,251, titled INTERACTIVE SURGICAL SYSTEM, now U.S. Pat. No. 11,278,281;

U.S. patent application Ser. No. 16/182,260, titled AUTOMATED DATA SCALING, ALIGNMENT, AND ORGANIZING BASED ON PREDEFINED PARAMETERS WITHIN SURGICAL NETWORKS, now U.S. Pat. No. 11,056,244;

U.S. patent application Ser. No. 16/182,267, titled SENSING THE PATIENT POSITION AND CONTACT UTILIZING THE MONO-POLAR RETURN PAD ELECTRODE TO PROVIDE SITUATIONAL AWARENESS TO THE HUB, now U.S. Patent Application Publication No. 2019/0201128;

U.S. patent application Ser. No. 16/182,249, titled POWERED SURGICAL TOOL WITH PREDEFINED ADJUSTABLE CONTROL ALGORITHM FOR CONTROLLING END EFFECTOR PARAMETER, now U.S. Pat. No. 11,234,756;

U.S. patent application Ser. No. 16/182,246, titled ADJUSTMENTS BASED ON AIRBORNE PARTICLE PROPERTIES, now U.S. Patent Application Publication No. 2019/0204201;

U.S. patent application Ser. No. 16/182,242, titled REAL-TIME ANALYSIS OF COMPREHENSIVE COST OF ALL INSTRUMENTATION USED IN SURGERY UTILIZING DATA FLUIDITY TO TRACK INSTRUMENTS THROUGH STOCKING AND IN-HOUSE PROCESSES, now U.S. Pat. No. 11,257,589;

U.S. patent application Ser. No. 16/182,255, titled USAGE AND TECHNIQUE ANALYSIS OF SURGEON/STAFF PERFORMANCE AGAINST A BASELINE TO OPTIMIZE DEVICE UTILIZATION AND PERFORMANCE FOR BOTH CURRENT AND FUTURE PROCEDURES, now U.S. Patent Application Publication No. 2019/0201126;

U.S. patent application Ser. No. 16/182,269, titled IMAGE CAPTURING OF THE AREAS OUTSIDE THE ABDOMEN TO IMPROVE PLACEMENT AND CONTROL OF A SURGICAL DEVICE IN USE, now U.S. Pat. No. 11,304,763;

U.S. patent application Ser. No. 16/182,278, titled COMMUNICATION OF DATA WHERE A SURGICAL NETWORK IS USING CONTEXT OF THE DATA AND REQUIREMENTS OF A RECEIVING SYSTEM/USER TO INFLUENCE INCLUSION OR LINKAGE OF DATA AND METADATA TO ESTABLISH CONTINUITY, now U.S. Patent Application Publication No. 2019/0201130;

U.S. patent application Ser. No. 16/182,290, titled SURGICAL NETWORK RECOMMENDATIONS FROM REAL TIME ANALYSIS OF PROCEDURE VARIABLES AGAINST A BASELINE HIGHLIGHTING DIFFERENCES FROM THE OPTIMAL SOLUTION, now U.S. Patent Application Publication No. 2019/0201102;

U.S. patent application Ser. No. 16/182,232, titled CONTROL OF A SURGICAL SYSTEM THROUGH A SURGICAL BARRIER, now U.S. Patent Application Publication No. 2019/0201158;

U.S. patent application Ser. No. 16/182,227, titled SURGICAL NETWORK DETERMINATION OF PRIORITIZATION OF COMMUNICATION, INTERACTION, OR PROCESSING BASED ON SYSTEM OR DEVICE NEEDS, now U.S. Pat. No. 10,892,995;

U.S. patent application Ser. No. 16/182,231, titled WIRELESS PAIRING OF A SURGICAL DEVICE WITH

ANOTHER DEVICE WITHIN A STERILE SURGICAL FIELD BASED ON THE USAGE AND SITUATIONAL AWARENESS OF DEVICES, now U.S. Pat. No. 10,758,310;

U.S. patent application Ser. No. 16/182,229, titled ADJUSTMENT OF STAPLE HEIGHT OF AT LEAST ONE ROW OF STAPLES BASED ON THE SENSED TISSUE THICKNESS OR FORCE IN CLOSING, now U.S. Pat. No. 11,096,693;

U.S. patent application Ser. No. 16/182,234, titled STAPLING DEVICE WITH BOTH COMPULSORY AND DISCRETIONARY LOCKOUTS BASED ON SENSED PARAMETERS, now U.S. Patent Application Publication No. 2019/0200997;

U.S. patent application Ser. No. 16/182,240, titled POWERED STAPLING DEVICE CONFIGURED TO ADJUST FORCE, ADVANCEMENT SPEED, AND OVERALL STROKE OF CUTTING MEMBER BASED ON SENSED PARAMETER OF FIRING OR CLAMPING, now U.S. Patent Application Publication No. 2019/0201034;

U.S. patent application Ser. No. 16/182,235, titled VARIATION OF RADIO FREQUENCY AND ULTRASONIC POWER LEVEL IN COOPERATION WITH VARYING CLAMP ARM PRESSURE TO ACHIEVE PREDEFINED HEAT FLUX OR POWER APPLIED TO TISSUE, now U.S. Pat. No. 11,446,052; and

U.S. patent application Ser. No. 16/182,238, titled ULTRASONIC ENERGY DEVICE WHICH VARIES PRESSURE APPLIED BY CLAMP ARM TO PROVIDE THRESHOLD CONTROL PRESSURE AT A CUT PROGRESSION LOCATION, now U.S. Pat. No. 11,419,667.

Applicant of the present application owns the following U.S. patent applications, filed on Sep. 10, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application No. 62/729,183, titled A CONTROL FOR A SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE THAT ADJUSTS ITS FUNCTION BASED ON A SENSED SITUATION OR USAGE;

U.S. Provisional Patent Application No. 62/729,177, titled AUTOMATED DATA SCALING, ALIGNMENT, AND ORGANIZING BASED ON PREDEFINED PARAMETERS WITHIN A SURGICAL NETWORK BEFORE TRANSMISSION;

U.S. Provisional Patent Application No. 62/729,176, titled INDIRECT COMMAND AND CONTROL OF A FIRST OPERATING ROOM SYSTEM THROUGH THE USE OF A SECOND OPERATING ROOM SYSTEM WITHIN A STERILE FIELD WHERE THE SECOND OPERATING ROOM SYSTEM HAS PRIMARY AND SECONDARY OPERATING MODES;

U.S. Provisional Patent Application No. 62/729,185, titled POWERED STAPLING DEVICE THAT IS CAPABLE OF ADJUSTING FORCE, ADVANCEMENT SPEED, AND OVERALL STROKE OF CUTTING MEMBER OF THE DEVICE BASED ON SENSED PARAMETER OF FIRING OR CLAMPING;

U.S. Provisional Patent Application No. 62/729,184, titled POWERED SURGICAL TOOL WITH A PREDEFINED ADJUSTABLE CONTROL ALGORITHM

FOR CONTROLLING AT LEAST ONE END EFFECTOR PARAMETER AND A MEANS FOR LIMITING THE ADJUSTMENT;

U.S. Provisional Patent Application No. 62/729,182, titled SENSING THE PATIENT POSITION AND CONTACT UTILIZING THE MONO POLAR RETURN PAD ELECTRODE TO PROVIDE SITUATIONAL AWARENESS TO THE HUB;

U.S. Provisional Patent Application No. 62/729,191, titled SURGICAL NETWORK RECOMMENDATIONS FROM REAL TIME ANALYSIS OF PROCEDURE VARIABLES AGAINST A BASELINE HIGHLIGHTING DIFFERENCES FROM THE OPTIMAL SOLUTION;

U.S. Provisional Patent Application No. 62/729,195, titled ULTRASONIC ENERGY DEVICE WHICH VARIES PRESSURE APPLIED BY CLAMP ARM TO PROVIDE THRESHOLD CONTROL PRESSURE AT A CUT PROGRESSION LOCATION; and

U.S. Provisional Patent Application No. 62/729,186, titled WIRELESS PAIRING OF A SURGICAL DEVICE WITH ANOTHER DEVICE WITHIN A STERILE SURGICAL FIELD BASED ON THE USAGE AND SITUATIONAL AWARENESS OF DEVICES.

Applicant of the present application owns the following U.S. patent applications, filed on Aug. 28, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 16/115,214, titled ESTIMATING STATE OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR;

U.S. patent application Ser. No. 16/115,205, titled TEMPERATURE CONTROL OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR;

U.S. patent application Ser. No. 16/115,233, titled RADIO FREQUENCY ENERGY DEVICE FOR DELIVERING COMBINED ELECTRICAL SIGNALS;

U.S. patent application Ser. No. 16/115,208, titled CONTROLLING AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO TISSUE LOCATION;

U.S. patent application Ser. No. 16/115,220, titled CONTROLLING ACTIVATION OF AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO THE PRESENCE OF TISSUE;

U.S. patent application Ser. No. 16/115,232, titled DETERMINING TISSUE COMPOSITION VIA AN ULTRASONIC SYSTEM;

U.S. patent application Ser. No. 16/115,239, titled DETERMINING THE STATE OF AN ULTRASONIC ELECTROMECHANICAL SYSTEM ACCORDING TO FREQUENCY SHIFT;

U.S. patent application Ser. No. 16/115,247, titled DETERMINING THE STATE OF AN ULTRASONIC END EFFECTOR;

U.S. patent application Ser. No. 16/115,211, titled SITUATIONAL AWARENESS OF ELECTROSURGICAL SYSTEMS;

U.S. patent application Ser. No. 16/115,226, titled MECHANISMS FOR CONTROLLING DIFFERENT ELECTROMECHANICAL SYSTEMS OF AN ELECTROSURGICAL INSTRUMENT;

U.S. patent application Ser. No. 16/115,240, titled DETECTION OF END EFFECTOR IMMERSION IN LIQUID;

U.S. patent application Ser. No. 16/115,249, titled INTERRUPTION OF ENERGY DUE TO INADVERTENT CAPACITIVE COUPLING;

U.S. patent application Ser. No. 16/115,256, titled INCREASING RADIO FREQUENCY TO CREATE PAD-LESS MONOPOLAR LOOP;

U.S. patent application Ser. No. 16/115,223, titled BIPOLAR COMBINATION DEVICE THAT AUTOMATICALLY ADJUSTS PRESSURE BASED ON ENERGY MODALITY; and

U.S. patent application Ser. No. 16/115,238, titled ACTIVATION OF ENERGY DEVICES.

Applicant of the present application owns the following U.S. patent applications, filed on Aug. 23, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application No. 62/721,995, titled CONTROLLING AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO TISSUE LOCATION;

U.S. Provisional Patent Application No. 62/721,998, titled SITUATIONAL AWARENESS OF ELECTROSURGICAL SYSTEMS;

U.S. Provisional Patent Application No. 62/721,999, titled INTERRUPTION OF ENERGY DUE TO INADVERTENT CAPACITIVE COUPLING;

U.S. Provisional Patent Application No. 62/721,994, titled BIPOLAR COMBINATION DEVICE THAT AUTOMATICALLY ADJUSTS PRESSURE BASED ON ENERGY MODALITY; and

U.S. Provisional Patent Application No. 62/721,996, titled RADIO FREQUENCY ENERGY DEVICE FOR DELIVERING COMBINED ELECTRICAL SIGNALS.

Applicant of the present application owns the following U.S. patent applications, filed on Jun. 30, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application No. 62/692,747, titled SMART ACTIVATION OF AN ENERGY DEVICE BY ANOTHER DEVICE;

U.S. Provisional Patent Application No. 62/692,748, titled SMART ENERGY ARCHITECTURE; and

U.S. Provisional Patent Application No. 62/692,768, titled SMART ENERGY DEVICES.

Applicant of the present application owns the following U.S. patent applications, filed on Jun. 29, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 16/024,090, titled CAPACITIVE COUPLED RETURN PATH PAD WITH SEPARABLE ARRAY ELEMENTS;

U.S. patent application Ser. No. 16/024,057, titled CONTROLLING A SURGICAL INSTRUMENT ACCORDING TO SENSED CLOSURE PARAMETERS;

U.S. patent application Ser. No. 16/024,067, titled SYSTEMS FOR ADJUSTING END EFFECTOR PARAMETERS BASED ON PERIOPERATIVE INFORMATION;

U.S. patent application Ser. No. 16/024,075, titled SAFETY SYSTEMS FOR SMART POWERED SURGICAL STAPLING;

U.S. patent application Ser. No. 16/024,083, titled SAFETY SYSTEMS FOR SMART POWERED SURGICAL STAPLING;

U.S. patent application Ser. No. 16/024,094, titled SURGICAL SYSTEMS FOR DETECTING END EFFECTOR TISSUE DISTRIBUTION IRREGULARITIES;
 U.S. patent application Ser. No. 16/024,138, titled SYSTEMS FOR DETECTING PROXIMITY OF SURGICAL END EFFECTOR TO CANCEROUS TISSUE;
 U.S. patent application Ser. No. 16/024,150, titled SURGICAL INSTRUMENT CARTRIDGE SENSOR ASSEMBLIES;
 U.S. patent application Ser. No. 16/024,160, titled VARIABLE OUTPUT CARTRIDGE SENSOR ASSEMBLY;
 U.S. patent application Ser. No. 16/024,124, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE ELECTRODE;
 U.S. patent application Ser. No. 16/024,132, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE CIRCUIT;
 U.S. patent application Ser. No. 16/024,141, titled SURGICAL INSTRUMENT WITH A TISSUE MARKING ASSEMBLY;
 U.S. patent application Ser. No. 16/024,162, titled SURGICAL SYSTEMS WITH PRIORITIZED DATA TRANSMISSION CAPABILITIES;
 U.S. patent application Ser. No. 16/024,066, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL;
 U.S. patent application Ser. No. 16/024,096, titled SURGICAL EVACUATION SENSOR ARRANGEMENTS;
 U.S. patent application Ser. No. 16/024,116, titled SURGICAL EVACUATION FLOW PATHS;
 U.S. patent application Ser. No. 16/024,149, titled SURGICAL EVACUATION SENSING AND GENERATOR CONTROL;
 U.S. patent application Ser. No. 16/024,180, titled SURGICAL EVACUATION SENSING AND DISPLAY;
 U.S. patent application Ser. No. 16/024,245, titled COMMUNICATION OF SMOKE EVACUATION SYSTEM PARAMETERS TO HUB OR CLOUD IN SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM;
 U.S. patent application Ser. No. 16/024,258, titled SMOKE EVACUATION SYSTEM INCLUDING A SEGMENTED CONTROL CIRCUIT FOR INTERACTIVE SURGICAL PLATFORM;
 U.S. patent application Ser. No. 16/024,265, titled SURGICAL EVACUATION SYSTEM WITH A COMMUNICATION CIRCUIT FOR COMMUNICATION BETWEEN A FILTER AND A SMOKE EVACUATION DEVICE; and
 U.S. patent application Ser. No. 16/024,273, titled DUAL IN-SERIES LARGE AND SMALL DROPLET FILTERS.

Applicant of the present application owns the following U.S. Provisional patent applications, filed on Jun. 28, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/691,228, titled A METHOD OF USING REINFORCED FLEX CIRCUITS WITH MULTIPLE SENSORS WITH ELECTROSURGICAL DEVICES;
 U.S. Provisional Patent Application Ser. No. 62/691,227, titled CONTROLLING A SURGICAL INSTRUMENT ACCORDING TO SENSED CLOSURE PARAMETERS;

U.S. Provisional Patent Application Ser. No. 62/691,230, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE ELECTRODE;
 U.S. Provisional Patent Application Ser. No. 62/691,219, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL;
 U.S. Provisional Patent Application Ser. No. 62/691,257, titled COMMUNICATION OF SMOKE EVACUATION SYSTEM PARAMETERS TO HUB OR CLOUD IN SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM;
 U.S. Provisional Patent Application Ser. No. 62/691,262, titled SURGICAL EVACUATION SYSTEM WITH A COMMUNICATION CIRCUIT FOR COMMUNICATION BETWEEN A FILTER AND A SMOKE EVACUATION DEVICE; and
 U.S. Provisional Patent Application Ser. No. 62/691,251, titled DUAL IN-SERIES LARGE AND SMALL DROPLET FILTERS.

Applicant of the present application owns the following U.S. Provisional patent application, filed on Apr. 19, 2018, the disclosure of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/659,900, titled METHOD OF HUB COMMUNICATION.

Applicant of the present application owns the following U.S. Provisional patent applications, filed on Mar. 30, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application No. 62/650,898 filed on Mar. 30, 2018, titled CAPACITIVE COUPLED RETURN PATH PAD WITH SEPARABLE ARRAY ELEMENTS;
 U.S. Provisional Patent Application Ser. No. 62/650,887, titled SURGICAL SYSTEMS WITH OPTIMIZED SENSING CAPABILITIES;
 U.S. Provisional Patent Application Ser. No. 62/650,882, titled SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM; and
 U.S. Provisional Patent Application Ser. No. 62/650,877, titled SURGICAL SMOKE EVACUATION SENSING AND CONTROLS.

Applicant of the present application owns the following U.S. patent applications, filed on Mar. 29, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 15/940,641, titled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES;
 U.S. patent application Ser. No. 15/940,648, titled INTERACTIVE SURGICAL SYSTEMS WITH CONDITION HANDLING OF DEVICES AND DATA CAPABILITIES;
 U.S. patent application Ser. No. 15/940,656, titled SURGICAL HUB COORDINATION OF CONTROL AND COMMUNICATION OF OPERATING ROOM DEVICES;
 U.S. patent application Ser. No. 15/940,666, titled SPATIAL AWARENESS OF SURGICAL HUBS IN OPERATING ROOMS;
 U.S. patent application Ser. No. 15/940,670, titled COOPERATIVE UTILIZATION OF DATA DERIVED FROM SECONDARY SOURCES BY INTELLIGENT SURGICAL HUBS;
 U.S. patent application Ser. No. 15/940,677, titled SURGICAL HUB CONTROL ARRANGEMENTS;

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U.S. patent application Ser. No. 15/940,632, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD;

U.S. patent application Ser. No. 15/940,640, titled COMMUNICATION HUB AND STORAGE DEVICE FOR STORING PARAMETERS AND STATUS OF A SURGICAL DEVICE TO BE SHARED WITH CLOUD BASED ANALYTICS SYSTEMS;

U.S. patent application Ser. No. 15/940,645, titled SELF DESCRIBING DATA PACKETS GENERATED AT AN ISSUING INSTRUMENT;

U.S. patent application Ser. No. 15/940,649, titled DATA PAIRING TO INTERCONNECT A DEVICE MEASURED PARAMETER WITH AN OUTCOME;

U.S. patent application Ser. No. 15/940,654, titled SURGICAL HUB SITUATIONAL AWARENESS;

U.S. patent application Ser. No. 15/940,663, titled SURGICAL SYSTEM DISTRIBUTED PROCESSING;

U.S. patent application Ser. No. 15/940,668, titled AGGREGATION AND REPORTING OF SURGICAL HUB DATA;

U.S. patent application Ser. No. 15/940,671, titled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER;

U.S. patent application Ser. No. 15/940,686, titled DISPLAY OF ALIGNMENT OF STAPLE CARTRIDGE TO PRIOR LINEAR STAPLE LINE;

U.S. patent application Ser. No. 15/940,700, titled STERILE FIELD INTERACTIVE CONTROL DISPLAYS;

U.S. patent application Ser. No. 15/940,629, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS;

U.S. patent application Ser. No. 15/940,704, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT;

U.S. patent application Ser. No. 15/940,722, titled CHARACTERIZATION OF TISSUE IRREGULARITIES THROUGH THE USE OF MONO-CHROMATIC LIGHT REFRACTIVITY;

U.S. patent application Ser. No. 15/940,742, titled DUAL CMOS ARRAY IMAGING;

U.S. patent application Ser. No. 15/940,636, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES;

U.S. patent application Ser. No. 15/940,653, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL HUBS;

U.S. patent application Ser. No. 15/940,660, titled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER;

U.S. patent application Ser. No. 15/940,679, titled CLOUD-BASED MEDICAL ANALYTICS FOR LINKING OF LOCAL USAGE TRENDS WITH THE RESOURCE ACQUISITION BEHAVIORS OF LARGER DATA SET;

U.S. patent application Ser. No. 15/940,694, titled CLOUD-BASED MEDICAL ANALYTICS FOR MEDICAL FACILITY SEGMENTED INDIVIDUALIZATION OF INSTRUMENT FUNCTION;

U.S. patent application Ser. No. 15/940,634, titled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES;

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U.S. patent application Ser. No. 15/940,706, titled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK;

U.S. patent application Ser. No. 15/940,675, titled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES;

U.S. patent application Ser. No. 15/940,627, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. patent application Ser. No. 15/940,637, titled COMMUNICATION ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. patent application Ser. No. 15/940,642, titled CONTROLS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. patent application Ser. No. 15/940,676, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. patent application Ser. No. 15/940,680, titled CONTROLLERS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. patent application Ser. No. 15/940,683, titled COOPERATIVE SURGICAL ACTIONS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. patent application Ser. No. 15/940,690, titled DISPLAY ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; and

U.S. patent application Ser. No. 15/940,711, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS.

Applicant of the present application owns the following U.S. Provisional patent applications, filed on Mar. 28, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/649,302, titled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES;

U.S. Provisional Patent Application Ser. No. 62/649,294, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD;

U.S. Provisional Patent Application Ser. No. 62/649,300, titled SURGICAL HUB SITUATIONAL AWARENESS;

U.S. Provisional Patent Application Ser. No. 62/649,309, titled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER;

U.S. Provisional Patent Application Ser. No. 62/649,310, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS;

U.S. Provisional Patent Application Ser. No. 62/649,291, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT;

U.S. Provisional Patent Application Ser. No. 62/649,296, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES;

U.S. Provisional Patent Application Ser. No. 62/649,333, titled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER;

U.S. Provisional Patent Application Ser. No. 62/649,327, titled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES;

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U.S. Provisional Patent Application Ser. No. 62/649,315, titled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK;
 U.S. Provisional Patent Application Ser. No. 62/649,313, titled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES;
 U.S. Provisional Patent Application Ser. No. 62/649,320, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;
 U.S. Provisional Patent Application Ser. No. 62/649,307, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; and
 U.S. Provisional Patent Application Ser. No. 62/649,323, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS.

Applicant of the present application owns the following U.S. Provisional patent applications, filed on Mar. 8, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR; and
 U.S. Provisional Patent Application Ser. No. 62/640,415, titled ESTIMATING STATE OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR.

Applicant of the present application owns the following U.S. Provisional patent applications, filed on Dec. 28, 2017, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM;
 U.S. Provisional Patent Application Ser. No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS; and
 U.S. Provisional Patent Application Ser. No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM.

Before explaining various aspects of surgical devices and generators in detail, it should be noted that the illustrative examples are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative examples may be implemented or incorporated in other aspects, variations and modifications, and may be practiced or carried out in various ways. Further, unless otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the illustrative examples for the convenience of the reader and are not for the purpose of limitation thereof. Also, it will be appreciated that one or more of the following-described aspects, expressions of aspects, and/or examples, can be combined with any one or more of the other following-described aspects, expressions of aspects and/or examples.

Surgical Hubs

Referring to FIG. 1, a computer-implemented interactive surgical system 100 includes one or more surgical systems 102 and a cloud-based system (e.g., the cloud 104 that may include a remote server 113 coupled to a storage device 105). Each surgical system 102 includes at least one surgical hub 106 in communication with the cloud 104 that may include a remote server 113. In one example, as illustrated in FIG. 1, the surgical system 102 includes a visualization system 108, a robotic system 110, and a handheld intelligent surgical instrument 112, which are configured to communicate with one another and/or the hub 106. In some aspects, a surgical system 102 may include an M number of hubs

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106, an N number of visualization systems 108, an O number of robotic systems 110, and a P number of handheld intelligent surgical instruments 112, where M, N, O, and P are integers greater than or equal to one.

In various aspects, the intelligent instruments 112 as described herein with reference to FIGS. 1-7 may be implemented as an energy device 213000 (FIG. 16), a combination energy instrument 213100 (FIG. 18), a surgical stapler 213102 (FIG. 18), a suction/irrigation device 213104 (FIG. 18), a sterile field display 213108 (FIG. 18), a surgical device 213204 (FIGS. 19A and 19B), a surgical evacuation system 50400 (FIG. 21), an electrosurgical instrument 50630 (FIG. 22), an evacuation system 50600 (FIG. 22), an evacuation system 50900 (FIG. 23), a scope 213501 (FIG. 25), and a surgical device 213608 (FIG. 26), for example. In such instances, the intelligent instruments 112 (e.g., devices 1_a-1_n), such as the combination energy instrument 213100 (FIG. 18), the surgical stapler 213102 (FIG. 18), the suction/irrigation device 213104 (FIG. 18), the sterile field display 213108 (FIG. 18), the surgical device 213204 (FIGS. 19A and 19B), the surgical evacuation system 50400 (FIG. 21), the electrosurgical instrument 50630 (FIG. 22), the evacuation system 50600 (FIG. 22), the evacuation system 50900 (FIG. 23), the scope 213501 (FIG. 25), and the surgical device 213608 (FIG. 26), are configured to operate in a surgical data network 201 as described with reference to FIG. 8.

FIG. 2 depicts an example of a surgical system 102 being used to perform a surgical procedure on a patient who is lying down on an operating table 114 in a surgical operating room 116. A robotic system 110 is used in the surgical procedure as a part of the surgical system 102. The robotic system 110 includes a surgeon's console 118, a patient side cart 120 (surgical robot), and a surgical robotic hub 122. The patient side cart 120 can manipulate at least one removably coupled surgical tool 117 through a minimally invasive incision in the body of the patient while the surgeon views the surgical site through the surgeon's console 118. An image of the surgical site can be obtained by a medical imaging device 124, which can be manipulated by the patient side cart 120 to orient the imaging device 124. The robotic hub 122 can be used to process the images of the surgical site for subsequent display to the surgeon through the surgeon's console 118.

Other types of robotic systems can be readily adapted for use with the surgical system 102. Various examples of robotic systems and surgical tools that are suitable for use with the present disclosure are described in U.S. Provisional Patent Application Ser. No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

Various examples of cloud-based analytics that are performed by the cloud 104, and are suitable for use with the present disclosure, are described in U.S. Provisional Patent Application Ser. No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

In various aspects, the imaging device 124 includes at least one image sensor and one or more optical components. Suitable image sensors include, but are not limited to, Charge-Coupled Device (CCD) sensors and Complementary Metal-Oxide Semiconductor (CMOS) sensors.

The optical components of the imaging device 124 may include one or more illumination sources and/or one or more lenses. The one or more illumination sources may be

directed to illuminate portions of the surgical field. The one or more image sensors may receive light reflected or refracted from the surgical field, including light reflected or refracted from tissue and/or surgical instruments.

The one or more illumination sources may be configured to radiate electromagnetic energy in the visible spectrum as well as the invisible spectrum. The visible spectrum, sometimes referred to as the optical spectrum or luminous spectrum, is that portion of the electromagnetic spectrum that is visible to (i.e., can be detected by) the human eye and may be referred to as visible light or simply light. A typical human eye will respond to wavelengths in air that are from about 380 nm to about 750 nm.

The invisible spectrum (i.e., the non-luminous spectrum) is that portion of the electromagnetic spectrum that lies below and above the visible spectrum (i.e., wavelengths below about 380 nm and above about 750 nm). The invisible spectrum is not detectable by the human eye. Wavelengths greater than about 750 nm are longer than the red visible spectrum, and they become invisible infrared (IR), micro-wave, and radio electromagnetic radiation. Wavelengths less than about 380 nm are shorter than the violet spectrum, and they become invisible ultraviolet, x-ray, and gamma ray electromagnetic radiation.

In various aspects, the imaging device **124** is configured for use in a minimally invasive procedure. Examples of imaging devices suitable for use with the present disclosure include, but not limited to, an arthroscope, angioscope, bronchoscope, choledochoscope, colonoscope, cytoscope, duodenoscope, enteroscope, esophagogastric-duodenoscope (gastroscope), endoscope, laryngoscope, nasopharyngoscopy, sigmoidoscope, thoracoscope, and ureteroscopy.

In one aspect, the imaging device employs multi-spectrum monitoring to discriminate topography and underlying structures. A multi-spectral image is one that captures image data within specific wavelength ranges across the electromagnetic spectrum. The wavelengths may be separated by filters or by the use of instruments that are sensitive to particular wavelengths, including light from frequencies beyond the visible light range, e.g., IR and ultraviolet. Spectral imaging can allow extraction of additional information the human eye fails to capture with its receptors for red, green, and blue. The use of multi-spectral imaging is described in greater detail under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. Multi-spectrum monitoring can be a useful tool in relocating a surgical field after a surgical task is completed to perform one or more of the previously described tests on the treated tissue.

It is axiomatic that strict sterilization of the operating room and surgical equipment is required during any surgery. The strict hygiene and sterilization conditions required in a “surgical theater,” i.e., an operating or treatment room, necessitate the highest possible sterility of all medical devices and equipment. Part of that sterilization process is the need to sterilize anything that comes in contact with the patient or penetrates the sterile field, including the imaging device **124** and its attachments and components. It will be appreciated that the sterile field may be considered a specified area, such as within a tray or on a sterile towel, that is considered free of microorganisms, or the sterile field may be considered an area, immediately around a patient, who has been prepared for a surgical procedure. The sterile field

may include the scrubbed team members, who are properly attired, and all furniture and fixtures in the area.

In various aspects, the visualization system **108** includes one or more imaging sensors, one or more image-processing units, one or more storage arrays, and one or more displays that are strategically arranged with respect to the sterile field, as illustrated in FIG. 2. In one aspect, the visualization system **108** includes an interface for HL7, PACS, and EMR. Various components of the visualization system **108** are described under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

As illustrated in FIG. 2, a primary display **119** is positioned in the sterile field to be visible to an operator at the operating table **114**. In addition, a visualization tower **111** is positioned outside the sterile field. The visualization tower **111** includes a first non-sterile display **107** and a second non-sterile display **109**, which face away from each other. The visualization system **108**, guided by the hub **106**, is configured to utilize the displays **107**, **109**, and **119** to coordinate information flow to operators inside and outside the sterile field. For example, the hub **106** may cause the visualization system **108** to display a snapshot of a surgical site, as recorded by an imaging device **124**, on a non-sterile display **107** or **109**, while maintaining a live feed of the surgical site on the primary display **119**. The snapshot on the non-sterile display **107** or **109** can permit a non-sterile operator to perform a diagnostic step relevant to the surgical procedure, for example.

In one aspect, the hub **106** is also configured to route a diagnostic input or feedback entered by a non-sterile operator at the visualization tower **111** to the primary display **119** within the sterile field, where it can be viewed by a sterile operator at the operating table. In one example, the input can be in the form of a modification to the snapshot displayed on the non-sterile display **107** or **109**, which can be routed to the primary display **119** by the hub **106**.

Referring to FIG. 2, a surgical instrument **112** is being used in the surgical procedure as part of the surgical system **102**. The hub **106** is also configured to coordinate information flow to a display of the surgical instrument **112**. For example, coordinate information flow is further described in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. A diagnostic input or feedback entered by a non-sterile operator at the visualization tower **111** can be routed by the hub **106** to the surgical instrument display **115** within the sterile field, where it can be viewed by the operator of the surgical instrument **112**. Example surgical instruments that are suitable for use with the surgical system **102** are described under the heading “Surgical Instrument Hardware” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety, for example.

Referring now to FIG. 3, a hub **106** is depicted in communication with a visualization system **108**, a robotic system **110**, and a handheld intelligent surgical instrument **112**. The hub **106** includes a hub display **135**, an imaging module **138**, a generator module **140** (which can include a monopolar generator **142**, a bipolar generator **144**, and/or an ultrasonic generator **143**), a communication module **130**, a processor module **132**, and a storage array **134**. In certain

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aspects, as illustrated in FIG. 3, the hub **106** further includes a smoke evacuation module **126**, a suction/irrigation module **128**, and/or an OR mapping module **133**.

During a surgical procedure, energy application to tissue, for sealing and/or cutting, is generally associated with smoke evacuation, suction of excess fluid, and/or irrigation of the tissue. Fluid, power, and/or data lines from different sources are often entangled during the surgical procedure. Valuable time can be lost addressing this issue during a surgical procedure. Detangling the lines may necessitate disconnecting the lines from their respective modules, which may require resetting the modules. The hub modular enclosure **136** offers a unified environment for managing the power, data, and fluid lines, which reduces the frequency of entanglement between such lines.

Aspects of the present disclosure present a surgical hub for use in a surgical procedure that involves energy application to tissue at a surgical site. The surgical hub includes a hub enclosure and a combo generator module slidably receivable in a docking station of the hub enclosure. The docking station includes data and power contacts. The combo generator module includes two or more of an ultrasonic energy generator component, a bipolar RF energy generator component, and a monopolar RF energy generator component that are housed in a single unit. In one aspect, the combo generator module also includes a smoke evacuation component, at least one energy delivery cable for connecting the combo generator module to a surgical instrument, at least one smoke evacuation component configured to evacuate smoke, fluid, and/or particulates generated by the application of therapeutic energy to the tissue, and a fluid line extending from the remote surgical site to the smoke evacuation component.

In one aspect, the fluid line is a first fluid line and a second fluid line extends from the remote surgical site to a suction and irrigation module slidably received in the hub enclosure. In one aspect, the hub enclosure comprises a fluid interface.

Certain surgical procedures may require the application of more than one energy type to the tissue. One energy type may be more beneficial for cutting the tissue, while another different energy type may be more beneficial for sealing the tissue. For example, a bipolar generator can be used to seal the tissue while an ultrasonic generator can be used to cut the sealed tissue. Aspects of the present disclosure present a solution where a hub modular enclosure **136** is configured to accommodate different generators, and facilitate an interactive communication therebetween. One of the advantages of the hub modular enclosure **136** is enabling the quick removal and/or replacement of various modules.

Aspects of the present disclosure present a modular surgical enclosure for use in a surgical procedure that involves energy application to tissue. The modular surgical enclosure includes a first energy-generator module, configured to generate a first energy for application to the tissue, and a first docking station comprising a first docking port that includes first data and power contacts, wherein the first energy-generator module is slidably movable into an electrical engagement with the power and data contacts and wherein the first energy-generator module is slidably movable out of the electrical engagement with the first power and data contacts,

Further to the above, the modular surgical enclosure also includes a second energy-generator module configured to generate a second energy, different than the first energy, for application to the tissue, and a second docking station comprising a second docking port that includes second data and power contacts, wherein the second energy-generator

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module is slidably movable into an electrical engagement with the power and data contacts, and wherein the second energy-generator module is slidably movable out of the electrical engagement with the second power and data contacts.

In addition, the modular surgical enclosure also includes a communication bus between the first docking port and the second docking port, configured to facilitate communication between the first energy-generator module and the second energy-generator module.

Referring to FIGS. 3-7, aspects of the present disclosure are presented for a hub modular enclosure **136** that allows the modular integration of a generator module **140**, a smoke evacuation module **126**, and a suction/irrigation module **128**. The hub modular enclosure **136** further facilitates interactive communication between the modules **140**, **126**, **128**. As illustrated in FIG. 5, the generator module **140** can be a generator module with integrated monopolar, bipolar, and ultrasonic components supported in a single housing unit **139** slidably insertable into the hub modular enclosure **136**. As illustrated in FIG. 5, the generator module **140** can be configured to connect to a monopolar device **146**, a bipolar device **147**, and an ultrasonic device **148**. Alternatively, the generator module **140** may comprise a series of monopolar, bipolar, and/or ultrasonic generator modules that interact through the hub modular enclosure **136**. The hub modular enclosure **136** can be configured to facilitate the insertion of multiple generators and interactive communication between the generators docked into the hub modular enclosure **136** so that the generators would act as a single generator.

In one aspect, the hub modular enclosure **136** comprises a modular power and communication backplane **149** with external and wireless communication headers to enable the removable attachment of the modules **140**, **126**, **128** and interactive communication therebetween.

In one aspect, the hub modular enclosure **136** includes docking stations, or drawers, **151**, herein also referred to as drawers, which are configured to slidably receive the modules **140**, **126**, **128**. FIG. 4 illustrates a partial perspective view of a surgical hub enclosure **136**, and a combo generator module **145** slidably receivable in a docking station **151** of the surgical hub enclosure **136**. A docking port **152** with power and data contacts on a rear side of the combo generator module **145** is configured to engage a corresponding docking port **150** with power and data contacts of a corresponding docking station **151** of the hub modular enclosure **136** as the combo generator module **145** is slid into position within the corresponding docking station **151** of the hub module enclosure **136**. In one aspect, the combo generator module **145** includes a bipolar, ultrasonic, and monopolar module and a smoke evacuation module integrated together into a single housing unit **139**, as illustrated in FIG. 5.

In various aspects, the smoke evacuation module **126** includes a fluid line **154** that conveys captured/collected smoke and/or fluid away from a surgical site and to, for example, the smoke evacuation module **126**. Vacuum suction originating from the smoke evacuation module **126** can draw the smoke into an opening of a utility conduit at the surgical site. The utility conduit, coupled to the fluid line, can be in the form of a flexible tube terminating at the smoke evacuation module **126**. The utility conduit and the fluid line define a fluid path extending toward the smoke evacuation module **126** that is received in the hub enclosure **136**.

In various aspects, the suction/irrigation module **128** is coupled to a surgical tool comprising an aspiration fluid line and a suction fluid line. In one example, the aspiration and

suction fluid lines are in the form of flexible tubes extending from the surgical site toward the suction/irrigation module 128. One or more drive systems can be configured to cause irrigation and aspiration of fluids to and from the surgical site.

In one aspect, the surgical tool includes a shaft having an end effector at a distal end thereof and at least one energy treatment associated with the end effector, an aspiration tube, and an irrigation tube. The aspiration tube can have an inlet port at a distal end thereof and the aspiration tube extends through the shaft. Similarly, an irrigation tube can extend through the shaft and can have an inlet port in proximity to the energy deliver implement. The energy deliver implement is configured to deliver ultrasonic and/or RF energy to the surgical site and is coupled to the generator module 140 by a cable extending initially through the shaft.

The irrigation tube can be in fluid communication with a fluid source, and the aspiration tube can be in fluid communication with a vacuum source. The fluid source and/or the vacuum source can be housed in the suction/irrigation module 128. In one example, the fluid source and/or the vacuum source can be housed in the hub enclosure 136 separately from the suction/irrigation module 128. In such example, a fluid interface can be configured to connect the suction/irrigation module 128 to the fluid source and/or the vacuum source.

In one aspect, the modules 140, 126, 128 and/or their corresponding docking stations on the hub modular enclosure 136 may include alignment features that are configured to align the docking ports of the modules into engagement with their counterparts in the docking stations of the hub modular enclosure 136. For example, as illustrated in FIG. 4, the combo generator module 145 includes side brackets 155 that are configured to slidably engage with corresponding brackets 156 of the corresponding docking station 151 of the hub modular enclosure 136. The brackets cooperate to guide the docking port contacts of the combo generator module 145 into an electrical engagement with the docking port contacts of the hub modular enclosure 136.

In some aspects, the drawers 151 of the hub modular enclosure 136 are the same, or substantially the same size, and the modules are adjusted in size to be received in the drawers 151. For example, the side brackets 155 and/or 156 can be larger or smaller depending on the size of the module. In other aspects, the drawers 151 are different in size and are each designed to accommodate a particular module.

Furthermore, the contacts of a particular module can be keyed for engagement with the contacts of a particular drawer to avoid inserting a module into a drawer with mismatching contacts.

As illustrated in FIG. 4, the docking port 150 of one drawer 151 can be coupled to the docking port 150 of another drawer 151 through a communications link 157 to facilitate an interactive communication between the modules housed in the hub modular enclosure 136. The docking ports 150 of the hub modular enclosure 136 may alternatively, or additionally, facilitate a wireless interactive communication between the modules housed in the hub modular enclosure 136. Any suitable wireless communication can be employed, such as for example Air Titan-Bluetooth.

FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing 160 configured to receive a plurality of modules of a surgical hub 206. The lateral modular housing 160 is configured to laterally receive and interconnect the modules 161. The modules 161 are slidably inserted into docking stations 162 of lateral modular housing 160, which includes

a backplane for interconnecting the modules 161. As illustrated in FIG. 6, the modules 161 are arranged laterally in the lateral modular housing 160. Alternatively, the modules 161 may be arranged vertically in a lateral modular housing.

FIG. 7 illustrates a vertical modular housing 164 configured to receive a plurality of modules 165 of the surgical hub 106. The modules 165 are slidably inserted into docking stations, or drawers, 167 of vertical modular housing 164, which includes a backplane for interconnecting the modules 165. Although the drawers 167 of the vertical modular housing 164 are arranged vertically, in certain instances, a vertical modular housing 164 may include drawers that are arranged laterally. Furthermore, the modules 165 may interact with one another through the docking ports of the vertical modular housing 164. In the example of FIG. 7, a display 177 is provided for displaying data relevant to the operation of the modules 165. In addition, the vertical modular housing 164 includes a master module 178 housing a plurality of sub-modules that are slidably received in the master module 178.

In various aspects, the imaging module 138 comprises an integrated video processor and a modular light source and is adapted for use with various imaging devices. In one aspect, the imaging device is comprised of a modular housing that can be assembled with a light source module and a camera module. The housing can be a disposable housing. In at least one example, the disposable housing is removably coupled to a reusable controller, a light source module, and a camera module. The light source module and/or the camera module can be selectively chosen depending on the type of surgical procedure. In one aspect, the camera module comprises a CCD sensor. In another aspect, the camera module comprises a CMOS sensor. In another aspect, the camera module is configured for scanned beam imaging. Likewise, the light source module can be configured to deliver a white light or a different light, depending on the surgical procedure.

During a surgical procedure, removing a surgical device from the surgical field and replacing it with another surgical device that includes a different camera or a different light source can be inefficient. Temporarily losing sight of the surgical field may lead to undesirable consequences. The module imaging device of the present disclosure is configured to permit the replacement of a light source module or a camera module midstream during a surgical procedure, without having to remove the imaging device from the surgical field.

In one aspect, the imaging device comprises a tubular housing that includes a plurality of channels. A first channel is configured to slidably receive the camera module, which can be configured for a snap-fit engagement with the first channel. A second channel is configured to slidably receive the light source module, which can be configured for a snap-fit engagement with the second channel. In another example, the camera module and/or the light source module can be rotated into a final position within their respective channels. A threaded engagement can be employed in lieu of the snap-fit engagement.

In various examples, multiple imaging devices are placed at different positions in the surgical field to provide multiple views. The imaging module 138 can be configured to switch between the imaging devices to provide an optimal view. In various aspects, the imaging module 138 can be configured to integrate the images from the different imaging device.

Various image processors and imaging devices suitable for use with the present disclosure are described in U.S. Pat. No. 7,995,045, titled COMBINED SBI AND CONVENTIONAL IMAGE PROCESSOR, which issued on Aug. 9,

2011, which is herein incorporated by reference in its entirety. In addition, U.S. Pat. No. 7,982,776, titled SBI MOTION ARTIFACT REMOVAL APPARATUS AND METHOD, which issued on Jul. 19, 2011, which is herein incorporated by reference in its entirety, describes various systems for removing motion artifacts from image data. Such systems can be integrated with the imaging module **138**. Furthermore, U.S. Patent Application Publication No. 2011/0306840, titled CONTROLLABLE MAGNETIC SOURCE TO FIXTURE INTRACORPOREAL APPARATUS, which published on Dec. 15, 2011, and U.S. Patent Application Publication No. 2014/0243597, titled SYSTEM FOR PERFORMING A MINIMALLY INVASIVE SURGICAL PROCEDURE, which published on Aug. 28, 2014, each of which is herein incorporated by reference in its entirety.

FIG. 8 illustrates a surgical data network **201** comprising a modular communication hub **203** configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to a cloud-based system (e.g., the cloud **204** that may include a remote server **213** coupled to a storage device **205**). In one aspect, the modular communication hub **203** comprises a network hub **207** and/or a network switch **209** in communication with a network router. The modular communication hub **203** also can be coupled to a local computer system **210** to provide local computer processing and data manipulation. The surgical data network **201** may be configured as passive, intelligent, or switching. A passive surgical data network serves as a conduit for the data, enabling it to go from one device (or segment) to another and to the cloud computing resources. An intelligent surgical data network includes additional features to enable the traffic passing through the surgical data network to be monitored and to configure each port in the network hub **207** or network switch **209**. An intelligent surgical data network may be referred to as a manageable hub or switch. A switching hub reads the destination address of each packet and then forwards the packet to the correct port.

Modular devices **1a-1n** located in the operating theater may be coupled to the modular communication hub **203**. The network hub **207** and/or the network switch **209** may be coupled to a network router **211** to connect the devices **1a-1n** to the cloud **204** or the local computer system **210**. Data associated with the devices **1a-1n** may be transferred to cloud-based computers via the router for remote data processing and manipulation. Data associated with the devices **1a-1n** may also be transferred to the local computer system **210** for local data processing and manipulation. Modular devices **2a-2m** located in the same operating theater also may be coupled to a network switch **209**. The network switch **209** may be coupled to the network hub **207** and/or the network router **211** to connect to the devices **2a-2m** to the cloud **204**. Data associated with the devices **2a-2n** may be transferred to the cloud **204** via the network router **211** for data processing and manipulation. Data associated with the devices **2a-2m** may also be transferred to the local computer system **210** for local data processing and manipulation.

It will be appreciated that the surgical data network **201** may be expanded by interconnecting multiple network hubs **207** and/or multiple network switches **209** with multiple network routers **211**. The modular communication hub **203** may be contained in a modular control tower configured to receive multiple devices **1a-1n/2a-2m**. The local computer system **210** also may be contained in a modular control tower. The modular communication hub **203** is connected to

a display **212** to display images obtained by some of the devices **1a-1n/2a-2m**, for example during surgical procedures. In various aspects, the devices **1a-1n/2a-2m** may include, for example, various modules such as an imaging module **138** coupled to an endoscope, a generator module **140** coupled to an energy-based surgical device, a smoke evacuation module **126**, a suction/irrigation module **128**, a communication module **130**, a processor module **132**, a storage array **134**, a surgical device coupled to a display, and/or a non-contact sensor module, among other modular devices that may be connected to the modular communication hub **203** of the surgical data network **201**.

In one aspect, the surgical data network **201** may comprise a combination of network hub(s), network switch(es), and network router(s) connecting the devices **1a-1n/2a-2m** to the cloud. Any one of or all of the devices **1a-1n/2a-2m** coupled to the network hub or network switch may collect data in real time and transfer the data to cloud computers for data processing and manipulation. It will be appreciated that cloud computing relies on sharing computing resources rather than having local servers or personal devices to handle software applications. The word “cloud” may be used as a metaphor for “the Internet,” although the term is not limited as such. Accordingly, the term “cloud computing” may be used herein to refer to “a type of Internet-based computing,” where different services—such as servers, storage, and applications—are delivered to the modular communication hub **203** and/or computer system **210** located in the surgical theater (e.g., a fixed, mobile, temporary, or field operating room or space) and to devices connected to the modular communication hub **203** and/or computer system **210** through the Internet. The cloud infrastructure may be maintained by a cloud service provider. In this context, the cloud service provider may be the entity that coordinates the usage and control of the devices **1a-1n/2a-2m** located in one or more operating theaters. The cloud computing services can perform a large number of calculations based on the data gathered by smart surgical instruments, robots, and other computerized devices located in the operating theater. The hub hardware enables multiple devices or connections to be connected to a computer that communicates with the cloud computing resources and storage.

Applying cloud computer data processing techniques on the data collected by the devices **1a-1n/2a-2m**, the surgical data network provides improved surgical outcomes, reduced costs, and improved patient satisfaction. At least some of the devices **1a-1n/2a-2m** may be employed to view tissue states to assess leaks or perfusion of sealed tissue after a tissue sealing and cutting procedure. At least some of the devices **1a-1n/2a-2m** may be employed to identify pathology, such as the effects of diseases, using the cloud-based computing to examine data including images of samples of body tissue for diagnostic purposes. This includes localization and margin confirmation of tissue and phenotypes. At least some of the devices **1a-1n/2a-2m** may be employed to identify anatomical structures of the body using a variety of sensors integrated with imaging devices and techniques such as overlaying images captured by multiple imaging devices. The data gathered by the devices **1a-1n/2a-2m**, including image data, may be transferred to the cloud **204** or the local computer system **210** or both for data processing and manipulation including image processing and manipulation. The data may be analyzed to improve surgical procedure outcomes by determining if further treatment, such as the application of endoscopic intervention, emerging technologies, a targeted radiation, targeted intervention, and precise robotics to tissue-specific sites and conditions, may be

pursued. Such data analysis may further employ outcome analytics processing, and using standardized approaches may provide beneficial feedback to either confirm surgical treatments and the behavior of the surgeon or suggest modifications to surgical treatments and the behavior of the surgeon.

In one implementation, the operating theater devices **1a-1n** may be connected to the modular communication hub **203** over a wired channel or a wireless channel depending on the configuration of the devices **1a-1n** to a network hub. The network hub **207** may be implemented, in one aspect, as a local network broadcast device that works on the physical layer of the Open System Interconnection (OSI) model. The network hub provides connectivity to the devices **1a-1n** located in the same operating theater network. The network hub **207** collects data in the form of packets and sends them to the router in half duplex mode. The network hub **207** does not store any media access control/Internet Protocol (MAC/IP) to transfer the device data. Only one of the devices **1a-1n** can send data at a time through the network hub **207**. The network hub **207** has no routing tables or intelligence regarding where to send information and broadcasts all network data across each connection and to a remote server **213** (FIG. 9) over the cloud **204**. The network hub **207** can detect basic network errors such as collisions, but having all information broadcast to multiple ports can be a security risk and cause bottlenecks.

In another implementation, the operating theater devices **2a-2m** may be connected to a network switch **209** over a wired channel or a wireless channel. The network switch **209** works in the data link layer of the OSI model. The network switch **209** is a multicast device for connecting the devices **2a-2m** located in the same operating theater to the network. The network switch **209** sends data in the form of frames to the network router **211** and works in full duplex mode. Multiple devices **2a-2m** can send data at the same time through the network switch **209**. The network switch **209** stores and uses MAC addresses of the devices **2a-2m** to transfer data.

The network hub **207** and/or the network switch **209** are coupled to the network router **211** for connection to the cloud **204**. The network router **211** works in the network layer of the OSI model. The network router **211** creates a route for transmitting data packets received from the network hub **207** and/or network switch **211** to cloud-based computer resources for further processing and manipulation of the data collected by any one of or all the devices **1a-1n/2a-2m**. The network router **211** may be employed to connect two or more different networks located in different locations, such as, for example, different operating theaters of the same healthcare facility or different networks located in different operating theaters of different healthcare facilities. The network router **211** sends data in the form of packets to the cloud **204** and works in full duplex mode. Multiple devices can send data at the same time. The network router **211** uses IP addresses to transfer data.

In one example, the network hub **207** may be implemented as a USB hub, which allows multiple USB devices to be connected to a host computer. The USB hub may expand a single USB port into several tiers so that there are more ports available to connect devices to the host system computer. The network hub **207** may include wired or wireless capabilities to receive information over a wired channel or a wireless channel. In one aspect, a wireless USB short-range, high-bandwidth wireless radio communication

protocol may be employed for communication between the devices **1a-1n** and devices **2a-2m** located in the operating theater.

In other examples, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via Bluetooth wireless technology standard for exchanging data over short distances (using short-wavelength UHF radio waves in the ISM band from 2.4 to 2.485 GHz) from fixed and mobile devices and building personal area networks (PANs). In other aspects, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via a number of wireless or wired communication standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long-term evolution (LTE), and Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, and Ethernet derivatives thereof, as well as any other wireless and wired protocols that are designated as 3G, 4G, 5G, and beyond. The computing module may include a plurality of communication modules. For instance, a first communication module may be dedicated to shorter-range wireless communications such as Wi-Fi and Bluetooth, and a second communication module may be dedicated to longer-range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

The modular communication hub **203** may serve as a central connection for one or all of the operating theater devices **1a-1n/2a-2m** and handles a data type known as frames. Frames carry the data generated by the devices **1a-1n/2a-2m**. When a frame is received by the modular communication hub **203**, it is amplified and transmitted to the network router **211**, which transfers the data to the cloud computing resources by using a number of wireless or wired communication standards or protocols, as described herein.

The modular communication hub **203** can be used as a standalone device or be connected to compatible network hubs and network switches to form a larger network. The modular communication hub **203** is generally easy to install, configure, and maintain, making it a good option for networking the operating theater devices **1a-1n/2a-2m**.

FIG. 9 illustrates a computer-implemented interactive surgical system **200**. The computer-implemented interactive surgical system **200** is similar in many respects to the computer-implemented interactive surgical system **100**. For example, the computer-implemented interactive surgical system **200** includes one or more surgical systems **202**, which are similar in many respects to the surgical systems **102**. Each surgical system **202** includes at least one surgical hub **206** in communication with a cloud **204** that may include a remote server **213**. In one aspect, the computer-implemented interactive surgical system **200** comprises a modular control tower **236** connected to multiple operating theater devices such as, for example, intelligent surgical instruments, robots, and other computerized devices located in the operating theater. As shown in FIG. 10, the modular control tower **236** comprises a modular communication hub **203** coupled to a computer system **210**. As illustrated in the example of FIG. 9, the modular control tower **236** is coupled to an imaging module **238** that is coupled to an endoscope **239**, a generator module **240** that is coupled to an energy device **241**, a smoke evacuator module **226**, a suction/irrigation module **228**, a communication module **230**, a processor module **232**, a storage array **234**, a smart device/instrument **235** optionally coupled to a display **237**, and a non-contact sensor module **242**. The operating theater devices are coupled to cloud computing resources and data

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storage via the modular control tower **236**. A robot hub **222** also may be connected to the modular control tower **236** and to the cloud computing resources. The devices/instruments **235**, visualization systems **208**, among others, may be coupled to the modular control tower **236** via wired or wireless communication standards or protocols, as described herein. The modular control tower **236** may be coupled to a hub display **215** (e.g., monitor, screen) to display and overlay images received from the imaging module, device/instrument display, and/or other visualization systems **208**. The hub display also may display data received from devices connected to the modular control tower in conjunction with images and overlaid images.

FIG. **10** illustrates a surgical hub **206** comprising a plurality of modules coupled to the modular control tower **236**. The modular control tower **236** comprises a modular communication hub **203**, e.g., a network connectivity device, and a computer system **210** to provide local processing, visualization, and imaging, for example. As shown in FIG. **10**, the modular communication hub **203** may be connected in a tiered configuration to expand the number of modules (e.g., devices) that may be connected to the modular communication hub **203** and transfer data associated with the modules to the computer system **210**, cloud computing resources, or both. As shown in FIG. **10**, each of the network hubs/switches in the modular communication hub **203** includes three downstream ports and one upstream port. The upstream network hub/switch is connected to a processor to provide a communication connection to the cloud computing resources and a local display **217**. Communication to the cloud **204** may be made either through a wired or a wireless communication channel.

The surgical hub **206** employs a non-contact sensor module **242** to measure the dimensions of the operating theater and generate a map of the surgical theater using either ultrasonic or laser-type non-contact measurement devices. An ultrasound-based non-contact sensor module scans the operating theater by transmitting a burst of ultrasound and receiving the echo when it bounces off the perimeter walls of an operating theater as described under the heading "Surgical Hub Spatial Awareness Within an Operating Room" in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, which is herein incorporated by reference in its entirety, in which the sensor module is configured to determine the size of the operating theater and to adjust Bluetooth-pairing distance limits. A laser-based non-contact sensor module scans the operating theater by transmitting laser light pulses, receiving laser light pulses that bounce off the perimeter walls of the operating theater, and comparing the phase of the transmitted pulse to the received pulse to determine the size of the operating theater and to adjust Bluetooth pairing distance limits, for example.

The computer system **210** comprises a processor **244** and a network interface **245**. The processor **244** is coupled to a communication module **247**, storage **248**, memory **249**, non-volatile memory **250**, and input/output interface **251** via a system bus. The system bus can be any of several types of bus structure(s) including the memory bus or memory controller, a peripheral bus or external bus, and/or a local bus using any variety of available bus architectures including, but not limited to, 9-bit bus, Industrial Standard Architecture (ISA), Micro-Chamel Architecture (MSA), Extended ISA (EISA), Intelligent Drive Electronics (IDE), VESA Local Bus (VLB), Peripheral Component Interconnect (PCI), USB, Advanced Graphics Port (AGP), Personal Computer

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Memory Card International Association bus (PCMCIA), Small Computer Systems Interface (SCSI), or any other proprietary bus.

The processor **244** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the processor may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), an internal read-only memory (ROM) loaded with StellarisWare® software, a 2 KB electrically erasable programmable read-only memory (EEPROM), and/or one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analogs, one or more 12-bit analog-to-digital converters (ADCs) with 12 analog input channels, details of which are available for the product datasheet.

In one aspect, the processor **244** may comprise a safety controller comprising two controller-based families such as TMS570 and RM4x, known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

The system memory includes volatile memory and non-volatile memory. The basic input/output system (BIOS), containing the basic routines to transfer information between elements within the computer system, such as during start-up, is stored in non-volatile memory. For example, the non-volatile memory can include ROM, programmable ROM (PROM), electrically programmable ROM (EPROM), EEPROM, or flash memory. Volatile memory includes random-access memory (RAM), which acts as external cache memory. Moreover, RAM is available in many forms such as SRAM, dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SL-DRAM), and direct Rambus RAM (DRRAM).

The computer system **210** also includes removable/non-removable, volatile/non-volatile computer storage media, such as for example disk storage. The disk storage includes, but is not limited to, devices like a magnetic disk drive, floppy disk drive, tape drive, Jaz drive, Zip drive, LS-60 drive, flash memory card, or memory stick. In addition, the disk storage can include storage media separately or in combination with other storage media including, but not limited to, an optical disc drive such as a compact disc ROM device (CD-ROM), compact disc recordable drive (CD-R Drive), compact disc rewritable drive (CD-RW Drive), or a digital versatile disc ROM drive (DVD-ROM). To facilitate the connection of the disk storage devices to the system bus, a removable or non-removable interface may be employed.

It is to be appreciated that the computer system **210** includes software that acts as an intermediary between users and the basic computer resources described in a suitable operating environment. Such software includes an operating system. The operating system, which can be stored on the disk storage, acts to control and allocate resources of the computer system. System applications take advantage of the management of resources by the operating system through program modules and program data stored either in the system memory or on the disk storage. It is to be appreciated

that various components described herein can be implemented with various operating systems or combinations of operating systems.

A user enters commands or information into the computer system **210** through input device(s) coupled to the I/O interface **251**. The input devices include, but are not limited to, a pointing device such as a mouse, trackball, stylus, touch pad, keyboard, microphone, joystick, game pad, satellite dish, scanner, TV tuner card, digital camera, digital video camera, web camera, and the like. These and other input devices connect to the processor through the system bus via interface port(s). The interface port(s) include, for example, a serial port, a parallel port, a game port, and a USB. The output device(s) use some of the same types of ports as input device(s). Thus, for example, a USB port may be used to provide input to the computer system and to output information from the computer system to an output device. An output adapter is provided to illustrate that there are some output devices like monitors, displays, speakers, and printers, among other output devices that require special adapters. The output adapters include, by way of illustration and not limitation, video and sound cards that provide a means of connection between the output device and the system bus. It should be noted that other devices and/or systems of devices, such as remote computer(s), provide both input and output capabilities.

The computer system **210** can operate in a networked environment using logical connections to one or more remote computers, such as cloud computer(s), or local computers. The remote cloud computer(s) can be a personal computer, server, router, network PC, workstation, micro-processor-based appliance, peer device, or other common network node, and the like, and typically includes many or all of the elements described relative to the computer system. For purposes of brevity, only a memory storage device is illustrated with the remote computer(s). The remote computer(s) is logically connected to the computer system through a network interface and then physically connected via a communication connection. The network interface encompasses communication networks such as local area networks (LANs) and wide area networks (WANs). LAN technologies include Fiber Distributed Data Interface (FDDI), Copper Distributed Data Interface (CDDI), Ethernet/IEEE 802.3, Token Ring/IEEE 802.5 and the like. WAN technologies include, but are not limited to, point-to-point links, circuit-switching networks like Integrated Services Digital Networks (ISDN) and variations thereon, packet-switching networks, and Digital Subscriber Lines (DSL).

In various aspects, the computer system **210** of FIG. **10**, the imaging module **238** and/or visualization system **208**, and/or the processor module **232** of FIGS. **9-10**, may comprise an image processor, image-processing engine, media processor, or any specialized digital signal processor (DSP) used for the processing of digital images. The image processor may employ parallel computing with single instruction, multiple data (SIMD) or multiple instruction, multiple data (MIMD) technologies to increase speed and efficiency. The digital image-processing engine can perform a range of tasks. The image processor may be a system on a chip with multicore processor architecture.

The communication connection(s) refers to the hardware/software employed to connect the network interface to the bus. While the communication connection is shown for illustrative clarity inside the computer system, it can also be external to the computer system **210**. The hardware/software necessary for connection to the network interface includes,

for illustrative purposes only, internal and external technologies such as modems, including regular telephone-grade modems, cable modems, and DSL modems, ISDN adapters, and Ethernet cards.

In various aspects, the devices/instruments **235** described with reference to FIGS. **9-10**, may be implemented as the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**). Accordingly, the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**) are configured to interface with the modular control tower **236** and the surgical hub **206**. Once connected to the surgical hub **206**, the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**) are configured to interface with the cloud **204**, the server **213**, other hub connected instruments, the hub display **215**, or the visualization system **209**, or combinations thereof. Further, once connected to hub **206**, the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**) may utilize the processing circuits available in the hub local computer system **210**.

FIG. **11** illustrates a functional block diagram of one aspect of a USB network hub **300** device, in accordance with at least one aspect of the present disclosure. In the illustrated aspect, the USB network hub device **300** employs a TUSB2036 integrated circuit hub by Texas Instruments. The USB network hub **300** is a CMOS device that provides an upstream USB transceiver port **302** and up to three downstream USB transceiver ports **304**, **306**, **308** in compliance with the USB 2.0 specification. The upstream USB transceiver port **302** is a differential root data port comprising a differential data minus (DM0) input paired with a differential data plus (DP0) input. The three downstream USB transceiver ports **304**, **306**, **308** are differential data ports where each port includes differential data plus (DP1-DP3) outputs paired with differential data minus (DM1-DM3) outputs.

The USB network hub **300** device is implemented with a digital state machine instead of a microcontroller, and no firmware programming is required. Fully compliant USB transceivers are integrated into the circuit for the upstream

USB transceiver port **302** and all downstream USB transceiver ports **304**, **306**, **308**. The downstream USB transceiver ports **304**, **306**, **308** support both full-speed and low-speed devices by automatically setting the slew rate according to the speed of the device attached to the ports. The USB network hub **300** device may be configured either in bus-powered or self-powered mode and includes a hub power logic **312** to manage power.

The USB network hub **300** device includes a serial interface engine **310** (SIE). The SIE **310** is the front end of the USB network hub **300** hardware and handles most of the protocol described in chapter 8 of the USB specification. The SIE **310** typically comprehends signaling up to the transaction level. The functions that it handles could include: packet recognition, transaction sequencing, SOP, EOP, RESET, and RESUME signal detection/generation, clock/data separation, non-return-to-zero invert (NRZI) data encoding/decoding and bit-stuffing, CRC generation and checking (token and data), packet ID (PID) generation and checking/decoding, and/or serial-parallel/parallel-serial conversion. The **310** receives a clock input **314** and is coupled to a suspend/resume logic and frame timer **316** circuit and a hub repeater circuit **318** to control communication between the upstream USB transceiver port **302** and the downstream USB transceiver ports **304**, **306**, **308** through port logic circuits **320**, **322**, **324**. The SIE **310** is coupled to a command decoder **326** via interface logic **328** to control commands from a serial EEPROM via a serial EEPROM interface **330**.

In various aspects, the USB network hub **300** can connect **127** functions configured in up to six logical layers (tiers) to a single computer. Further, the USB network hub **300** can connect to all peripherals using a standardized four-wire cable that provides both communication and power distribution. The power configurations are bus-powered and self-powered modes. The USB network hub **300** may be configured to support four modes of power management: a bus-powered hub, with either individual-port power management or ganged-port power management, and the self-powered hub, with either individual-port power management or ganged-port power management. In one aspect, using a USB cable, the USB network hub **300**, the upstream USB transceiver port **302** is plugged into a USB host controller, and the downstream USB transceiver ports **304**, **306**, **308** are exposed for connecting USB compatible devices, and so forth.

Additional details regarding the structure and function of the surgical hub and/or surgical hub networks can be found in U.S. Provisional Patent Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed Apr. 19, 2018, which is hereby incorporated by reference herein in its entirety.

Cloud System Hardware and Functional Modules

FIG. 12 is a block diagram of the computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure. In one aspect, the computer-implemented interactive surgical system is configured to monitor and analyze data related to the operation of various surgical systems that include surgical hubs, surgical instruments, robotic devices and operating theaters or healthcare facilities. The computer-implemented interactive surgical system comprises a cloud-based analytics system. Although the cloud-based analytics system is described as a surgical system, it is not necessarily limited as such and could be a cloud-based medical system generally. As illustrated in FIG. 12, the cloud-based analytics system comprises a plurality of

surgical instruments **7012** (may be the same or similar to instruments **112**), a plurality of surgical hubs **7006** (may be the same or similar to hubs **106**), and a surgical data network **7001** (may be the same or similar to network **201**) to couple the surgical hubs **7006** to the cloud **7004** (may be the same or similar to cloud **204**). Each of the plurality of surgical hubs **7006** is communicatively coupled to one or more surgical instruments **7012**. The hubs **7006** are also communicatively coupled to the cloud **7004** of the computer-implemented interactive surgical system via the network **7001**. The cloud **7004** is a remote centralized source of hardware and software for storing, manipulating, and communicating data generated based on the operation of various surgical systems. As shown in FIG. 12, access to the cloud **7004** is achieved via the network **7001**, which may be the Internet or some other suitable computer network. Surgical hubs **7006** that are coupled to the cloud **7004** can be considered the client side of the cloud computing system (i.e., cloud-based analytics system). Surgical instruments **7012** are paired with the surgical hubs **7006** for control and implementation of various surgical procedures or operations as described herein.

In addition, surgical instruments **7012** may comprise transceivers for data transmission to and from their corresponding surgical hubs **7006** (which may also comprise transceivers). Combinations of surgical instruments **7012** and corresponding hubs **7006** may indicate particular locations, such as operating theaters in healthcare facilities (e.g., hospitals), for providing medical operations. For example, the memory of a surgical hub **7006** may store location data. As shown in FIG. 12, the cloud **7004** comprises central servers **7013** (which may be same or similar to remote server **113** in FIG. 1 and/or remote server **213** in FIG. 9), hub application servers **7002**, data analytics modules **7034**, and an input/output (“I/O”) interface **7007**. The central servers **7013** of the cloud **7004** collectively administer the cloud computing system, which includes monitoring requests by client surgical hubs **7006** and managing the processing capacity of the cloud **7004** for executing the requests. Each of the central servers **7013** comprises one or more processors **7008** coupled to suitable memory devices **7010** which can include volatile memory such as random-access memory (RAM) and non-volatile memory such as magnetic storage devices. The memory devices **7010** may comprise machine executable instructions that when executed cause the processors **7008** to execute the data analytics modules **7034** for the cloud-based data analysis, operations, recommendations and other operations described below. Moreover, the processors **7008** can execute the data analytics modules **7034** independently or in conjunction with hub applications independently executed by the hubs **7006**. The central servers **7013** also comprise aggregated medical data databases **2212**, which can reside in the memory **2210**.

Based on connections to various surgical hubs **7006** via the network **7001**, the cloud **7004** can aggregate data from specific data generated by various surgical instruments **7012** and their corresponding hubs **7006**. Such aggregated data may be stored within the aggregated medical data databases **7011** of the cloud **7004**. In particular, the cloud **7004** may advantageously perform data analysis and operations on the aggregated data to yield insights and/or perform functions that individual hubs **7006** could not achieve on their own. To this end, as shown in FIG. 12, the cloud **7004** and the surgical hubs **7006** are communicatively coupled to transmit and receive information. The I/O interface **7007** is connected to the plurality of surgical hubs **7006** via the network **7001**. In this way, the I/O interface **7007** can be configured to

transfer information between the surgical hubs **7006** and the aggregated medical data databases **7011**. Accordingly, the I/O interface **7007** may facilitate read/write operations of the cloud-based analytics system. Such read/write operations may be executed in response to requests from hubs **7006**. These requests could be transmitted to the hubs **7006** through the hub applications. The I/O interface **7007** may include one or more high speed data ports, which may include universal serial bus (USB) ports, IEEE 1394 ports, as well as Wi-Fi and Bluetooth I/O interfaces for connecting the cloud **7004** to hubs **7006**. The hub application servers **7002** of the cloud **7004** are configured to host and supply shared capabilities to software applications (e.g. hub applications) executed by surgical hubs **7006**. For example, the hub application servers **7002** may manage requests made by the hub applications through the hubs **7006**, control access to the aggregated medical data databases **7011**, and perform load balancing. The data analytics modules **7034** are described in further detail with reference to FIG. 13.

The particular cloud computing system configuration described in the present disclosure is specifically designed to address various issues arising in the context of medical operations and procedures performed using medical devices, such as the surgical instruments **7012**, **112**. In particular, the surgical instruments **7012** may be digital surgical devices configured to interact with the cloud **7004** for implementing techniques to improve the performance of surgical operations. Various surgical instruments **7012** and/or surgical hubs **7006** may comprise touch controlled user interfaces such that clinicians may control aspects of interaction between the surgical instruments **7012** and the cloud **7004**. Other suitable user interfaces for control such as auditory controlled user interfaces can also be used.

FIG. 13 is a block diagram which illustrates the functional architecture of the computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure. The cloud-based analytics system includes a plurality of data analytics modules **7034** that may be executed by the processors **7008** of the cloud **7004** for providing data analytic solutions to problems specifically arising in the medical field. As shown in FIG. 13, the functions of the cloud-based data analytics modules **7034** may be assisted via hub applications **7014** hosted by the hub application servers **7002** that may be accessed on surgical hubs **7006**. The cloud processors **7008** and hub applications **7014** may operate in conjunction to execute the data analytics modules **7034**. Application program interfaces (APIs) **7016** define the set of protocols and routines corresponding to the hub applications **7014**. Additionally, the APIs **7016** manage the storing and retrieval of data into and from the aggregated medical data databases **7011** for the operations of the applications **7014**. The caches **7018** also store data (e.g., temporarily) and are coupled to the APIs **7016** for more efficient retrieval of data used by the applications **7014**. The data analytics modules **7034** in FIG. 13 include modules for resource optimization **7020**, data collection and aggregation **7022**, authorization and security **7024**, control program updating **7026**, patient outcome analysis **7028**, recommendations **7030**, and data sorting and prioritization **7032**. Other suitable data analytics modules could also be implemented by the cloud **7004**, according to some aspects. In one aspect, the data analytics modules are used for specific recommendations based on analyzing trends, outcomes, and other data.

For example, the data collection and aggregation module **7022** could be used to generate self-describing data (e.g., metadata) including identification of notable features or configuration (e.g., trends), management of redundant data

sets, and storage of the data in paired data sets which can be grouped by surgery but not necessarily keyed to actual surgical dates and surgeons. In particular, pair data sets generated from operations of surgical instruments **7012** can comprise applying a binary classification, e.g., a bleeding or a non-bleeding event. More generally, the binary classification may be characterized as either a desirable event (e.g., a successful surgical procedure) or an undesirable event (e.g., a misfired or misused surgical instrument **7012**). The aggregated self-describing data may correspond to individual data received from various groups or subgroups of surgical hubs **7006**. Accordingly, the data collection and aggregation module **7022** can generate aggregated metadata or other organized data based on raw data received from the surgical hubs **7006**. To this end, the processors **7008** can be operationally coupled to the hub applications **7014** and aggregated medical data databases **7011** for executing the data analytics modules **7034**. The data collection and aggregation module **7022** may store the aggregated organized data into the aggregated medical data databases **2212**.

The resource optimization module **7020** can be configured to analyze this aggregated data to determine an optimal usage of resources for a particular or group of healthcare facilities. For example, the resource optimization module **7020** may determine an optimal order point of surgical stapling instruments **7012** for a group of healthcare facilities based on corresponding predicted demand of such instruments **7012**. The resource optimization module **7020** might also assess the resource usage or other operational configurations of various healthcare facilities to determine whether resource usage could be improved. Similarly, the recommendations module **7030** can be configured to analyze aggregated organized data from the data collection and aggregation module **7022** to provide recommendations. For example, the recommendations module **7030** could recommend to healthcare facilities (e.g., medical service providers such as hospitals) that a particular surgical instrument **7012** should be upgraded to an improved version based on a higher than expected error rate, for example. Additionally, the recommendations module **7030** and/or resource optimization module **7020** could recommend better supply chain parameters such as product reorder points and provide suggestions of different surgical instrument **7012**, uses thereof, or procedure steps to improve surgical outcomes. The healthcare facilities can receive such recommendations via corresponding surgical hubs **7006**. More specific recommendations regarding parameters or configurations of various surgical instruments **7012** can also be provided. Hubs **7006** and/or surgical instruments **7012** each could also have display screens that display data or recommendations provided by the cloud **7004**.

The patient outcome analysis module **7028** can analyze surgical outcomes associated with currently used operational parameters of surgical instruments **7012**. The patient outcome analysis module **7028** may also analyze and assess other potential operational parameters. In this connection, the recommendations module **7030** could recommend using these other potential operational parameters based on yielding better surgical outcomes, such as better sealing or less bleeding. For example, the recommendations module **7030** could transmit recommendations to a surgical hub **7006** regarding when to use a particular cartridge for a corresponding stapling surgical instrument **7012**. Thus, the cloud-based analytics system, while controlling for common variables, may be configured to analyze the large collection of raw data and to provide centralized recommendations over multiple healthcare facilities (advantageously determined

based on aggregated data). For example, the cloud-based analytics system could analyze, evaluate, and/or aggregate data based on type of medical practice, type of patient, number of patients, geographic similarity between medical providers, which medical providers/facilities use similar types of instruments, etc., in a way that no single healthcare facility alone would be able to analyze independently.

The control program updating module 7026 could be configured to implement various surgical instrument 7012 recommendations when corresponding control programs are updated. For example, the patient outcome analysis module 7028 could identify correlations linking specific control parameters with successful (or unsuccessful) results. Such correlations may be addressed when updated control programs are transmitted to surgical instruments 7012 via the control program updating module 7026. Updates to instruments 7012 that are transmitted via a corresponding hub 7006 may incorporate aggregated performance data that was gathered and analyzed by the data collection and aggregation module 7022 of the cloud 7004. Additionally, the patient outcome analysis module 7028 and recommendations module 7030 could identify improved methods of using instruments 7012 based on aggregated performance data.

The cloud-based analytics system may include security features implemented by the cloud 7004. These security features may be managed by the authorization and security module 7024. Each surgical hub 7006 can have associated unique credentials such as username, password, and other suitable security credentials. These credentials could be stored in the memory 7010 and be associated with a permitted cloud access level. For example, based on providing accurate credentials, a surgical hub 7006 may be granted access to communicate with the cloud to a predetermined extent (e.g., may only engage in transmitting or receiving certain defined types of information). To this end, the aggregated medical data databases 7011 of the cloud 7004 may comprise a database of authorized credentials for verifying the accuracy of provided credentials. Different credentials may be associated with varying levels of permission for interaction with the cloud 7004, such as a predetermined access level for receiving the data analytics generated by the cloud 7004.

Furthermore, for security purposes, the cloud could maintain a database of hubs 7006, instruments 7012, and other devices that may comprise a "black list" of prohibited devices. In particular, a surgical hub 7006 listed on the black list may not be permitted to interact with the cloud, while surgical instruments 7012 listed on the black list may not have functional access to a corresponding hub 7006 and/or may be prevented from fully functioning when paired to its corresponding hub 7006. Additionally or alternatively, the cloud 7004 may flag instruments 7012 based on incompatibility or other specified criteria. In this manner, counterfeit medical devices and improper reuse of such devices throughout the cloud-based analytics system can be identified and addressed.

The surgical instruments 7012 may use wireless transceivers to transmit wireless signals that may represent, for example, authorization credentials for access to corresponding hubs 7006 and the cloud 7004. Wired transceivers may also be used to transmit signals. Such authorization credentials can be stored in the respective memory devices of the surgical instruments 7012. The authorization and security module 7024 can determine whether the authorization credentials are accurate or counterfeit. The authorization and security module 7024 may also dynamically generate autho-

rization credentials for enhanced security. The credentials could also be encrypted, such as by using hash based encryption. Upon transmitting proper authorization, the surgical instruments 7012 may transmit a signal to the corresponding hubs 7006 and ultimately the cloud 7004 to indicate that the instruments 7012 are ready to obtain and transmit medical data. In response, the cloud 7004 may transition into a state enabled for receiving medical data for storage into the aggregated medical data databases 7011. This data transmission readiness could be indicated by a light indicator on the instruments 7012, for example. The cloud 7004 can also transmit signals to surgical instruments 7012 for updating their associated control programs. The cloud 7004 can transmit signals that are directed to a particular class of surgical instruments 7012 (e.g., electro-surgical instruments) so that software updates to control programs are only transmitted to the appropriate surgical instruments 7012. Moreover, the cloud 7004 could be used to implement system wide solutions to address local or global problems based on selective data transmission and authorization credentials. For example, if a group of surgical instruments 7012 are identified as having a common manufacturing defect, the cloud 7004 may change the authorization credentials corresponding to this group to implement an operational lockout of the group.

The cloud-based analytics system may allow for monitoring multiple healthcare facilities (e.g., medical facilities like hospitals) to determine improved practices and recommend changes (via the recommendations module 2030, for example) accordingly. Thus, the processors 7008 of the cloud 7004 can analyze data associated with an individual healthcare facility to identify the facility and aggregate the data with other data associated with other healthcare facilities in a group. Groups could be defined based on similar operating practices or geographical location, for example. In this way, the cloud 7004 may provide healthcare facility group wide analysis and recommendations. The cloud-based analytics system could also be used for enhanced situational awareness. For example, the processors 7008 may predictively model the effects of recommendations on the cost and effectiveness for a particular facility (relative to overall operations and/or various medical procedures). The cost and effectiveness associated with that particular facility can also be compared to a corresponding local region of other facilities or any other comparable facilities.

The data sorting and prioritization module 7032 may prioritize and sort data based on criticality (e.g., the severity of a medical event associated with the data, unexpectedness, suspiciousness). This sorting and prioritization may be used in conjunction with the functions of the other data analytics modules 7034 described above to improve the cloud-based analytics and operations described herein. For example, the data sorting and prioritization module 7032 can assign a priority to the data analysis performed by the data collection and aggregation module 7022 and patient outcome analysis modules 7028. Different prioritization levels can result in particular responses from the cloud 7004 (corresponding to a level of urgency) such as escalation for an expedited response, special processing, exclusion from the aggregated medical data databases 7011, or other suitable responses. Moreover, if necessary, the cloud 7004 can transmit a request (e.g. a push message) through the hub application servers for additional data from corresponding surgical instruments 7012. The push message can result in a notification displayed on the corresponding hubs 7006 for requesting supporting or additional data. This push message may be required in situations in which the cloud detects a

significant irregularity or outlier and the cloud cannot determine the cause of the irregularity. The central servers **7013** may be programmed to trigger this push message in certain significant circumstances, such as when data is determined to be different from an expected value beyond a predetermined threshold or when it appears security has been comprised, for example.

In various aspects, the surgical instrument(s) **7012** described above with reference to FIGS. **12** and **13**, may be implemented as the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**). Accordingly, the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**) are configured to interface with the surgical hub **7006** and the network **2001**, which is configured to interface with cloud **7004**. Accordingly, the processing power provided by the central servers **7013** and the data analytics module **7034** are configured to process information (e.g., data and control) from the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**).

Additional details regarding the cloud analysis system can be found in U.S. Provisional Patent Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed Apr. 19, 2018, which is hereby incorporated by reference herein in its entirety.

Situational Awareness

Although an “intelligent” device including control algorithms that respond to sensed data can be an improvement over a “dumb” device that operates without accounting for sensed data, some sensed data can be incomplete or inconclusive when considered in isolation, i.e., without the context of the type of surgical procedure being performed or the type of tissue that is being operated on. Without knowing the procedural context (e.g., knowing the type of tissue being operated on or the type of procedure being performed), the control algorithm may control the modular device incorrectly or suboptimally given the particular context-free sensed data. For example, the optimal manner for a control algorithm to control a surgical instrument in response to a particular sensed parameter can vary according to the particular tissue type being operated on. This is due to the fact that different tissue types have different properties (e.g., resistance to tearing) and thus respond differently to actions taken by surgical instruments. Therefore, it may be desirable for a surgical instrument to take different actions even when

the same measurement for a particular parameter is sensed. As one specific example, the optimal manner in which to control a surgical stapling and cutting instrument in response to the instrument sensing an unexpectedly high force to close its end effector will vary depending upon whether the tissue type is susceptible or resistant to tearing. For tissues that are susceptible to tearing, such as lung tissue, the instrument’s control algorithm would optimally ramp down the motor in response to an unexpectedly high force to close to avoid tearing the tissue. For tissues that are resistant to tearing, such as stomach tissue, the instrument’s control algorithm would optimally ramp up the motor in response to an unexpectedly high force to close to ensure that the end effector is clamped properly on the tissue. Without knowing whether lung or stomach tissue has been clamped, the control algorithm may make a suboptimal decision.

One solution utilizes a surgical hub including a system that is configured to derive information about the surgical procedure being performed based on data received from various data sources and then control the paired modular devices accordingly. In other words, the surgical hub is configured to infer information about the surgical procedure from received data and then control the modular devices paired to the surgical hub based upon the inferred context of the surgical procedure. FIG. **14** illustrates a diagram of a situationally aware surgical system **5100**, in accordance with at least one aspect of the present disclosure. In some exemplifications, the data sources **5126** include, for example, the modular devices **5102** (which can include sensors configured to detect parameters associated with the patient and/or the modular device itself), databases **5122** (e.g., an EMR database containing patient records), and patient monitoring devices **5124** (e.g., a blood pressure (BP) monitor and an electrocardiography (EKG) monitor).

A surgical hub **5104**, which may be similar to the hub **106** in many respects, can be configured to derive the contextual information pertaining to the surgical procedure from the data based upon, for example, the particular combination(s) of received data or the particular order in which the data is received from the data sources **5126**. The contextual information inferred from the received data can include, for example, the type of surgical procedure being performed, the particular step of the surgical procedure that the surgeon is performing, the type of tissue being operated on, or the body cavity that is the subject of the procedure. This ability by some aspects of the surgical hub **5104** to derive or infer information related to the surgical procedure from received data can be referred to as “situational awareness.” In one exemplification, the surgical hub **5104** can incorporate a situational awareness system, which is the hardware and/or programming associated with the surgical hub **5104** that derives contextual information pertaining to the surgical procedure from the received data.

The situational awareness system of the surgical hub **5104** can be configured to derive the contextual information from the data received from the data sources **5126** in a variety of different ways. In one exemplification, the situational awareness system includes a pattern recognition system, or machine learning system (e.g., an artificial neural network), that has been trained on training data to correlate various inputs (e.g., data from databases **5122**, patient monitoring devices **5124**, and/or modular devices **5102**) to corresponding contextual information regarding a surgical procedure. In other words, a machine learning system can be trained to accurately derive contextual information regarding a surgical procedure from the provided inputs. In another exemplification, the situational awareness system can include a

lookup table storing pre-characterized contextual information regarding a surgical procedure in association with one or more inputs (or ranges of inputs) corresponding to the contextual information. In response to a query with one or more inputs, the lookup table can return the corresponding contextual information for the situational awareness system for controlling the modular devices **5102**. In one exemplification, the contextual information received by the situational awareness system of the surgical hub **5104** is associated with a particular control adjustment or set of control adjustments for one or more modular devices **5102**. In another exemplification, the situational awareness system includes a further machine learning system, lookup table, or other such system, which generates or retrieves one or more control adjustments for one or more modular devices **5102** when provided the contextual information as input.

A surgical hub **5104** incorporating a situational awareness system provides a number of benefits for the surgical system **5100**. One benefit includes improving the interpretation of sensed and collected data, which would in turn improve the processing accuracy and/or the usage of the data during the course of a surgical procedure. To return to a previous example, a situationally aware surgical hub **5104** could determine what type of tissue was being operated on; therefore, when an unexpectedly high force to close the surgical instrument's end effector is detected, the situationally aware surgical hub **5104** could correctly ramp up or ramp down the motor of the surgical instrument for the type of tissue.

As another example, the type of tissue being operated can affect the adjustments that are made to the compression rate and load thresholds of a surgical stapling and cutting instrument for a particular tissue gap measurement. A situationally aware surgical hub **5104** could infer whether a surgical procedure being performed is a thoracic or an abdominal procedure, allowing the surgical hub **5104** to determine whether the tissue clamped by an end effector of the surgical stapling and cutting instrument is lung (for a thoracic procedure) or stomach (for an abdominal procedure) tissue. The surgical hub **5104** could then adjust the compression rate and load thresholds of the surgical stapling and cutting instrument appropriately for the type of tissue.

As yet another example, the type of body cavity being operated in during an insufflation procedure can affect the function of a smoke evacuator. A situationally aware surgical hub **5104** could determine whether the surgical site is under pressure (by determining that the surgical procedure is utilizing insufflation) and determine the procedure type. As a procedure type is generally performed in a specific body cavity, the surgical hub **5104** could then control the motor rate of the smoke evacuator appropriately for the body cavity being operated in. Thus, a situationally aware surgical hub **5104** could provide a consistent amount of smoke evacuation for both thoracic and abdominal procedures.

As yet another example, the type of procedure being performed can affect the optimal energy level for an ultrasonic surgical instrument or radio frequency (RF) electro-surgical instrument to operate at. Arthroscopic procedures, for example, require higher energy levels because the end effector of the ultrasonic surgical instrument or RF electro-surgical instrument is immersed in fluid. A situationally aware surgical hub **5104** could determine whether the surgical procedure is an arthroscopic procedure. The surgical hub **5104** could then adjust the RF power level or the ultrasonic amplitude of the generator (i.e., "energy level") to compensate for the fluid filled environment. Relatedly, the type of tissue being operated on can affect the optimal

energy level for an ultrasonic surgical instrument or RF electro-surgical instrument to operate at. A situationally aware surgical hub **5104** could determine what type of surgical procedure is being performed and then customize the energy level for the ultrasonic surgical instrument or RF electro-surgical instrument, respectively, according to the expected tissue profile for the surgical procedure. Furthermore, a situationally aware surgical hub **5104** can be configured to adjust the energy level for the ultrasonic surgical instrument or RF electro-surgical instrument throughout the course of a surgical procedure, rather than just on a procedure-by-procedure basis. A situationally aware surgical hub **5104** could determine what step of the surgical procedure is being performed or will subsequently be performed and then update the control algorithms for the generator and/or ultrasonic surgical instrument or RF electro-surgical instrument to set the energy level at a value appropriate for the expected tissue type according to the surgical procedure step.

As yet another example, data can be drawn from additional data sources **5126** to improve the conclusions that the surgical hub **5104** draws from one data source **5126**. A situationally aware surgical hub **5104** could augment data that it receives from the modular devices **5102** with contextual information that it has built up regarding the surgical procedure from other data sources **5126**. For example, a situationally aware surgical hub **5104** can be configured to determine whether hemostasis has occurred (i.e., whether bleeding at a surgical site has stopped) according to video or image data received from a medical imaging device. However, in some cases the video or image data can be inconclusive. Therefore, in one exemplification, the surgical hub **5104** can be further configured to compare a physiologic measurement (e.g., blood pressure sensed by a BP monitor communicably connected to the surgical hub **5104**) with the visual or image data of hemostasis (e.g., from a medical imaging device **124** (FIG. 2) communicably coupled to the surgical hub **5104**) to make a determination on the integrity of the staple line or tissue weld. In other words, the situational awareness system of the surgical hub **5104** can consider the physiological measurement data to provide additional context in analyzing the visualization data. The additional context can be useful when the visualization data may be inconclusive or incomplete on its own.

Another benefit includes proactively and automatically controlling the paired modular devices **5102** according to the particular step of the surgical procedure that is being performed to reduce the number of times that medical personnel are required to interact with or control the surgical system **5100** during the course of a surgical procedure. For example, a situationally aware surgical hub **5104** could proactively activate the generator to which an RF electro-surgical instrument is connected if it determines that a subsequent step of the procedure requires the use of the instrument. Proactively activating the energy source allows the instrument to be ready for use as soon as the preceding step of the procedure is completed.

As another example, a situationally aware surgical hub **5104** could determine whether the current or subsequent step of the surgical procedure requires a different view or degree of magnification on the display according to the feature(s) at the surgical site that the surgeon is expected to need to view. The surgical hub **5104** could then proactively change the displayed view (supplied by, e.g., a medical imaging device for the visualization system **108**) accordingly so that the display automatically adjusts throughout the surgical procedure.

As yet another example, a situationally aware surgical hub **5104** could determine which step of the surgical procedure is being performed or will subsequently be performed and whether particular data or comparisons between data will be required for that step of the surgical procedure. The surgical hub **5104** can be configured to automatically call up data screens based upon the step of the surgical procedure being performed, without waiting for the surgeon to ask for the particular information.

Another benefit includes checking for errors during the setup of the surgical procedure or during the course of the surgical procedure. For example, a situationally aware surgical hub **5104** could determine whether the operating theater is setup properly or optimally for the surgical procedure to be performed. The surgical hub **5104** can be configured to determine the type of surgical procedure being performed, retrieve the corresponding checklists, product location, or setup needs (e.g., from a memory), and then compare the current operating theater layout to the standard layout for the type of surgical procedure that the surgical hub **5104** determines is being performed. In one exemplification, the surgical hub **5104** can be configured to compare the list of items for the procedure scanned by a suitable scanner, for example, and/or a list of devices paired with the surgical hub **5104** to a recommended or anticipated manifest of items and/or devices for the given surgical procedure. If there are any discontinuities between the lists, the surgical hub **5104** can be configured to provide an alert indicating that a particular modular device **5102**, patient monitoring device **5124**, and/or other surgical item is missing. In one exemplification, the surgical hub **5104** can be configured to determine the relative distance or position of the modular devices **5102** and patient monitoring devices **5124** via proximity sensors, for example. The surgical hub **5104** can compare the relative positions of the devices to a recommended or anticipated layout for the particular surgical procedure. If there are any discontinuities between the layouts, the surgical hub **5104** can be configured to provide an alert indicating that the current layout for the surgical procedure deviates from the recommended layout.

As another example, a situationally aware surgical hub **5104** could determine whether the surgeon (or other medical personnel) was making an error or otherwise deviating from the expected course of action during the course of a surgical procedure. For example, the surgical hub **5104** can be configured to determine the type of surgical procedure being performed, retrieve the corresponding list of steps or order of equipment usage (e.g., from a memory), and then compare the steps being performed or the equipment being used during the course of the surgical procedure to the expected steps or equipment for the type of surgical procedure that the surgical hub **5104** determined is being performed. In one exemplification, the surgical hub **5104** can be configured to provide an alert indicating that an unexpected action is being performed or an unexpected device is being utilized at the particular step in the surgical procedure.

Overall, the situational awareness system for the surgical hub **5104** improves surgical procedure outcomes by adjusting the surgical instruments (and other modular devices **5102**) for the particular context of each surgical procedure (such as adjusting to different tissue types) and validating actions during a surgical procedure. The situational awareness system also improves surgeons' efficiency in performing surgical procedures by automatically suggesting next steps, providing data, and adjusting displays and other modular devices **5102** in the surgical theater according to the specific context of the procedure.

In one aspect, as described hereinbelow with reference to FIGS. **24-40**, the modular device **5102** is implemented as the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**). Accordingly, the modular device **5102** implemented as the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**) are configured to operate as a data source **5126** and to interact with the database **5122** and patient monitoring devices **5124**. The modular device **5102** implemented as the combination energy instrument **213100** (FIG. **18**), the surgical stapler **213102** (FIG. **18**), the suction/irrigation device **213104** (FIG. **18**), the sterile field display **213108** (FIG. **18**), the surgical device **213204** (FIGS. **19A** and **19B**), the surgical evacuation system **50400** (FIG. **21**), the electrosurgical instrument **50630** (FIG. **22**), the evacuation system **50600** (FIG. **22**), the evacuation system **50900** (FIG. **23**), the scope **213501** (FIG. **25**), and the surgical device **213608** (FIG. **26**) are further configured to interact with the surgical hub **5104** to provide information (e.g., data and control) to the surgical hub **5104** and receive information (e.g., data and control) from the surgical hub **5104**.

Referring now to FIG. **15**, a timeline **5200** depicting situational awareness of a hub, such as the surgical hub **106** or **206** (FIGS. **1-11**), for example, is depicted. The timeline **5200** is an illustrative surgical procedure and the contextual information that the surgical hub **106**, **206** can derive from the data received from the data sources at each step in the surgical procedure. The timeline **5200** depicts the typical steps that would be taken by the nurses, surgeons, and other medical personnel during the course of a lung segmentectomy procedure, beginning with setting up the operating theater and ending with transferring the patient to a post-operative recovery room.

The situationally aware surgical hub **106**, **206** receives data from the data sources throughout the course of the surgical procedure, including data generated each time medical personnel utilize a modular device that is paired with the surgical hub **106**, **206**. The surgical hub **106**, **206** can receive this data from the paired modular devices and other data sources and continually derive inferences (i.e., contextual information) about the ongoing procedure as new data is received, such as which step of the procedure is being performed at any given time. The situational awareness system of the surgical hub **106**, **206** is able to, for example, record data pertaining to the procedure for generating reports, verify the steps being taken by the medical personnel, provide data or prompts (e.g., via a display screen) that may be pertinent for the particular procedural step, adjust modular devices based on the context (e.g., activate monitors, adjust the field of view (FOV) of the medical imaging device, or change the energy level of an ultrasonic surgical instrument or RF electrosurgical instrument), and take any other such action described above.

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As the first step S202 in this illustrative procedure, the hospital staff members retrieve the patient's EMR from the hospital's EMR database. Based on select patient data in the EMR, the surgical hub 106, 206 determines that the procedure to be performed is a thoracic procedure.

Second step S204, the staff members scan the incoming medical supplies for the procedure. The surgical hub 106, 206 cross-references the scanned supplies with a list of supplies that are utilized in various types of procedures and confirms that the mix of supplies corresponds to a thoracic procedure. Further, the surgical hub 106, 206 is also able to determine that the procedure is not a wedge procedure (because the incoming supplies either lack certain supplies that are necessary for a thoracic wedge procedure or do not otherwise correspond to a thoracic wedge procedure).

Third step S206, the medical personnel scan the patient band via a scanner that is communicably connected to the surgical hub 106, 206. The surgical hub 106, 206 can then confirm the patient's identity based on the scanned data.

Fourth step S208, the medical staff turns on the auxiliary equipment. The auxiliary equipment being utilized can vary according to the type of surgical procedure and the techniques to be used by the surgeon, but in this illustrative case they include a smoke evacuator, insufflator, and medical imaging device. When activated, the auxiliary equipment that are modular devices can automatically pair with the surgical hub 106, 206 that is located within a particular vicinity of the modular devices as part of their initialization process. The surgical hub 106, 206 can then derive contextual information about the surgical procedure by detecting the types of modular devices that pair with it during this pre-operative or initialization phase. In this particular example, the surgical hub 106, 206 determines that the surgical procedure is a VATS procedure based on this particular combination of paired modular devices. Based on the combination of the data from the patient's EMR, the list of medical supplies to be used in the procedure, and the type of modular devices that connect to the hub, the surgical hub 106, 206 can generally infer the specific procedure that the surgical team will be performing. Once the surgical hub 106, 206 knows what specific procedure is being performed, the surgical hub 106, 206 can then retrieve the steps of that procedure from a memory or from the cloud and then cross-reference the data it subsequently receives from the connected data sources (e.g., modular devices and patient monitoring devices) to infer what step of the surgical procedure the surgical team is performing.

Fifth step S210, the staff members attach the EKG electrodes and other patient monitoring devices to the patient. The EKG electrodes and other patient monitoring devices are able to pair with the surgical hub 106, 206. As the surgical hub 106, 206 begins receiving data from the patient monitoring devices, the surgical hub 106, 206 thus confirms that the patient is in the operating theater.

Sixth step S212, the medical personnel induce anesthesia in the patient. The surgical hub 106, 206 can infer that the patient is under anesthesia based on data from the modular devices and/or patient monitoring devices, including EKG data, blood pressure data, ventilator data, or combinations thereof, for example. Upon completion of the sixth step S212, the pre-operative portion of the lung segmentectomy procedure is completed and the operative portion begins.

Seventh step S214, the patient's lung that is being operated on is collapsed (while ventilation is switched to the contralateral lung). The surgical hub 106, 206 can infer from the ventilator data that the patient's lung has been collapsed, for example. The surgical hub 106, 206 can infer that the

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operative portion of the procedure has commenced as it can compare the detection of the patient's lung collapsing to the expected steps of the procedure (which can be accessed or retrieved previously) and thereby determine that collapsing the lung is the first operative step in this particular procedure.

Eighth step S216, the medical imaging device (e.g., a scope) is inserted and video from the medical imaging device is initiated. The surgical hub 106, 206 receives the medical imaging device data (i.e., video or image data) through its connection to the medical imaging device. Upon receipt of the medical imaging device data, the surgical hub 106, 206 can determine that the laparoscopic portion of the surgical procedure has commenced. Further, the surgical hub 106, 206 can determine that the particular procedure being performed is a segmentectomy, as opposed to a lobectomy (note that a wedge procedure has already been discounted by the surgical hub 106, 206 based on data received at the second step S204 of the procedure). The data from the medical imaging device 124 (FIG. 2) can be utilized to determine contextual information regarding the type of procedure being performed in a number of different ways, including by determining the angle at which the medical imaging device is oriented with respect to the visualization of the patient's anatomy, monitoring the number or medical imaging devices being utilized (i.e., that are activated and paired with the surgical hub 106, 206), and monitoring the types of visualization devices utilized. For example, one technique for performing a VATS lobectomy places the camera in the lower anterior corner of the patient's chest cavity above the diaphragm, whereas one technique for performing a VATS segmentectomy places the camera in an anterior intercostal position relative to the segmental fissure. Using pattern recognition or machine learning techniques, for example, the situational awareness system can be trained to recognize the positioning of the medical imaging device according to the visualization of the patient's anatomy. As another example, one technique for performing a VATS lobectomy utilizes a single medical imaging device, whereas another technique for performing a VATS segmentectomy utilizes multiple cameras. As yet another example, one technique for performing a VATS segmentectomy utilizes an infrared light source (which can be communicably coupled to the surgical hub as part of the visualization system) to visualize the segmental fissure, which is not utilized in a VATS lobectomy. By tracking any or all of this data from the medical imaging device, the surgical hub 106, 206 can thereby determine the specific type of surgical procedure being performed and/or the technique being used for a particular type of surgical procedure.

Ninth step S218, the surgical team begins the dissection step of the procedure. The surgical hub 106, 206 can infer that the surgeon is in the process of dissecting to mobilize the patient's lung because it receives data from the RF or ultrasonic generator indicating that an energy instrument is being fired. The surgical hub 106, 206 can cross-reference the received data with the retrieved steps of the surgical procedure to determine that an energy instrument being fired at this point in the process (i.e., after the completion of the previously discussed steps of the procedure) corresponds to the dissection step. In certain instances, the energy instrument can be an energy tool mounted to a robotic arm of a robotic surgical system.

Tenth step S220, the surgical team proceeds to the ligation step of the procedure. The surgical hub 106, 206 can infer that the surgeon is ligating arteries and veins because it receives data from the surgical stapling and cutting instru-

ment indicating that the instrument is being fired. Similarly to the prior step, the surgical hub **106**, **206** can derive this inference by cross-referencing the receipt of data from the surgical stapling and cutting instrument with the retrieved steps in the process. In certain instances, the surgical instrument can be a surgical tool mounted to a robotic arm of a robotic surgical system.

Eleventh step **S222**, the segmentectomy portion of the procedure is performed. The surgical hub **106**, **206** can infer that the surgeon is transecting the parenchyma based on data from the surgical stapling and cutting instrument, including data from its cartridge. The cartridge data can correspond to the size or type of staple being fired by the instrument, for example. As different types of staples are utilized for different types of tissues, the cartridge data can thus indicate the type of tissue being stapled and/or transected. In this case, the type of staple being fired is utilized for parenchyma (or other similar tissue types), which allows the surgical hub **106**, **206** to infer that the segmentectomy portion of the procedure is being performed.

Twelfth step **S224**, the node dissection step is then performed. The surgical hub **106**, **206** can infer that the surgical team is dissecting the node and performing a leak test based on data received from the generator indicating that an RF or ultrasonic instrument is being fired. For this particular procedure, an RF or ultrasonic instrument being utilized after parenchyma was transected corresponds to the node dissection step, which allows the surgical hub **106**, **206** to make this inference. It should be noted that surgeons regularly switch back and forth between surgical stapling/cutting instruments and surgical energy (i.e., RF or ultrasonic) instruments depending upon the particular step in the procedure because different instruments are better adapted for particular tasks. Therefore, the particular sequence in which the stapling/cutting instruments and surgical energy instruments are used can indicate what step of the procedure the surgeon is performing. Moreover, in certain instances, robotic tools can be utilized for one or more steps in a surgical procedure and/or handheld surgical instruments can be utilized for one or more steps in the surgical procedure. The surgeon(s) can alternate between robotic tools and handheld surgical instruments and/or can use the devices concurrently, for example. Upon completion of the twelfth step **S224**, the incisions are closed up and the post-operative portion of the procedure begins.

Thirteenth step **S226**, the patient's anesthesia is reversed. The surgical hub **106**, **206** can infer that the patient is emerging from the anesthesia based on the ventilator data (i.e., the patient's breathing rate begins increasing), for example.

Lastly, the fourteenth step **S228** is that the medical personnel remove the various patient monitoring devices from the patient. The surgical hub **106**, **206** can thus infer that the patient is being transferred to a recovery room when the hub loses EKG, BP, and other data from the patient monitoring devices. As can be seen from the description of this illustrative procedure, the surgical hub **106**, **206** can determine or infer when each step of a given surgical procedure is taking place according to data received from the various data sources that are communicably coupled to the surgical hub **106**, **206**.

In various aspects, the combination energy instrument **213100** (FIG. 18), the surgical stapler **213102** (FIG. 18), the suction/irrigation device **213104** (FIG. 18), the sterile field display **213108** (FIG. 18), the surgical device **213204** (FIGS. 19A and 19B), the surgical evacuation system **50400** (FIG. 21), the electrosurgical instrument **50630** (FIG. 22), the

evacuation system **50600** (FIG. 22), the evacuation system **50900** (FIG. 23), the scope **213501** (FIG. 25), and the surgical device **213608** (FIG. 26) are configured to operate in a situational awareness in a hub environment, such as the surgical hub **106** or **206** (FIGS. 1-11), for example, as depicted by the timeline **5200**. Situational awareness is further described in U.S. Provisional Patent Application Ser. No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed Apr. 19, 2018, which is herein incorporated by reference in its entirety. In certain instances, operation of a robotic surgical system, including the various robotic surgical systems disclosed herein, for example, can be controlled by the hub **106**, **206** based on its situational awareness and/or feedback from the components thereof and/or based on information from the cloud **104**.

Adaptive Control Schemes and Interactive System Controls

In various aspects, a computer system, such as the computer system of a surgical hub, for example, can be configured to extract deeper context around captured data and then utilize this deeper context to improve indications of actionable items for the operator. Deeper context can include metadata and data combinations, in which data from different sources and/or obtained at different times (e.g., preoperatively, intraoperatively, postoperatively) is synthesized to determine relationships and/or correlations. In one instance, the context of the data can be informed by the situational awareness of the surgical hub during a surgical procedure, which can indicate actionable items, such as step-specific or timed adjustments and/or recommendations for the clinician. The system can be configured to implement an adjustable controls scheme with interdependent control aspects. For example, the system can be configured to provide an adaptive control scheme with interactive system controls based on the captured data.

In one aspect, a control for a surgical hub or hub-connected device can be configured to adjust its function (or have its function adjusted) based on a sensed situation or usage. The function can be adjusted by adjusting a scale or degree of the surgical function (e.g. a power level or clamping force), adjusting one or more thresholds of the surgical function (e.g. maximum and minimum power level limits), adjusting a resulting action of the surgical function (e.g. changing an energy modality), adjusting a type of measurement for affecting the surgical function (e.g. from an on-off activation to an operator-controlled variable activation), and/or adjusting a complementary action for the surgical function (e.g. adjusting a clamping force based on a selected energy modality), for example. Additional examples of adjustments to surgical functions are described herein.

In one aspect, the type of measurement can change between an on-off activation detection to a measurement of degree, such as with a strain gauge or Hall Effect sensor. For example, an actuator for a surgical function can act as an on-off actuator in a first mode corresponding to a first surgical state and a degree-based actuator in a second mode corresponding to a second surgical state. The degree-based actuator can be configured to adjust a firing stroke length and/or force, an end effector closure amount and/or clamping force, and/or an energy activation type and/or level, for example. In one aspect, an activation and/or deactivation threshold can be adjusted based on the time in the procedure and/or whether it is a precision-usage surgical step or a

gross-usage surgical step, which can be informed by the situational awareness of the system, for example.

In one aspect, a surgical hub can be configured to prioritize a visual indication on a display within the operating room or surgical theater of actionable contextual secondary information gathered from the data available during the procedure. The prioritization can be based on the certainty and/or importance of the actionable information, for example. In one aspect, an activation or toggling icon for the actionable information can include highlighting where the normal graphical interface is less highlighted. The display can be on a surgical device or a main display (e.g., a display connected to the surgical hub), for example.

In certain instances, a clinician may want to use a surgical device for multiple surgical functions. For example, a surgical device can be used to coagulate blood vessels, cut tissue, and/or seal tissue. In order to achieve these different functions, the control scheme for the surgical device can be adjusted. For example, different power levels or thresholds can be utilized. Additionally or alternatively, different energy modalities (e.g., ultrasonic, bipolar, and monopolar) can be utilized. It can be desirable to switch between the different surgical functions quickly and without relying on complex controls on the surgical device and/or on a remote console. In certain instances, it can be desirable to implement such control scheme adjustments and adaptations with existing surgical devices and/or using surgical devices having minimal input controls, i.e. actuators.

In certain aspects, a surgical system can be configured to implement automatic changes in controls based on monitored conditions within the procedure. A single input control or actuator can be utilized to effect different surgical functions. The available surgical functions actuatable with the input control can depend on the surgical step or state of a surgical procedure. In various instances, a situationally-aware surgical hub or hub-connected surgical device can determine the surgical step and/or state from the situational awareness thereof. Accordingly, the system can be configured adjust devices' control schemes based on situational awareness of the user's intent. In one instance, in a first surgical state (determined by the situational awareness of the surgical hub), the power level of a transducer can be defined by a first range and, in a second surgical state (determined by the situational awareness of the surgical hub), the power level of the transducer can be defined by a second, different range. Additionally or alternatively, in a first surgical state (determined by the situational awareness of the surgical hub), a combination energy device can operate in a combination mode and, in a second surgical state (determined by the situational awareness of the surgical hub), the combination energy device can operate in an ultrasonic mode without the application of electrosurgical energy. The different surgical functions can be implemented by a single actuator, which can comprise a smart or intelligent actuator that is informed by the situational awareness of a surgical hub.

In one aspect, a surgical device can comprise an actuator configured to receive an input. The surgical device can further comprise a control circuit configured to receive a signal from a situationally-aware surgical hub indicative of a surgical state, receive an actuation signal from the actuator in response to the input, and implement a surgical function in response to the actuation signal. The surgical function can comprise a first surgical function when the surgical state corresponds to a first surgical state and a second surgical function when the surgical state corresponds to a second surgical state. The second surgical state can be different than the first surgical state and the second surgical function can

be different than the first surgical function. In another aspect, a non-transitory medium storing computer readable instructions can, when executed, cause a machine, like an intelligent surgical device, to receive a signal from a situationally-aware surgical hub indicative of a surgical state, receive an actuation signal in response to an input applied to an actuator of a surgical device, and implement a surgical function in response to the actuation signal. The surgical function can comprise a first surgical function when the surgical state corresponds to a first surgical state and a second surgical function when the surgical state corresponds to a second surgical state. The second surgical state can be different than the first surgical state and the second surgical function can be different than the first surgical function. Alternative control scheme adjustments and adaptations are further described herein.

As further described herein, the function of a control, such as an actuator, for example, can be adjusted based on one or more factors, including the other instruments currently in use in the surgical theater, the step of the surgical procedure, and/or the input to a secondary control of the surgical device. Such factors can be determined, at least in part, from situational awareness, as further described herein.

In some applications, the degree of displacement of an actuator and/or the input force on the actuator of a surgical device can affect the power (e.g. amplitude) limits of the surgical device. Referring now to FIG. 16, an energy device **213000** can include a first actuation button **213002** and a second actuation button **213004**. In various instances, the first actuation button **213002** can correspond to a first power level, and the second actuation button **213004** can correspond to a second power level. For example, the first actuation button **213002** can comprise a higher, or maximum, power level, which can correspond to a cutting mode, and the second actuation button **213004** can comprise a lower, or minimum, power level, which can correspond to a coagulating mode. In such instances, the first actuation button **213002** can be referred to as a "cut button", and the second actuation button **213004** can be referred to as a "coagulation button."

The actuation buttons **213002**, **213004** can have a spring bias aspect that allows the actuation buttons **213002**, **213004** to act as on-off switches and also detect the force that an operator applies thereto. In one aspect, the actuation buttons **213002**, **213004** can have a first or normal activation that accounts for a particular portion of the button's range of motion and a second or modified activation that accounts for another portion of the button's range of motion. For example, the first 70-80% of the button's range of motion can effect activation of energy and the remaining portion of the button's range of motion can adjust the energy level. In such instances, if the clinician were to compress either actuation button **213002**, **213004** farther by overcoming the higher bias force, then the button could change function and not only apply the cut or coagulation energy level, but increase the power level from the level set on the generator. For example, the power level could be increased one level when the actuation buttons moves into the remaining portion of the button's range of motion. In this operation, the high-force actuation would act to both activate the energy and increase the power level of the energy.

In one aspect, the surgical functions controlled by the actuation buttons **213002**, **213004** can be subject to the position of an end effector **213006** and/or a closure trigger **213008**. For example, referring still to FIG. 16, movement of the closure trigger **213008** through a first range of motion (i.e. through a first angle θ_1), can effect closure of the end

effector **213006** a first amount, a first distance a . Movement of the closure trigger **213008** through a second range of motion (i.e. through a second angle θ_2), can affect closure of the end effector **213006** a second amount, a second distance b . In certain instances, the first amount can correspond to a closure motion of jaws **213010**, **213012**, and the second amount can correspond to a clamping motion of the jaws **213010**, **213012**. The first jaw **213010** can be a clamp arm, which can move relative to the second jaw **213012** in certain instances. In other instances, the second jaw **213012** alone and/or both jaws **213010**, **213012** can be moved by the closure trigger **213008** to close and/or clamp the end effector **213006**.

The first actuation button **213002** and/or the second actuation button **213004** can implement different surgical functions based on the position of the closure trigger **213008** and/or the clamp arm **213010**. For example, if the actuation buttons **213002**, **213004** were activated when the jaws **213010**, **213012** of the end effector **213006** are between the fully open and a partially-closed threshold (e.g., only closed $\frac{2}{3}$ of the entire closure range), the actuation buttons **213002**, **213004** can have respective first energy levels. Further, if the jaws **213010**, **213012** are moved beyond the partially-closed threshold (e.g., over $\frac{2}{3}$ closed), the first and second actuation buttons **213002**, **213004** can have respective second energy levels. The second energy level for each button **213002**, **213004** can be higher than the corresponding first energy levels for the respective actuation buttons **213002**, **213004**. For example, the energy levels can be higher when the jaws **213010**, **213012** are more closed, such as when the jaws **213010**, **213012** are actively clamping tissue therebetween.

A pair of graphs **213020**, **213022** depicting jaw closure and transducer amplitude relative to trigger angle (0) for the energy device **213000** are depicted in FIG. 17. Referring to the graph **213020**, a control algorithm for the energy device **213000** is configured to adjust the amplitude of an ultrasonic transducer according to the trigger angle (0), i.e. the degree to which the closure trigger **213008** (FIG. 16) is depressed. The amplitude can be adjusted at one or more thresholds, such as the trigger angle threshold **213024** indicated in FIG. 17.

More specifically, during a first range of motion of the closure trigger **213008** (e.g. through the first trigger displacement angle θ_1) the first actuation button **213002** can activate energy at a first maximum threshold Max_a and the second actuation button **213004** can activate energy at a first minimum threshold Min_a . During a second range of motion of the closure trigger **213008** (e.g. through the second trigger displacement angle θ_2) the first actuation button **213002** can activate energy at a second maximum threshold Max_b and the second actuation button **213004** can activate energy at a second minimum threshold Min_b . In the depicted graph **213022**, the first minimum threshold Min_a is different than the second minimum threshold Min_b and the first maximum threshold Max_a is different than the second maximum threshold Max_b . More specifically, the first minimum threshold Min_a is less than the second minimum threshold Min_b and the first maximum threshold Max_a is less than the second maximum threshold Max_b . As shown in FIG. 17, the first minimum threshold Min_a comprises 30% of the transducer amplitude limit, whereas the second minimum threshold Min_b comprises 75% of the transducer amplitude limit. Additionally, the second maximum threshold Max_b comprises 100% of the transducer amplitude limit, whereas the first maximum threshold Max_a only comprises 40% of the transducer amplitude limit. Other threshold values are con-

templated. For example, the second minimum threshold Min_b can be less than the first maximum threshold Max_a .

In still other instances, the first minimum threshold Min_a can be greater than the second minimum threshold Min_b and/or the first maximum threshold Max_a can be greater than the second maximum threshold Max_b . In still other instances, only one of the limits can be adjusted when the trigger displacement angle exceeds the trigger angle threshold **213024**. For example, the second actuation button **213004** can maintain the first minimum threshold Min_a during the entire range of motion of the closure trigger **213008**, including the angles θ_1 and θ_2 . In other words, the amplitude of the transducer can be maintained at 30% when the second actuation button **213004** is depressed regardless of the position of the first jaw **213010** and/or the closure trigger **213008**. Additionally or alternatively, one or more of the limits can be adjusted at additional trigger angle thresholds.

Referring now to the graph **213020**, a closure rate for the closure trigger **213008** can also be adjusted based on a detected surgical state and/or surgical step. For example, the closure trigger **213008** can be closed at a first rate during a first range of motion through the first trigger displacement range θ_1 , and the closure trigger **213008** can be closed at a second rate during a second range of motion through the second trigger displacement range θ_2 . More specifically, the first rate, which corresponds to a slope $a/\text{ZONE A}$, can continue until the closure trigger **213008** meets the trigger angle threshold **213024**. In the depicted arrangements, the trigger angle threshold **213024** corresponds to 75% end effector closure. In other instances, the trigger angle threshold **213024** can define less than or more than 75% end effector closure, such as $\frac{2}{3}$ rd closed and/or 90% closed, for example. Upon reaching the trigger angle threshold **213024**, the closure rate can be increased to a second rate in ZONE B until the closure trigger **213008** is 100% closed. In other instances, the closure rate can be decreased upon reaching the trigger angle threshold **213024**. Additionally or alternatively, the closure rate can be adjusted at additional trigger angle thresholds.

In various instances, the position of the closure trigger **213008** can indicate a surgical state of the surgical procedure. For example, the closure trigger **213008** can remain open during a first surgical step, be closed to a first degree during a second surgical step, and be closed to a second degree during a third surgical step. For example, open jaws can be used to cauterize blood vessels, partially closed jaws can be used to coagulate tissue, and/or completely closed jaws can be used to cut tissue. By monitoring the position of the closure trigger **213008**, the control circuit is configured to determine the surgical state and/or step, and adjust a surgical function, i.e. a power level, of the hub-connected device based on the detected state and/or step. Situational awareness can further inform the determination. For example, during a certain type of surgical procedure and/or when treating a certain type of tissue, the maximum and/or minimum threshold(s) and/or trigger angle threshold(s) can be adjusted in a first manner and, during another type of surgical procedure and/or when treating another type of tissue, the maximum and/or minimum threshold(s) and/or trigger angle threshold(s) can be adjusted in a second manner. For example, lung tissue and stomach tissue can necessitate different responses. Additionally or alternatively, arteries and veins can necessitate different thresholds. Situational awareness can determine the surgical procedure and/or tissue type.

In various instances, one or more controls and/or actuators for a surgical device can be adjusted based on another control/actuator on the surgical device. Actuating a first control/actuator can cause a second control/actuator to have an adjusted function. Referring again to FIG. 16, for example, if the jaws **213010**, **213012** of the energy device **213000** are open wide and energy is activated via one of the activation buttons **213002**, **213004**, a surgical system can sense or otherwise determine that the surgeon intends to use the instrument for a first surgical function, such as back cutting, for example. Accordingly, the minimum and maximum power levels (e.g., amplitude) can be set to their normal levels for the first surgical function. If the jaws **213010**, **213012** of the energy device **213000** are more than half open, the energy actuation button(s) **213002**, **213004** can drive an ultrasonic response only; however, if the jaws **213010**, **213012** are more than half closed, the energy device **213000** can function in a standard combination ultrasonic-electrosurgical mode. As yet another example, if the jaws **213010**, **213012** of the energy device **213000** are fully clamped at the highest compression level as ultrasonic energy is being applied and the user requests more pressure via the closure trigger **213008** while still actuating the maximum energy activation button **213002**, the energy modality can automatically shift from ultrasonic to electrosurgical RF, and the energy device **213000** can further increase the pressure applied with the clamping jaw **213010**.

In one aspect, controls can be adjusted according to the other instruments currently in use in the same surgical theater. For example, if a secondary energy device is within the field of view (e.g., the field of view of a scope) and a predefined key and/or sequence of control actuators are actuated, the surgical hub can adjust the controls for a primary energy device also within the field of view. In one instance, when the closure trigger and energy button are actuated on a primary energy device and a secondary device is within the field of view, a scalable adjustment control on the secondary energy device can be used to control the power level for the primary energy device. In such instances, the generator powering the primary energy device can be turned up or down with a control on the secondary energy device. Such remote control of a surgical function can only be enabled when a predefined sequence of surgical actions have been performed and/or during a particular step in a surgical procedure in various instances.

In another example, touching or contact between surgical devices can allow one of the surgical devices to adjust the controls of the other surgical device. For example, if a second surgical device touches a first surgical device, then a particular action or combination of actions performed by or on the second surgical device can cause a control algorithm to adjust a function of the first surgical device. For example, when the jaws of the second surgical device are in a fully open configuration, the jaws can press an actuation button on the first surgical device, and then an adjustment control on the second surgical device can be used to change the power level of the first surgical device, such as the power provided by a generator, for example. Once the desired level is set, the second surgical device can be removed from contact with the first surgical device. Thereafter, the first and second surgical devices can operate as normal devices at the newly-set power level. In one instance, continuity sensors can be configured to detect the contact between the surgical devices.

In one aspect, controls for a surgical device can be adjusted according to the step of the surgical procedure being performed. Situational awareness can inform the sur-

gical hub of the current and/or next step of the surgical procedure. For example, based on the previous surgical actions and/or the order of usage of the surgical device(s) and/or generator(s), a surgical hub can determine what particular step of a particular surgical procedure is being performed, such as whether the procedure is currently in a nodal dissection step, vessel transecting step, and so on. If the surgical hub determines that the energy device being used is in a fine dissection mode, the system can automatically adjust the function of the energy device's buttons to different settings, e.g., a lower power setting for use with this procedural step. In one implementation, the surgical hub and/or generator can determine the procedural specific step or context.

In another implementation, the system can learn and anticipate the procedural specific step or context by analyzing the particular clinician's most common usage at each stage of the surgical procedure and/or after a particular number or type of surgical instrument exchanges. In such a learned mode, the generator, in one aspect, does not initially implement any automatic adjustments to the controls. Rather, after monitoring the same clinician's behavior over a predetermined number of procedures that include the same steps, the surgical device can automatically change the button usage based on the monitored and past usage by the clinician. In various instances, the surgical device can provide notice to the clinician when the function of the controls is adjusted. For example, the surgical device can provide an auditory notice (e.g., a beep or verbal explanation), a visual cue (e.g. a flashing light and/or words on a screen), and/or a tactile warning (e.g. vibrations and/or movement of the surgical device or a portion thereof, such as the actuator button itself). In other instances, the surgical hub can recommend an adjustment to the controls. Recommendations from a surgical hub are further described herein.

In various aspects, controls for a surgical hub, surgical instruments, and other devices can be adjusted based on a screen in operation on a sterile field display. The controls of the surgical devices can be adjusted based on the displayed information. In one aspect, a control that normally controls panning or adjusting the focus of a visualization device (e.g., a scope) can be configured to adjust magnification if a hyperspectral imaging overlay is active, for example. Hyperspectral imaging is further described in U.S. patent application Ser. No. 15/940,722, titled CHARACTERIZATION OF TISSUE IRREGULARITIES THROUGH THE USE OF MONO-CHROMATIC LIGHT REFRACTIVITY, filed Mar. 29, 2018, which is incorporated by reference herein in its entirety.

As another example, the on-handle controls for a surgical instrument in the field of view of a sterile field display can be adjusted by selections on the sterile field display. Moreover, the adjustments can be based on situational awareness in various instances. For example, the system can determine that a particular surgical device is being utilized and permit the functions of that surgical device to be controlled from a second device, such as a display screen within the sterile field. Referring now to FIG. 18, a surgical procedure involving a combination energy instrument **213100**, a surgical stapler **213102**, a suction/irrigation device **213104**, and a sterile field display **213108** is shown. The combination energy device **213100** includes a single energy button **213110**, which is configured to implement different energy modalities. For example, the combination energy device **213100** can operate in a bipolar mode, monopolar mode, ultrasonic mode, and/or various combinations thereof (e.g. a blended mode). In one instance, the combination energy

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device **213100** can operate in a monopolar-bipolar mode. In another instance, the combination energy device **213100** can operate in an ultrasonic-bipolar mode and, in still other instances, the combination energy device can operate in a ultrasonic-monopolar mode. The various energy modalities can be applied sequentially and/or blended together simultaneously.

In one aspect, the controls on the handle of a surgical instrument in the field of view of a sterile field display **213108** can be adjusted by selections on the sterile field display **213108**. For example, the combination energy instrument **213100** includes a single button **213110**, which can be configured to apply energy to selectively cut and coagulate tissue. The energy modality activated by the button **213110** can be adjusted with a selection on the sterile field display **213108**.

In one aspect, the energy activation button **213110** can have an indicator that changes or otherwise indicates when the sterile field display **213108** is being used to change the function of the button **213110** and/or that the sterile field display **213108** has shared control of the button **213110**. For example, the combination energy device **213100** can include an LED on the handle and/or the energy activation button **213110** itself that can change color and/or otherwise changes its appearance (e.g. change the color of the energy activation button **213110** and/or become illuminated with flashing and/or pulsing light) when adjusted from a secondary device, such as the sterile field display **213108**. In one instance, the button **213110** can have a normal red hue projected through light channels from a multicolor LED for a cutting function, whereas a blue hue can be projected through the light channels from the multicolor LED for a coagulation function. When the sterile field display **213108** changes the surgical function to a blended energy modality, for example, the button **213110**, which was red, can be changed to purple. In various instances, the interface area on the sterile field display **213108** can also include an indicator in a corresponding color (e.g. purple words and/or highlighting when the button **213110** is illuminated with a purple hue) to indicate that the functions between the displayed controls on the surgical field display **213108** and the combination energy device **213100** are linked.

Various surgical instruments can be controlled in different ways depending on the number and type of buttons/actuators thereon, as well as the desired surgical functions for the surgical instrument. For example, if a surgical instrument includes two buttons (e.g., a first button corresponding to a first power level that is suitable for coagulation and a second button corresponding to a second power level that is suitable for cutting) the sterile field display **213108** can be configured to adjust the functions of the first and second buttons. For example, one of the buttons can be utilized to activate monopolar energy and the other button can be configured to activate a blended energy. In such instances, the buttons can include an indicator (e.g. a color LED) that changes when the sterile field display **213108** is being used to change the function of the buttons and/or to indicate that the sterile field display **213108** has shared control of the buttons. The indicator can communicate the selected energy modality and/or how the energy modality was selected. For example, the first button can include a red-hued LED, and the second button can include a blue-hued LED. When the first button is configured to activate a blended energy modality, the red-hued LED can become purple-hued. Additionally, the interface area on the sterile field display **213108** can include a purple indicator.

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In various instances, the sterile field display **213108** can allow a clinician to switch the functions of the buttons on the handle of the combination energy device **213100**. In such instances, the color of the two buttons could be reversed. In another instance, the functions of the buttons can be changed. For instance, the second button (which previously corresponded to a power level suitable for cutting tissue, for example) can be changed to an energy activation button, and the first button (which previously corresponded to a power level suitable for coagulating tissue, for example) can be changed to a power level adjustment control. In such instances, each time the power level adjustment control is activated, the sterile field display **213108** can show the newly-selected power level. For example, the sterile field display **213108** can utilize a flashing icon to indicate a change to the generator power level. Thereafter, activation of the second button (currently acting as the energy activation button) can be used to activate and deactivate energy at the newly-set power level.

In still other instances, the button **213110** and/or other buttons on the combination energy device **213100** can be changed from on-off activation functions to a pressure-sensitive activation function. For example, a higher level of force applied to the button **213110** can be configured to increase the intensity of the energy activation such as by turning up the power level based on the force on the button **213110**. Further, the sterile field display **213108** can be used to change the discrete on/off function of the button **213110** to a pressure-sensitive adjustment function. Further, the sterile field display **213108** and the button(s) **213110** can utilize different indications when the power level is changed. For instance, the light emitted from the button(s) could vary in intensity as the pressure-sensitive activation adjusts the power level of the combination energy device **213100**. Additionally, the sterile field display **213108** can indicate a color that is synchronized with the color of the button **213110** to indicate the current function and pairing.

In various instances, a surgical hub can be configured to provide recommendations to a clinician during a surgical procedure. For example, the surgical hub can aggregate data from multiple sources to determine a recommended surgical action. In various instances, the recommendations can be based on the situational awareness of the surgical hub. The relevancy or importance of a recommendation can depend on the surgical context or scenario. For example, a general recommendation can be escalated to a higher level based on data obtained by various data sources and/or sensors. Because a surgical hub is able to aggregate significant volumes of information, it can be helpful to prioritize recommendations for the clinician to avoid overwhelming the clinician with too much data at a given time. For example, if a clinician is constantly being interrupted with recommendations—of which some or many are non-relevant and/or unhelpful—the clinician may ignore the hub's recommendations entirely and/or disable a recommendation setting on the surgical hub. By prioritizing recommendations based on situational awareness, the clinician can be well-informed with respect to actionable and/or time-sensitive items.

In various instances, the importance of the recommendation can depend on a position of a surgical device. For example, actionable aspects can be highlighted or emphasized to the clinician. Additionally, non-actionable and/or non-critical aspects can be displayed discretely and/or selectively hidden from view. If the position of the surgical device indicates that a recommendation is actionable by the clinician, the surgical hub can emphasize the recommendation

whereas the non-actionable recommendations can be displayed more discretely and/or selectively hidden from view. In such instances, actionable recommendations are selectively communicated to the clinician, which enables the clinician to consider the value of such a recommendation in a timely manner.

In various instances, for example, a surgical system can include a screen configured to selectively display a recommendation to a clinician in an operating room. The surgical system can also include a control circuit communicatively coupled to the screen and a situationally-aware surgical hub, wherein the control circuit is configured to receive a signal from the situationally-aware surgical hub indicative of a surgical state, determine a priority level of the recommendation based on the surgical state, and communicate the priority level of the recommendation via the display. In various instances, the control circuit comprises a processor and a memory communicatively coupled to the processor, the memory storing instructions executable by the processor to receive a signal from the situationally-aware surgical hub indicative of a surgical state, determine a priority level of the recommendation based on the surgical state, and communicate the priority level of the recommendation via the display. In various instances, a non-transitory computer readable medium stores computer readable instructions which, when executed, cause a machine, like an intelligent surgical device, to receive a signal from the situationally-aware surgical hub indicative of a surgical state, determine a priority level of the recommendation based on the surgical state, and communicate the priority level of the recommendation via the display.

In various aspects, a surgical system can be configured to provide a notification to a clinician when a specific aspect of contextual information is actionable. For example, FIGS. 19A and 19B depict a display screen **213200** including a recommendation **213202**, **213202'** for a clinician based on input from a situationally-aware surgical hub, such as the surgical hub **106** (FIGS. 1-3) or surgical hub **206** (FIG. 10) or surgical hub **7006** (FIG. 12). In various instances, the display screen **213200** includes a user interface overlay on the display screen **213200**, which provides context-dependent recommendations. For example, the surgical system can highlight a data feed with an actionable icon to indicate that the underlying contextual information has an actionable aspect. In various instances, the recommendation feed can always be present; however, the recommendation is only emphasized when the recommendation is actionable. A recommendation can be emphasized with highlighting, flashing, and/or markings, for example. In other instances, the recommendations feed is only displayed when it includes an actionable item.

As shown in FIG. 19B, the recommendation **213202'** is highlighted to indicate an elevated priority level based on an anticipated surgical act and the input from the situationally-aware surgical hub. More specifically, in FIG. 19A, a surgical device **213204** is shown in a first position, which is in a general vicinity of a diseased area **213206** of a lung **213208**. In FIG. 19B, the surgical device **213204** has moved toward the diseased area **213206**. Based on the trajectory of the surgical device **213204** and/or the position of the surgical device **213204**, the surgical system can determine that the recommendation **213202** from FIG. 19A is now actionable and, thus, indicate the recommendation **213202'** with highlighting as shown in FIG. 19B. In other words, a position change of the surgical device **213204** changes the confidence of the context-based recommendation. The increased

confidence level corresponds to an elevated priority level for the recommendation **213202'**.

The actionable aspect, or actionable item, can be a user-selected change in an instrument functional parameter in certain instances. For example, if friable tissue is detected by the system, the system can recommend an adjustment to protect the tissue. In one instance, the surgical system can recommend a reduction of the targeted clamping load of a stapler. Additionally or alternatively, the surgical system can recommend use of a surgical adjunct material, such as a buttress or tissue thickness compensator, for example. As another example, if the system detects comorbidity, such as the comorbidity of the patient and a mendicant, for example, that affects healing, the system can be configured to prompt or propose that the user adjust the compression wait time, utilize an adjunct, and/or utilize a reduced staple size.

In one aspect, the actionable aspect can be the automatic overlay of procedure-critical information during device placement. For example, referring again to FIGS. 19A and 19B, the display screen **213200** is configured to label and/or call-out the diseased area **213206** and/or nodules in proximity of the surgical devices. In various aspects, the automatic feature can be toggled between on and off depending on a user's preferences. Similarly, the recommendation feed can be toggled between on and off. In various instances, recommendations can only be displayed when they are actionable and/or elevated to a priority level above a pre-defined threshold that is preprogrammed and/or adjusted by an operator.

In one aspect, the system can be configured to highlight a data feed that has direct implications to a current surgical step or action. The relevant data within the surgical context might impact how the clinician conducts one or more of the subsequent functional tasks. For example, the data feed related to blood pressure measurements can be highlighted in certain instances, such as when the blood pressure measurement is unusually high or low. Unusually high or low blood pressure can affect how a clinician performs a transection of a large vessel with an energy device. Therefore, in various instances, when the situational awareness indicates that the clinician is preparing to perform the transection, the blood pressure data can be highlighted. As another example, the system can be configured to monitor the return path linkage to the patient. When the return path linkage is limited and a monopolar electrosurgical instrument is being utilized for dissection or mobilization, for example, the return path linkage data can be highlighted for the clinician.

In yet another example, near-infrared (NIR) fluorescence of indocyanine green (ICG) imaging can be highlighted based on actionable items determined by the situational awareness and/or context clues detected by one or more sensors at the surgical site. NIR fluorescence can be used to visualize anatomic structures, perfusion and perfusion defects as well as the lymphatic system, for example. The high penetration depth of NIR light can allow the visualization of the distribution of ICG up to a depth of 10 mm below the tissue surface. In liver surgery, the system can be used to visualize liver metastases or primary tumors of the liver. Exemplary applications include:

Visualization of perfusion, in which ischemic areas can be displayed in real-time more easily and efficiently, and/or perfusion can be examined, for instance in anastomoses, which can enable the surgeon to take intraoperative action, such as for rapid perfusion assessment of a planned resection zone, as well as of the subsequent anastomosis in a colon or esophageal resection and gastric bypass, for example, perfusion assessment of

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flap plasty in open surgery, visualization of liver segments, and multidisciplinary use in laparoscopic, endoscopic, and open surgery;

Visualization of the biliary tree anatomy, which can allow rapid and reliable identification of the biliary anatomy, for instance in cholecystectomy for reduced surgical duration and facilitated differentiation between cystic duct and common bile duct, as well as display of intraoperative bile leakage with ICG, such as following a partial hepatectomy, for example;

Visualization of hepatocellular carcinoma and metastases, including intraoperative visualization of metastases and hepatocellular carcinoma on or below the liver surface, diagnostics of superficial or near-surface micrometastases down to the millimeter range, easier definition of resection borders, and visualization of liver segments, for example; and

Visualization of the lymphatic system, which allows real-time visualization of the entire lymphatic system draining a tumor, for example, and can avoid nuclear medicine imaging with detection rates comparing favorably with established methods of lymph node visualization, localization of lymphatic leakages, and multidisciplinary use, for instance in gynecology, urology, and general surgery.

NIR/ICG Visualization can be provided by OPAL1® technology from KARL STORZ, which is described at www.karlstorz.com/ie/en/nir-icg-near-infrared-fluorescence.htm, for example. The NIR/ICG data can be emphasized or highlighted to the clinician when the visualization is important for how the surgeon performs the subsequent step based on the situational awareness and anticipated surgical steps.

In another example, a surgical system can be configured to highlight an alternative imaging or analysis system when the surgical system detects that the imaging or analysis system is detecting something relevant to the current step or current task of a surgical device. For example, the system can highlight the laser Doppler scanning array function when a dissector is about to intersect with a significant vessel or underlining structure. As another example, the system can highlight computed tomography (CT) imaging when a stapler appears to be placed within the limits of a marginal tissue area. The various recommendations to the clinician can be communicated on a display screen within the sterile field. In various instances, one or more recommendations can be further emphasized with an auditory and/or tactile indication in the sterile field, such as a verbal warning and/or beep when the recommendation is actionable and/or time-sensitive.

In various instances, a recommendation is actionable when the recommendation can be implemented within a predefined time and/or number of surgical steps. For example, a recommendation can be actionable if it should be implemented within the upcoming minutes (e.g. 1-2 minutes) of the procedure. In certain instances, a recommendation can be actionable if it should be implemented within the next 5, 10, or 30 seconds of the procedure, for example. The time frame can be determined by a clinician's reaction skills, for example. In certain instances, a first type of emphasis for the recommendation can be provided at a first time, and a second type of emphasis can be provided at a later time when the recommendation is more time-sensitive/urgent. In still other instances, the recommendation can be provided when it relates to the next surgical step expected based on the situational awareness of the clinician. In other instances, recommendations can be provided two steps before it should

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be implemented. The reader will appreciate that the various time-based and step-based settings for actionable recommendations can be predefined and/or adjusted by a clinician and/or operator. In certain instances, actionable recommendations can be required, such as during a surgical training session, for example. In other instances, actionable recommendations can be optional and/or customizable by the clinician.

In various aspects, a surgical system can be configured to implement a data bridge for the clinician. Various techniques for interfacing the data within the surgical hub to a display for use by a clinician are described herein. For example, intraoperative surgical data can be contributed to the implementation and tracking of post-operative care metrics.

FIG. 20 depicts a block diagram 213300 illustrating intraoperative surgical data 213302 being correlated with predicted outcomes 213304 and corresponding recommendations 213306 being provided. For example, intraoperative surgical data 213302 can be correlated with prior outcomes to determine the predicted outcomes 213304 based on the intraoperative surgical data 213302 to inform the clinician(s) about key concerns, recommendations, and potential best practices. Further, the intraoperative surgical data 213302 can be correlated to prior case history to predict outcomes of care metrics flagged by an operator or organization, such as a hospital or health system. Predicted outcomes 213304 that are based on the data correlations can include, for example, the patient's length of stay, re-admission rates, re-operation rates, time to chest tube removal, time to colon motility, and the number, type, and severity of insurance or other clinically-coded events (e.g., ICD-9 or ICD-10 codes for bleeding related complications, transfusions, and so on). Further, the intraoperative data set can, in some aspects, be comprehensive of all available primary and secondary data sources from the procedure (e.g., EMR, situational awareness learnings, video, and so on).

The block diagram 213300 refers to a Lower Anterior Resection (LAR) procedure, in which a tumor is located deep in the patient's pelvis. Firing information, which can be obtained from a video and/or various sensors, for example, indicates the margin is suboptimal, which corresponds to an increased chance of cancer reoccurrence. The firing information also indicates that significant tension is placed on the colon, which corresponds to an increased risk of leaking. Furthermore, the firing information indicates that the firing line is not perpendicular, which corresponds to an increased time to colon mobility. The intraoperative surgical data 213302 also monitors the procedure time. In the example of FIG. 20, the procedure time is characterized as "very long" with extensive dissection and significant colon manipulation. In combination, these factors also correspond to an increased chance of cancer reoccurrence. The data bridge can acquire additional data as well, such as the medications delivered and/or patient demographics, for example, which can further correspond to various predicted outcomes and/or comorbidities. Based on the various predicted outcomes 213304, the surgical hub can provide various recommendations for post-operative care, including aggressive oncology follow-up treatment, placement of a rectal drain, increased dietary restrictions, a hospital cost analysis, and an anticipated length-of-stay (LOS) can inform the hospital bed assignments and/or requirements.

In various aspects, a surgical system can be configured to detect movement, velocity, acceleration, and/or orientation of elements, such as particulate matter and/or aerosol particles, for example, within a scope's field of view (FOV). By detecting one or more of these properties, the surgical

system can increase the depth of data that is extractable during a surgical procedure and/or can illuminate aspects of the surgical procedure for a clinician. For example, the system can be configured to determine active actionable events for the clinician's attention.

Properties of fluid flow at a surgical site can be an indicator of an actionable item, or adjustment, for a surgical hub and/or clinician. Fluid flow includes the flow of smoke, insufflation fluids, blood, contrast agents, and/or dye, for example. In various instances, properties of fluid flow can correspond to different surgical events, such as increased smoke generation during an energy treatment, detection of bleeding, perfusion assessment, and/or detection of a leaking anastomosis. Based on the actionable item determined by a detected property, or properties, of fluid flow, a recommendation can be provided. However, without the detection and/or monitoring of one or more properties of fluid flow, it can be difficult to determine various actionable items and corresponding recommendations.

In various instances, by detecting and/or monitoring one or more properties of fluid flow, an increased depth of data can be extracted for analysis by a surgical hub. As a result, actionable items can be identified and/or identified more expediently and corresponding recommendations can be provided and/or implemented in a timely manner. In certain instances, a surgical hub can automatically implement an adjustment and/or alert the clinician to an actionable item.

In one aspect, a surgical system comprises a sensor configured to detect a property of airborne particles in a fluid within a patient's abdominal cavity, a surgical device configured to implement a surgical function, and a control circuit comprising a processor and a memory communicatively coupled to the processor, wherein the control circuit is communicatively coupled to the sensor and the surgical device. The control circuit can be configured to receive an input signal from the sensor indicative of the property of airborne particles in the fluid and, in response to the input signal, provide an output signal to the surgical device indicative of an adjustment to the surgical function. The surgical system can be configured to detect airborne particulates and aerosols within the insufflation gases within a body (e.g., the abdomen), for example. In one aspect, the detection methods include, for example, optical, laser refractive, ultrasonic, and/or magnetic resonance. In one aspect, functional adjustments can include, for example, affecting the power level of the generator, the venting of the space, the degree of smoke evacuation, the degree of filtration, and/or application of a secondary condenser to induce condensation. In one aspect, the adjustment can result in a proportionate increase in a system that was already active, the addition of a second effect or function to the already active one, and/or the swapping out of one actuation or function for another system entirely. Various adjustments are further described herein.

Detecting one or more properties of fluid flow at a surgical site can encompass one or more measurement systems and/or methods further described herein. For example, optical measurement systems and/or ultrasonic measurement systems can be employed. Optical measurement systems include laser Doppler velocimetry (LDV) or laser Doppler flowmetry (LDF), particle image velocimetry (PIV), and/or near-infrared (NIR) fluorescence of indocyanine green (ICG), for example, which are further described herein. Ultrasonic measurement systems include non-contact pass-through ultrasound detection and non-contact reflection ultrasound detection, for example, which are also further described herein. The various techniques for detecting and/

or imaging particle flow can be employed in a number of surgical uses including smoke detection and control of evacuation, aerosol and particulate differentiation, bleeding detection, perfusion assessment to identify anatomical structures, dye identification, differentiation of broken superficial surface reflection, and identification of flowing, leaking, and/or oozing fluids on a surface, for example. Various surgical uses are further described herein.

LDV, which is often referred to as LDF in a clinical or medical application, is a technique in which the Doppler shift in a laser beam is used to measure the velocity in transparent or semi-transparent fluid flows and/or the linear or vibratory motion of opaque, reflecting surfaces. In one instance, two beams of collimated, monochromatic, and coherent laser light cross in the flow of the fluid being measured. The two beams are usually obtained by splitting a single beam, thus ensuring coherence between the two beams. Lasers with wavelengths in the visible spectrum (390-750 nm) can be used, such as He—Ne, Argon ion, and/or laser diode, which allow the beam path to be observed. In other instances, wavelengths outside of the visible spectrum can be utilized, which can avoid distracting a clinician and/or interference with other imaging/visualization systems, for example. Transmitting optics focus the two beams to intersect at the focal point of the beam. The intersecting beams can interfere and generate a set of straight fringes, for example. As particles entrained in the fluid pass through the fringes, they reflect light that is then collected by a receiving optics and focused on a photodetector. The reflected light fluctuates in intensity, the frequency of which is equivalent to the Doppler shift between the incident and scattered light, and is thus proportional to the component of particle velocity that lies in the plane of the two beams. If the LDV sensor is aligned with the flow such that the fringes are perpendicular to the flow direction, the electrical signal from the photodetector is proportional to the full particle velocity. By combining three devices (e.g.; He—Ne, Argon ion, and laser diode lasers) with different wavelengths, all three flow velocity components can be simultaneously measured in various instances.

In various instances, a PIV system can be utilized to determine two-dimensional or three-dimensional vector fields for a fluid flow. PIV is an optical method of flow visualization that can be used to obtain instantaneous velocity measurements and related properties in fluids. The fluid includes tracer particles that are assumed to faithfully follow the flow dynamics of the fluid. The fluid with entrained particles is illuminated so that the particles are visible, and the motion of the particles is used to calculate speed and direction (the velocity field) of the flow. A PIV sensor can include a camera, such as a digital camera with a Charge-Coupled Device (CCD) chip, a strobe or laser with an optical arrangement to limit the physical region illuminated, such as a cylindrical lens used to convert a light beam to a line or sheet, a synchronizer to act as an external trigger for control of the camera and the laser, and seeding or tracer particles within the fluid under investigation. A fiber optic cable or liquid light guide can connect the laser to the lens setup, and PIV software can be used to post-process the optical images, for example.

In a two-dimensional PIV application, a laser light sheet of light scattering particles is illuminated through two consecutive frames, which are captured to create a two-dimensional map of particle movement and the vectors of the movement. For stereo PIV, or three-dimensional PIV, two or more cameras can be utilized at predefined angles to one another to capture the two consecutive frames and another

camera can be configured to enable out-of-plane measurement. In one aspect, time-resolved particle tracking can enable the evaluation of particle tracks or paths, as well as a determination of the volume and velocity of the particles.

Surgical uses for the optical particle detection and tracking techniques and systems described herein can include, for example, smoke detection applications, bleeding detection applications, perfusion assessment applications, and dye identification applications. Various example applications are further described herein.

In one application, detection of the presence and/or concentration of smoke in an internal cavity, such as an abdomen cavity, for example, can be utilized for rate control of insufflation equipment, rate control or activation of the smoke evacuation equipment, and/or adaptation of generator algorithm parameters to minimize smoke generation. Smoke evacuation systems are further described herein.

In various instances, a surgical hub can be communicatively coupled to a smoke evacuation system. Smoke is often generated during a surgical procedure that utilizes one or more energy devices. Energy devices use energy to affect tissue. In an energy device, the energy is supplied by a generator, as further described herein. Energy devices include devices with tissue-contacting electrodes, such as an electrosurgical device having one or more radio frequency (RF) electrodes, and devices with vibrating surfaces, such as an ultrasonic device having an ultrasonic blade. For an electrosurgical device, a generator is configured to generate oscillating electric currents to energize the electrodes. For an ultrasonic device, a generator is configured to generate ultrasonic vibrations to energize the ultrasonic blade.

Many surgical systems employ a surgical evacuation system that captures the resultant smoke from a surgical procedure, and directs the captured smoke through a filter and an exhaust port away from the clinician(s) and/or from the patient(s). For example, an evacuation system can be configured to evacuate smoke that is generated during an electrosurgical procedure. The reader will appreciate that such an evacuation system can be referred to as a “smoke evacuation system” though such evacuation systems can be configured to evacuate more than just smoke from a surgical site. Throughout the present disclosure, the “smoke” evacuated by an evacuation system is not limited to just smoke. Rather, the smoke evacuation systems disclosed herein can be used to evacuate a variety of fluids, including liquids, gases, vapors, smoke, steam, or combinations thereon. The fluids can be biologic in origin and/or can be introduced to the surgical site from an external source during a procedure. The fluids can include water, saline, lymph, blood, exudate, and/or pyogenic discharge, for example. Moreover, the fluids can include particulates or other matter (e.g. cellular matter or debris) that is evacuated by the evacuation system. For example, such particulates can be suspended in the fluid. Example smoke evacuation systems, as well as various subassemblies and/or components thereof, are depicted in FIGS. 21-23. Smoke evacuation systems are further described in U.S. patent application Ser. No. 16/024,066, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL, filed Jun. 29, 2018, which is incorporated by reference herein in its entirety.

In certain instances, a processor can be located within an evacuator housing of a surgical evacuation system. For example, referring to FIG. 21, a processor 50408 and a memory 50410 therefor are positioned within an evacuator housing 50440 of a surgical evacuation system 50400. The processor 50408 is in signal communication with a motor driver 50428, various internal sensors 50430, a display

50442, the memory 50410, and a communication device 50418. The communication device 50418 can include a transceiver configured to communicate over physical wires or wirelessly. The communication device 50418 may further include one or more additional transceivers. The transceivers may include, but are not limited to cellular modems, wireless mesh network transceivers, Wi-Fi® transceivers, low power wide area (LPWA) transceivers, and/or near field communications transceivers (NFC). The communication device 50418 may include or may be configured to communicate with a mobile telephone, a sensor system (e.g., environmental, position, motion, etc.) and/or a sensor network (wired and/or wireless), a computing system (e.g., a server, a workstation computer, a desktop computer, a laptop computer, a tablet computer (e.g., iPad®, GalaxyTab® and the like), an ultraportable computer, an ultramobile computer, a netbook computer and/or a subnotebook computer; etc. In at least one aspect of the present disclosure, one of the devices may be a coordinator node.

The transceivers may be configured to receive serial transmit data via respective universal asynchronous receiver-transmitters (UARTs) from the processor 50408 to modulate the serial transmit data onto an RF carrier to produce a transmit RF signal and to transmit the transmit RF signal via respective antennas. The transceiver(s) are further configured to receive a receive RF signal via respective antennas that includes an RF carrier modulated with serial receive data, to demodulate the receive RF signal to extract the serial receive data and to provide the serial receive data to respective UARTs for provision to the processor 50408. Each RF signal has an associated carrier frequency and an associated channel bandwidth. The channel bandwidth is associated with the carrier frequency, the transmit data and/or the receive data. Each RF carrier frequency and channel bandwidth is related to the operating frequency range(s) of the transceiver(s). Each channel bandwidth is further related to the wireless communication standard and/or protocol with which the transceiver(s) may comply. In other words, each transceiver may correspond to an implementation of a selected wireless communication standard and/or protocol, e.g., IEEE 802.11 a/b/g/n for Wi-Fi® and/or IEEE 802.15.4 for wireless mesh networks using Zigbee routing.

The communication device 50418 can allow the processor 50408 in the surgical evacuation system 50400 to communicate with other devices within a surgical system. For example, the communication device 50418 can allow wired and/or wireless communication to one or more external sensors 50432, one or more surgical devices 50444, one or more hubs 50448, one or more clouds 50446, and/or one or more additional surgical systems and/or tools. The reader will readily appreciate that the surgical evacuation system 50400 of FIG. 21 can be incorporated into an intelligent electrosurgical system in certain instances. The surgical evacuation system 50400 also includes a pump 50450, including a pump motor 50451 thereof, an evacuation conduit 50436, and an exhaust 50452. Various pumps, evacuation conduits and exhausts are further described herein. The surgical evacuation system 50400 can also include a sensing and intelligent controls device. The sensing and intelligent controls device includes sensor algorithms and communication algorithms that facilitate communication between the smoke evacuation system and other devices to adapt their control programs.

The surgical evacuation system 50400 can be programmed to monitor one or more parameters of a surgical system and can affect a surgical function based on one or

more algorithms stored in a memory in signal communication with a processor of a surgical hub and/or hub-connected surgical device and/or the processor **50408** within the surgical evacuation system **50400**. Various exemplary aspects disclosed herein can be implemented by such algorithms, for example. For example, intelligent surgical evacuation systems and adjustment algorithms are further described in U.S. patent application Ser. No. 16/024,066, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL, filed Jun. 29, 2018, which is incorporated by reference herein in its entirety.

An electrosurgical system can include a signal generator, an electrosurgical instrument, a return electrode, and a surgical evacuation system. The generator may be an RF wave generator that produces RF electrical energy. Connected to the electrosurgical instrument is a utility conduit. The utility conduit includes a cable that communicates electrical energy from the signal generator to the electrosurgical instrument. The utility conduit also includes a vacuum hose that conveys captured/collected smoke and/or fluid away from a surgical site. Such an exemplary electrosurgical system **50601** is shown in FIG. 22. More specifically, the electrosurgical system **50601** includes a generator **50640**, an electrosurgical instrument **50630**, a return electrode **50646**, and an evacuation system **50600**. The electrosurgical instrument **50630** includes a handle **50632** and a distal conduit opening **50634** that is fluidically coupled to a suction hose **50636** of the evacuation system **50600**. The electrosurgical instrument **50630** also includes an electrode that is powered by the generator **50640**. A first electrical connection **50642**, e.g., a wire, extends from the electrosurgical instrument **50630** to the generator **50640**. A second electrical connection **50644**, e.g., a wire, extends from the generator **50646** to an electrode, i.e., the return electrode **50646**. In other instances, the electrosurgical instrument **50630** can be a bipolar electrosurgical instrument. The distal conduit opening **50634** on the electrosurgical instrument **50630** is fluidically coupled to the suction hose **50636** that extends to a filter end cap **50603** of a filter that is installed in an evacuator housing **50618** of the evacuation system **50600**.

In other instances, the distal conduit opening **50634** for the evacuation system **50600** can be on a handpiece or tool that is separate from the electrosurgical instrument **50630**. For example, the evacuation system **50600** can include a surgical tool that is not coupled to the generator **50640** and/or does not include tissue-energizing surfaces. In certain instances, the distal conduit opening **50634** for the evacuation system **50600** can be releasably attached to an electrosurgical tool. For example, the evacuation system **50600** can include a clip-on or snap-on conduit terminating at a distal conduit opening, which can be releasably attached to a surgical tool.

The electrosurgical instrument **50630** is configured to deliver electrical energy to target tissue of a patient to cut the tissue and/or cauterize blood vessels within and/or near the target tissue, as described herein. Specifically, an electrical discharge is provided by the electrode tip to the patient in order to cause heating of cellular matter of the patient that is in close contact with or adjacent to electrode tip. The tissue heating takes place at an appropriately high temperature to allow the electrosurgical instrument **50630** to be used to perform electrosurgery. The return electrode **50646**, or return pad, can be placed under the patient during the surgical procedure, in order to complete the circuit and provide a return electrical path to the generator **50640** for energy that passes into the patient's body.

The heating of cellular matter of the patient by the electrode tip, or cauterization of blood vessels to prevent bleeding, often results in smoke being released where the cauterization takes place, as further described herein. In such instances, because the evacuation conduit opening **50634** is near the electrode tip, the evacuation system **50600** is configured to capture the smoke that is released during a surgical procedure. Vacuum suction may draw the smoke into the conduit opening **50634**, through the electrosurgical instrument **50630**, and into the suction hose **50636** toward the evacuator housing **50618** of the evacuation system **50600**. In various instances, the electrosurgical system **50601** and/or the evacuation system **50600** can include a control circuit including a processor and memory, as described herein with respect to the control schematic in FIG. 21.

Referring now to FIG. 23, another evacuator housing **50918** for an evacuation system **50900** is depicted. The evacuator housing **50918** defines a flow path **50904** between an inlet **50922** to the evacuator housing **50918** and an outlet **50924** to the evacuator housing **50918**. Intermediate the inlet **50922** and the outlet **50924**, a fluid trap **50960**, a filter **50970**, and a pump **50906** are sequentially arranged. The evacuator housing **50918** can include a socket or a receptacle **50971** dimensioned to receive a modular fluid trap and/or a replaceable filter **50970**. At a diverter valve **50934**, fluid can be directed into a condenser **50935** of the fluid trap **50960** and the smoke can continue toward the filter **50970**. In certain instances, the fluid trap **50960** can include baffles, such as the baffles **50964**, and/or splatter screens, such as the screen **50962**, for example, for preventing the captured fluid from splashing out of the fluid trap **50960**. The filter **50970** includes a pleated ultra-low penetration air (ULPA) filter **50942** and a charcoal filter **50944**. A sealed conduit or tube **50905** extends between the various in-line components. The evacuator housing **50918** also includes sensors **50830**, **50832**, **50836**, **50838**, **50840**, **50846**, **50848**, **50850**, **50852**, and **50854** which are further described herein and in U.S. patent application Ser. No. 16/024,066, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL, filed Jun. 29, 2018, which is incorporated by reference herein in its entirety.

Referring still to FIG. 23, the evacuator housing **50918** also includes a centrifugal blower arrangement **50980** and a recirculating valve **50990**. The recirculating valve **50990** can selectively open and close to recirculate fluid through the fluid trap **50960**. For example, if the fluid detection sensor **50836** detects a fluid, the recirculating valve **50990** can be opened such that the fluid is directed back away from the filter **50970** and back into the fluid trap **50960**. If the fluid detection sensor **50836** does not detect a fluid, the valve **50990** can be closed such that the smoke is directed into the filter **50970**. When fluid is recirculated via the recirculating valve **50990**, the fluid can be drawn through a recirculation conduit **50982**. The centrifugal blower arrangement **50980** is engaged with the recirculation conduit **50982** to generate a recirculating suction force in the recirculation conduit **50982**. More specifically, when the recirculating valve **50990** is open and the pump **50906** is activated, the suction force generated by the pump **50906** downstream of the filter **50970** can generate rotation of the first centrifugal blower, or first squirrel cage, **50984**, which can be transferred to the second centrifugal blower, or second squirrel cage, **50986**, which draws the recirculated fluid through the recirculating valve **50990** and into the fluid trap **50960**.

In various aspects of the present disclosure, the control schematics of FIG. 21 can be utilized with the various

surgical systems, evacuation systems, sensor systems and evacuator housings of FIGS. 22 and 23.

Smoke evacuated from a surgical site can include liquids, aerosols, and/or gases, and/or can include material of different chemical and/or physical properties, such as particulate matter and particles of different sizes and/or densities, for example. The different types of materials evacuated from a surgical site can affect the efficiency of the surgical evacuation system and the pump thereof. Moreover, certain types of material can require the pump to draw excessive power and/or can risk damaging the motor for the pump.

The power supplied to the pump can be modulated to control the flowrate of smoke through the evacuation system based on input from one or more sensors along the flow path. Output from the sensors can be indicative of a state or quality of the smoke evacuation system and/or one or more properties of the evacuated smoke such as the type(s) and ratios of matter, chemical properties, density, and/or size of particulates, for example. In one aspect of the present disclosure, a pressure differential between two pressure sensors in the evacuation system can indicate the state of the region therebetween such as the state of a filter, a fluid trap, and/or the overall system, for example. Based on the sensor input, an operational parameter of the motor for the pump can be adjusted by changing the current supplied to the motor and/or the duty cycle, which is configured to change the motor speed.

In one aspect of the present disclosure, by modulating the flowrate of smoke through the evacuation system, the efficiency of the filter can be improved and/or the motor can be protected from burnout.

A surgical evacuation system can include one or more particle counters, or particle sensors, for detecting the size and/or concentration of particulate within the smoke. Referring again to FIG. 23, the particle sensors 50838 and 50848 are depicted. The reader will readily appreciate that various particle measurement means are possible. For example, a particle sensor can be an optical sensor, a laser sensor, a photoelectric sensor, an ionization sensor, an electrostatic sensor, and/or combinations thereof. Various particle sensors are further described herein.

In various instances, the speed of the motor and, thus, the speed of the pump can be adjusted based on the particulate concentration detected by the one or more particle sensors in a surgical evacuation system. For example, when the particle sensor(s) detect an increased concentration of particulate in the flow path, which can correspond to an increased quantity of smoke in the flow path, the speed of the motor can be increased to increase the speed of the pump and to draw more fluid into the smoke evacuation system from the surgical site. Similarly, when the particle sensor(s) detects a decreased concentration of particulate in the flow path, which can correspond to a decreased quantity of smoke in the flow path, the speed of the motor can be decreased to decrease the speed of the pump and to reduce suction from the surgical site. Additional and alternative adjustment algorithms for the surgical evacuation system are further described herein. Moreover, in certain instances, based on the sensor data from the smoke evacuation system, a generator in the surgical system can be controlled to adjust the amount of smoke generated at the surgical site, as further described herein.

In addition to particle sensors positioned along the flow path of the surgical evacuation system, the system can include one or more sensors for detecting the particulate concentration in the ambient room, for example, in the operating room or surgical theater. Referring again to FIG.

23, the air quality particle sensor 50852 is installed on an external surface of the evacuator housing 50918. Alternative locations for the air quality particle sensor 50852 are also envisioned.

In at least one instance, a particle sensor can be positioned downstream of the filter and, in certain instances, can be positioned at or near the outlet of the filter. For example, the particle sensor 50848 is positioned downstream of the filter 50970 and the pump 50906 in the smoke evacuation system 50900. Because the particle sensor 50848 is positioned downstream of the filter(s) 50970, the particle sensor is configured to confirm that the filter(s) 50970 have removed sufficient particulate from the smoke. In various instances, such a sensor can be adjacent to the exhaust outlet 50924 of the evacuator housing 50918. In one aspect of the present disclosure, an electrostatic particle sensor can be utilized. For example, the exhaust outlet 50924 can include an electrostatic particulate sensor that the exhaust flows past downstream of the filtration system and prior to being exhaust into the surgical theater.

The particulate concentration detected by one or more sensors of the surgical evacuation system can be communicated to a clinician in a number of different ways. For example, the evacuator housing 50918 and/or the evacuation device can include an indicator, such as one or more lights and/or display screens. For example, an LED on the evacuator housing 50918 may change color (e.g. from blue to red) depending on the volume of particulate detected by the sensor(s). In other instances, the indicator can include an alarm or warning, which can be tactile, auditory, and/or visual, for example. In such instances, when the particulate concentration in the ambient air detected by the air quality sensor (e.g. the particle sensor 50852) exceeds a threshold amount, the clinician(s) in the surgical theater can be notified by the indicator(s).

In certain instances, a surgical evacuation system can include an optical sensor. The optical sensor can include an electronic sensor that converts light, or a change in the light, into an electronic signal. The optical sensor can utilize a light scattering method to detect and count particles in the smoke to determine the concentration of particles in the smoke. In various instances, the light is laser-based. For example, in one instance, a laser light source is configured to illuminate particles as the particles move through a detection chamber. As the particles pass through the laser's beam, the light source is refracted, obscured, redirected, and/or absorbed. The scattered light is recorded by a photo detector, and the recorded light is analyzed. For example, the recorded light can be converted to an electrical signal indicative of the size and quantity of the particles, which corresponds to the particulate concentration in the smoke. The particulate concentration in the smoke can be calculated in real-time by a laser optical sensor, for example. In one aspect of the present disclosure, at least one of the particle sensors 50838, 50848, 50852 are laser optical sensors. Various alternative sensing techniques and sensor arrangements are further described herein and in U.S. patent application Ser. No. 16/024,066, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL, filed Jun. 29, 2018, which is incorporated by reference herein in its entirety.

Based on the presence and concentration of smoke in an abdomen cavity, the generator algorithm for minimizing or controlling the smoke generation can be adapted. For example, information/knowledge gained from previous surgical events, such as an earlier firing and/or energy activation, can be used to adjust one or more parameters of a

generator control algorithm. Referring now to FIG. 24, a series of graphs 213400, 213402, and 213404 are depicted in which particle count 213406, fan speed 213408, and energy device power 213410 are shown relative to time (t). The particle count 213406 can be detected with the various sensors and/or detection techniques disclosed herein. One or more surgical functions and/or one or more devices can be adjusted in response to the detected particle count 213406, which corresponds to the smoke concentration at the surgical site (e.g. in an internal cavity of the patient). Additionally, the type and/or degree of adjustment can depend on knowledge gained during previous firings and/or activations.

The energy device power 213410 is provided during a first activation 213412 from time t_0 to time t_2 and during a second activation 213414 from time t_5 to time t_7 . Knowledge gained during the first activation 213412 can be utilized to adjust the control algorithm of a control circuit applied during the second activation 213414. The control circuit can be encompassed in a surgical hub and/or smoke evacuator as depicted in the control schematic of FIG. 21, for example

Referring still to FIG. 24, when the energy device is activated for the first activation 213412, the energy device power 213410 is set to a predefined power level 213430, and the particle count 213406 begins to increase toward a first threshold 213416. The predefined power level 213430 can be a clinician-selected power level. For example, the predefined power level 213430 can be preprogrammed on the surgical device and/or at the surgical hub. In certain instances, the surgical device can be configured to operate at different predefined power levels depending on the actuators and/or control buttons activated by the clinician.

At time t_1 , as the first activation 213412 continues, the particle count 213406 increases above the first threshold 213416 to a peak value 213418 at time t_2 when the first activation 213412 ends. In response to the particle count 213406 exceeding the first threshold 213416 at time t_1 , a control circuit in signal communication with the smoke evacuator, is configured to implement a first adjustment 213420. For example, the control circuit is configured to increase the fan speed 213408 from a low setting 213426 to a high setting 213428.

After the first activation 213412 ends at time t_2 , the particle count 213406 can begin to decrease from the peak value 213418; however, the smoke evacuator can continue to operate at the high setting 213428 until time t_3 when the particle count 213406 drops below the first threshold 213416. In response to the particle count 213406 decreasing below the first threshold 213416, the fan speed 213408 can be decreased to the low setting 213426 and, in various instances, can be turned off when the particle count 213406 drops to zero at time t_4 . In other instances, the smoke evacuator can include additional speeds which can correspond to additional particle count thresholds and/or can be scaled in proportion to the particle count 213406, for example. In various instances, the smoke evacuator can operate at a ultra-low setting and/or cycle on-and-off when in a sleep or resting mode when the particle count is zero and/or below a minimum threshold value.

At time t_5 , when the second activation 213414 is initiated, the particle count 213406 again begins to increase and the fan speed 213408 can be set at the low setting 213426 to extract the smoke from the surgical site. In various instances, before any smoke is detected, the smoke evacuator can be activated in response to an activation of the energy device. As shown in FIG. 24, in certain instances, the particle count 213406 can continue to increase during the second activation 213414 while the fan speed 213408 is set to the low

setting 213426 and, at time t_6 , as the second activation 213414 continues, the particle count 213406 again increases above the first threshold 213416. If the control circuit were to implement the first adjustment 213420 again, the particle count 213406 would again follow the hypothetical curve 213424 to the peak value 213418 at time t_7 when the second activation 213414 ends. However, in response to the particle count 213406 again exceeding the first threshold 213416 at time t_6 in the second activation 213416, the control circuit is configured to adapt the control algorithm and implement a second adjustment 213422 that is different than the first adjustment 213420.

For example, in addition to increasing the fan speed 213408 from the low setting 213426 to the high setting 213428 at time t_6 , the control circuit is configured to decrease the energy device power 213410 below the predefined power level 213430. For example, the control actuated by the clinician can correspond to the predefined power level 213430; however, a control circuit for the surgical hub can adjust the energy device power 213430 to an adjusted power level 213432. The adjusted power level 213432 in FIG. 24 is less than the predefined power level 213430. In other instances, the hub-controlled adjustment can increase the energy device power 213410 and/or otherwise adjust the energy device power, such as by applying pulses of varying duration and/or frequency (e.g. pulse width modulation and/or pulse frequency modulation) and/or adjusting the energy modality, for example.

The second adjustment 213422 can be learned or developed based on the system's response to the first adjustment 213420. For example, if the peak value 213418 obtained when implementing the first adjustment 213420 is still considered to be too high and/or associated with undesirable consequences, such as obstructing the clinician's view and/or interfering with the application of energy to tissue, for example, the control circuit can seek a different adjustment. As described herein, the second adjustment 213422 includes the first adjustment 213422 (i.e. adjusting the fan speed 213408 from the low setting 213426 to the high setting 213428) and the additional adjustment to the energy device power 213410. In other instances, the second adjustment 213422 can replace the first adjustment 213420.

In certain instances, the second adjustment 213422 can adjust a threshold and/or degree of the first adjustment 213420. For example, the second adjustment 213422 can include increasing the fan speed 213408 to a level that is greater than the high setting 213428. In various instances, the second adjustment 213422 can include a modification to the first adjustment 213420 that is proportional to a desired effect. For example, if the first adjustment 213410 increased the fan speed 213408 by 10% and obtained a 10% reduction in smoke concentration, the second adjustment 213422 could be a 15% reduction in fan speed 213408 to obtain a desired 15% reduction in smoke concentration.

The first and second adjustments described herein are example adjustments. Adjustments can include changing the power level of a generator, power level of a smoke evacuator, venting of a space, and degree and/or type of filtration and/or condensation-induction in a smoke evacuation system, for example.

As described herein, the techniques for improved identification and tracking of particles, which can determine the directionality and velocity of a fluid, for example, can be utilized in various surgical applications. For example, the improved techniques can be used for smoke detection applications, as further described herein. Additionally or alternatively, based on the directionality and/or velocity of the

smoke, the surgical system can be configured to adjust the smoke evacuation motor control to clear the smoke more quickly and then slow down to minimize noise in the operating room (OR). As another example, the system can increase insufflation recirculation to evacuate the particulate away from the active field of operation. For example, if a sensor detects the smoke moving toward a treatment region where a clinician is actively working and/or treating, it can be desirable to increase the circulation within the cavity to quickly move the obstructing particulate (i.e. smoke) away from the treatment region.

In one application, improved techniques for the identification and tracking of particles can be utilized for bleeding detection. Based on the directionality and velocity of the flow, the control circuit and/or surgical hub can determine and/or recommend a subsequent step to address the bleeding. In another application, the improved techniques for identification and tracking of particles can be utilized for perfusion assessment in tissue. Perfusion assessment can be utilized to identify anatomical structures, for example. In one example, NIR illumination of ICG can be used to image perfusion in a lung segment. Visualization of such perfusion can be used to identify/demarcate an intersegmental plane for segmentectomy procedures, for example. Thereafter, the control circuit and/or surgical hub can determine and/or recommend a subsequent step. For example, the control circuit and/or surgical hub can recommend a particular surgical device and/or location. In certain instances, the control circuit and/or surgical hub can guide the surgical device to the desired location.

In one application, the improved techniques for the identification and detection of particles can be used with migration of dye, which can enable the system to highlight a leak path. The size and location of the leak can be determined, for example. In one instance, when leak testing an anastomosis, a leak path may be identified from the colon into the abdomen. In various instances, the identification of a leak path can inform the clinician's decision-making process. For example, a recommendation regarding how and/or when to address the leak path can be provided to the clinician. In certain instances, contextual information provided by the hub can affect the recommendation provided to the clinician. Contextual information can indicate the probability of a functional leak taking into account the current pressure within the colon dye fluid, the viability of the tissue from computerized topography (CT) imaging and/or instrument manipulation forces, and/or patient-specific complicating factors, which can be identified in an electronic health record (EHR), for example. The recommendation can include suggestions for viable products to reinforce the anastomosis based on the identification of detected implications (e.g., the size of leak, the location of the leak with respect to "dog ears" or lateral intersection margins/stapled corners, and/or compression loads detected in the circular stapler). Additionally or alternatively, the recommendations can include the proposal of an overlaying of reinforcement agents like fibrin, thrombin, and/or oxidized regenerated cellulose (ORC), for example. Additionally or alternatively, the recommendations can include a proposal to over-sew or re-staple a portion of the line due to tissue tension or collateral blood supply issues, for example.

In various instances, the detection of one or more properties of fluid flow at a surgical site can encompass one or more of the various measurements systems and/or techniques further described herein. For example, optical measurement systems, such as laser Doppler velocimetry (LDV) or laser Doppler flowmetry (LDF), particle image veloci-

metry (PIV), and/or near-infrared (NIR) fluorescence of indocyanine green (ICG), for example, can be utilized. In other instances, instead of a camera/optical measurement system, detection and/or measurements can utilize an alternative sensing array to track particles. For example, ultrasonic measurements systems, such as non-contact pass-through ultrasound detection and non-contact reflection ultrasound detection, for example, can be utilized.

In one aspect, non-contact pass-through ultrasound can be utilized for particle detection. Referring to an ultrasonic pass-through particle detection system **213500** in FIG. **25**, an ultrasound emitter **213502** can be positioned within the patient's body (e.g. within an abdominal cavity) similar to an optical scope. In various instances, the emitter **213502** can be positioned on a distal end of a scope **213501** for an imaging system. The scope can include a camera that views the surgical site and transmits the view(s) to a display that is viewable by a clinician. Scopes include, but are not limited to, arthroscopes, angioscopes, bronchoscopes, choledochoscopes, colonoscopes, cytosopes, duodenoscopes, enteroscopes, esophagogastric-duodenoscopes (gastroscopies), endoscopes, laryngoscopes, nasopharyngo-neprosopes, sigmoidoscopes, thoracoscopes, ureterscopes, and exoscopes. In other instances, the emitter **213502** can be a separate and distinct device from a scope. The ultrasound emitter **213502** is configured to transmit a curtain of homogeneous sound, which passes through a fluid **213504** (e.g. moving particles such as smoke particulate and/or steam aerosols) and is received on the opposite side of the fluid **213504** by a receiver **213506**. The receiver **213506** can be positioned on a surgical device, such as an end effector and/or a distal nose of a staple cartridge **213508** received within an elongate channel of an end effector, for example. In other instances, the receiver **213506** can be positioned on a separate and distinct surgical tool. As various sound waves pass from the emitter **213502** through the fluid **213504**, certain waves can be blocked by the fluid **213504**. Based on the sound waves received by the receiver **213506**, the ultrasonic pass-through particle detection system **213500** is configured to determine the Doppler shift, i.e., the speeding up or slowing down of the sound waves, which corresponds to the properties of the fluid's flow.

Surgical uses for the ultrasonic pass-through particle detection system **213500** include, for example, smoke detection applications and control of evacuation, including detection of the velocity, directionality, and concentrations of particulates in the flow and detection of flow patterns and/or dispersion rates, for example. Surgical uses also include the differentiation between aerosol particles and particulates (e.g. distinguishing smoke from steam). Such distinctions can be helpful when adjusting a smoke evacuation system, such as changing the fluid flow path to address aerosols within the flow path, which may damage a particulate matter filter if not properly extracted, for example. Additional surgical applications for particle detection and monitoring, such as ultrasound techniques utilizing the ultrasonic pass-through particle detection system **213500**, for example, are further described herein. Moreover, a surgical system and/or control circuit therefor can be configured to utilize information from such an ultrasonic pass-through particle detection system to adjust and/or suggest a surgical function during a surgical procedure.

In one aspect, non-contact reflection sound detection can be utilized for particle detection. Referring now to FIG. **26**, an ultrasonic reflection particle detection system **213600** is shown. Similar to the ultrasonic pass-through technique described above with respect to FIG. **25**, the ultrasonic

reflection particle detection system **213600** includes an emitter, or transmitter/sound source, **213602** and a receiver **213606**. However, this ultrasonic reflection particle detection system **213600** differs from the ultrasonic pass-through particle detection system **213500** in that the emitter **213602** and the receiver **213606** are co-located on the same surgical device **213608**. In other instances, the emitter **213602** and the receiver **213606** can be positioned on separate devices but located adjacently to one other. In various instances, the surgical device **213608** can be a scope like the device **213502** (FIG. 25), for example.

The ultrasonic reflection particle detection system **213600** is configured to detect reflected sound waves. The reflection and refraction of the sound waves will depend on the surface properties of the tissue T that the surgical device **213608** is targeting. Since the surgical device **213608** is not in contact with the tissue T, it is configured to sense the superficial aspects of the tissue T, e.g. the surface features of the tissue T and not properties/features deep within the tissue T. For example, referring to the flowchart **213650** in FIG. 27, data from the emitter **213602** and the receiver **213606** regarding the sound wave properties is obtained at block **213652** and provided to a control circuit for analysis at block **213654**. The control circuit is configured to determine tissue surface information at block **213656**. Various control circuits for receiving input signals, calculating tissue surface information, and providing outputs can be incorporated into the various control circuits described herein. For example, algorithms for determining the tissue surface information from the input signals can be stored in the memory of the various control circuits described herein.

Surgical uses for the ultrasonic reflection particle detection systems include, for example, differentiation of broken superficial surface reflection and the detection of flowing or oozing fluids on a tissue surface. For example, sound reflection that indicates broken or interrupted superficial tissue features can be used to detect unhealthy tissue features, such as infection, adhesions, scarring, remodeling, and cancer. Such tissue features can correspond to a tissue surface that is more interrupted and, thus, that causes increased refraction and dampening of the ultrasound waves in comparison to healthy tissue, which typically defines a continuous, uninterrupted surface. Flowing or oozing fluids on the surface of the tissue can also create different Doppler effects on the sound waves that are directed toward the surface while producing a sound reflection that is different than the reflection for a solid tissue surface. Furthermore, the more fluid that is present, the greater the sound absorption that the fluid can induce due to absorbing a portion of the energy of the sound wave. Additionally, Doppler effects on the sound wave would result in a Doppler wave shift in the sound wave depending on whether the fluid is moving towards or away from the transmitter and receiver. Directionality of the fluid flow can be detected. In various instances, varying the range of frequencies emitted by the transmitter would allow the system to further refine the surface tension, modulus, and or viscosity of the detected fluid, for example. A surgical system or control circuit therefor can be configured to utilize information from such an ultrasonic reflection particle detection system to adjust and/or suggest a surgical function during a surgical procedure.

In various aspects, the system can be configured to monitor the material in the flow being removed/evacuated from the abdomen by a smoke evacuation system or filtered and replaced by an irrigation/insufflation system, for example, with the various techniques described herein for

the identification and detection of particles along a flow path. In certain instances, the system can be configured to detect particles within the extracted gasses and directly measure their flow rate. In one aspect, an ultrasonic pass-through array can be placed at a crossing between the passage that is extracting the insufflation gasses or filtering the insufflation gasses, for example. Such an array can be configured to detect particles passing through and, thus, can be used to determine the type and concentration of the particles flowing therethrough. An alternative to the ultrasonic pass-through array is a laser counter array which can be placed similarly. The laser counter array can measure a blocked laser light wave passing from one side of the passage to the other or the refraction of light off of the particles.

In various instances, vortex-shedding can be used in combination with either the ultrasonic pass-through array or laser counter array, described above. Vortex-shedding is configured to detect the size of the particles and their flow rate, rather than merely using the presence or concentration of the particles given a predefined flow rate. When utilizing a vortex-shedding flowmeter, there are no moving parts or elements within the flow that are necessary to measure the flow. Advantageously, this can improve the ability to clean the detection system and reuse it, and/or can minimize the buildup of bodily fluids within the detection system, for example.

EXAMPLES

Various aspects of the subject matter described herein are set out in the following numbered examples:

Example 1—A surgical system, comprising a sensor configured to detect a property of airborne particles in a fluid within a patient's abdominal cavity; a surgical device configured to implement a surgical function; and a control circuit comprising a processor and a memory storing instructions executable by the processor to: receive an input signal from the sensor indicative of the property of airborne particles in the fluid; and in response to the input signal, provide an output signal to the surgical device indicative of an adjustment to the surgical function.

Example 2—The surgical system of Example 1, wherein the sensor comprises at least one of an optical sensor, a laser sensor, an ultrasonic sensor, a magnetic resonance sensor, and a vortex shedding sensor.

Example 3—The surgical system of any one of Examples 1 and 2, wherein the property of airborne particles in the fluid comprises at least one of a particle type, particle size, particle concentration, particle velocity, and particle direction.

Example 4—The surgical system of any one of Examples 1-3, wherein the adjustment to the surgical function comprises at least one of proportionately increasing the surgical function based on the property of airborne particles in the fluid, adding a supplemental surgical function to the surgical function, and replacing the surgical function with an alternative surgical function.

Example 5—The surgical system of any one of Examples 1-4, wherein the surgical device comprises a generator configured to provide power to an energy device at a power level, and wherein the adjustment to the surgical function comprises adjusting the power level provided to the energy device from the generator.

Example 6—The surgical system of any one of Examples 1-4, wherein the surgical device comprises a smoke evacuator comprising a pump configured to operate at a speed, and

wherein the adjustment to the surgical function comprises adjusting the speed of the pump.

Example 7—The surgical system of any one of Examples 1-4, wherein the surgical device comprises a smoke evacuator comprising a filtering system, and wherein the adjustment to the surgical function comprises adjusting a flow path through the filtering system of the smoke evacuator.

Example 8—The surgical system of Example 7, wherein the filtering system comprises a particulate filter, and wherein the flow path is diverted through the particulate filter when the input signal indicates an increase in particulate concentration of airborne particles in the fluid.

Example 9—The surgical system of any one of Examples 7 and 8, wherein the filtering system comprises a condenser, and wherein the flow path is diverted through the condenser when the input signal indicates an increase in aerosol concentration of airborne particles in the fluid.

Example 10—The surgical system of any one of Examples 1-4, wherein the surgical device comprises an operating room vent, and wherein the adjustment to the surgical function comprises adjusting the operating room vent to increase ventilation therethrough.

Example 11—The surgical system of any one of Examples 1-10, wherein the control circuit is further configured to: receive a plurality of input signals indicative of a plurality of properties of airborne particles in the fluid, wherein the plurality of input signals comprises the input signal; and in response to the plurality of input signals, provide the output signal to the surgical device indicative of the adjustment to the surgical function.

Example 12—A non-transitory medium storing computer readable instructions which, when executed, cause a surgical device to: receive an input signal from a sensor indicative of a property of airborne particles in a fluid within a patient's abdominal cavity; and in response to the input signal, provide an output signal to a surgical device indicative of an adjustment to a surgical function.

Example 13—The non-transitory medium of Example 12, wherein the computer readable instructions, when executed, cause the surgical device to provide the output signal to a generator to adjust a power level of the generator.

Example 14—The non-transitory medium of Example 12, wherein the computer readable instructions, when executed, cause the surgical device to provide the output signal to a smoke evacuator to adjust a speed of a pump of the smoke evacuator.

Example 15—The non-transitory medium of Example 12, wherein the computer readable instructions, when executed, cause the surgical device to provide the output signal to a smoke evacuator to adjust a flow path through the smoke evacuator.

Example 16—The non-transitory medium of any one of Examples 12-15, wherein the computer readable instructions, when executed, cause the surgical device to adjust the surgical function by changing the surgical function in proportion to a change in the property of airborne particles in the fluid.

Example 17—The non-transitory medium of any one of Examples 12-16, wherein the computer readable instructions, when executed, cause the surgical device to adjust the surgical function by adding a supplemental surgical function in combination with the surgical function.

Example 18—The non-transitory medium of any one of Examples 12-17, wherein the computer readable instructions, when executed, cause the surgical device to adjust the surgical function by replacing the surgical function with an alternate surgical function.

Example 19—A method, comprising: receiving an input signal from a sensor indicative of a property of airborne particles in a fluid within a patient's abdominal cavity; and in response to the input signal, automatically providing an output signal to a surgical device indicative of an adjustment to a surgical function of the surgical device.

Example 20—The method of Example 19, further comprising: receiving a second input signal from the sensor indicative of the property of airborne particles in the fluid at a time subsequent to receiving the input signal; and in response to the second input signal, automatically providing a second output signal to a second surgical device indicative of an adjustment to a surgical function of the second surgical device, wherein the second surgical device is different than the first surgical device.

Example 21—A surgical device, comprising: an actuator configured to receive an input; and a control circuit configured to: receive a signal from a situationally-aware surgical hub indicative of a surgical state; receive an actuation signal from the actuator in response to the input; and implement a surgical function in response to the actuation signal, wherein the surgical function comprises a first surgical function when the surgical state corresponds to a first surgical state, wherein the surgical function comprises a second surgical function when the surgical state corresponds to a second surgical state, wherein the second surgical state is different than the first surgical state, and wherein the second surgical function is different than the first surgical function.

Example 22—The surgical device of Example 21, wherein the situationally-aware surgical hub comprises a situational awareness module, and wherein the surgical state comprises a step in a surgical procedure.

Example 23—The surgical device of any one of Examples 21 and 22, wherein the surgical state comprises identification of a suite of surgical devices currently in use in a surgical theater.

Example 24—The surgical device of any one of Examples 21-23, wherein the surgical state comprises a position of a portion of the surgical device.

Example 25—The surgical device of any one of Examples 21-24, wherein the surgical state comprises a position of a jaw of an end effector of the surgical device.

Example 26—The surgical device of any one of Examples 21-25, wherein the actuator is movable through a first range of motion from a first position to a second position and through a second range of motion from the second position to a third position, wherein movement of the actuator in the first range of motion is configured to activate the surgical function, and wherein movement of the actuator in the second range of motion is configured to scale a degree of the surgical function.

Example 27—The surgical device of any one of Examples 21-26, further comprising a Hall Effect sensor configured to determine the position of the actuator, wherein the first surgical function comprises an activation, and wherein the second surgical function comprises adjusting a degree of the activation based on the position of the actuator detected by the Hall Effect sensor.

Example 28—The surgical device of any one of Examples 21-26, further comprising a strain gauge operably coupled to the actuator, wherein the strain gauge is configured to determine a force of the input on the actuator, wherein the first surgical function comprises an activation, and wherein the second surgical function comprises adjusting a degree of the activation based on the force of the input detected by the strain gauge.

Example 29—The surgical device of any one of Examples 21-28, wherein the first surgical function is confined by a first maximum threshold and a first minimum threshold, wherein the second surgical function comprises a second maximum threshold and a second minimum threshold, wherein the first maximum threshold is different than the second maximum threshold, and wherein the first minimum threshold is different than the second minimum threshold.

Example 30—The surgical device of any one of Examples 21-29, wherein the first surgical state corresponds to a gross usage surgical step, and wherein the second surgical state corresponds to a precision usage surgical step.

Example 31—The surgical device of any one of Examples 21-30, wherein the actuator comprises at least one of a button, a switch, a toggle, a trigger, a lever, a dial, and a knob.

Example 32—The surgical device of any one of Examples 21-31, wherein the control circuit comprises a processor and a memory communicatively coupled to the processor.

Example 33—A non-transitory medium storing computer readable instructions which, when executed, cause a surgical device to: receive a signal from a situationally-aware surgical hub indicative of a surgical state; receive an actuation signal in response to an input applied to an actuator of a surgical device; and implement a surgical function in response to the actuation signal, wherein the surgical function comprises a first surgical function when the surgical state corresponds to a first surgical state, wherein the surgical function comprises a second surgical function when the surgical state corresponds to a second surgical state, wherein the second surgical state is different than the first surgical state, and wherein the second surgical function is different than the first surgical function.

Example 34—A system, comprising: a screen configured to selectively display a recommendation to a clinician in an operating room; and a control circuit comprising a processor and a memory in signal communication with the processor, wherein the memory stores instructions executable by the processor to: receive a signal from a situationally-aware surgical hub indicative of a surgical state; determine a priority level of the recommendation based on the surgical state; and communicate the priority level of the recommendation via the screen.

Example 35—The system of Example 34, wherein the control circuit is further configured to select one or more recommendations from a plurality of possible recommendations based on the surgical state.

Example 36—The system of Example 35, wherein the control circuit is further configured to adjust the priority level of the recommendation based on an anticipated surgical action, and wherein the anticipated surgical action is based on the surgical state.

Example 37—The system of Example 36, wherein the anticipated surgical action is based on a position of a surgical device at a surgical site.

Example 38—The system of any one of Examples 34-37, wherein an elevated priority level is communicated with at least one of marking, emphasizing, highlighting, and flashing.

Example 39—The system of any one of Examples 34-38, wherein the screen is positioned on a surgical device.

Example 40—The system of any one of Examples 34-39, further comprising an imaging system comprising a video monitor positionable in the operating room, wherein the screen is positioned on the video monitor.

While several forms have been illustrated and described, it is not the intention of Applicant to restrict or limit the

scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a means for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. Those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and/or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution.

Instructions used to program logic to perform various disclosed aspects can be stored within a memory in the system, such as dynamic random access memory (DRAM), cache, flash memory, or other storage. Furthermore, the instructions can be distributed via a network or by way of other computer readable media. Thus a machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer), but is not limited to, floppy diskettes, optical disks, compact disc, read-only memory (CD-ROMs), and magneto-optical disks, read-only memory (ROMs), random access memory (RAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic or optical cards, flash memory, or a tangible, machine-readable storage used in the transmission of information over the Internet via electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Accordingly, the non-transitory computer-readable medium includes any type of tangible machine-readable medium suitable for storing or transmitting electronic instructions or information in a form readable by a machine (e.g., a computer).

As used in any aspect herein, the term “control circuit” may refer to, for example, hardwired circuitry, programmable circuitry (e.g., a computer processor including one or more individual instruction processing cores, processing unit, processor, microcontroller, microcontroller unit, controller, digital signal processor (DSP), programmable logic device (PLD), programmable logic array (PLA), or field programmable gate array (FPGA)), state machine circuitry, firmware that stores instructions executed by programmable circuitry, and any combination thereof. The control circuit may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), an application-specific integrated circuit (ASIC), a system on-chip (SoC), desktop computers, laptop computers, tablet computers, servers, smart phones, etc. Accordingly, as used herein “control circuit” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

As used in any aspect herein, the term “logic” may refer to an app, software, firmware and/or circuitry configured to perform any of the aforementioned operations. Software may be embodied as a software package, code, instructions, instruction sets and/or data recorded on non-transitory computer readable storage medium. Firmware may be embodied as code, instructions or instruction sets and/or data that are hard-coded (e.g., nonvolatile) in memory devices.

As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

As used in any aspect herein, an “algorithm” refers to a self-consistent sequence of steps leading to a desired result, where a “step” refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

A network may include a packet switched network. The communication devices may be capable of communicating with each other using a selected packet switched network communications protocol. One example communications protocol may include an Ethernet communications protocol which may be capable permitting communication using a Transmission Control Protocol/Internet Protocol (TCP/IP). The Ethernet protocol may comply or be compatible with the Ethernet standard published by the Institute of Electrical and Electronics Engineers (IEEE) titled “IEEE 802.3 Stan-

dard”, published in December, 2008 and/or later versions of this standard. Alternatively or additionally, the communication devices may be capable of communicating with each other using an X.25 communications protocol. The X.25 communications protocol may comply or be compatible with a standard promulgated by the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T). Alternatively or additionally, the communication devices may be capable of communicating with each other using a frame relay communications protocol. The frame relay communications protocol may comply or be compatible with a standard promulgated by Consultative Committee for International Telegraph and Telephone (CCITT) and/or the American National Standards Institute (ANSI). Alternatively or additionally, the transceivers may be capable of communicating with each other using an Asynchronous Transfer Mode (ATM) communications protocol. The ATM communications protocol may comply or be compatible with an ATM standard published by the ATM Forum titled “ATM-MPLS Network Interworking 2.0” published August 2001, and/or later versions of this standard. Of course, different and/or after-developed connection-oriented network communication protocols are equally contemplated herein.

Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system’s registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

One or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” refers to the portion closest to the clinician and the term “distal” refers to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical,” “horizontal,” “up,” and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to

introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

It is worthy to note that any reference to “one aspect,” “an aspect,” “an exemplification,” “one exemplification,” and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases “in one aspect,” “in an aspect,” “in an exemplification,” and “in one exemplification” in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

What is claimed is:

1. A surgical system, comprising:

a display; and

a control circuit comprising a processor and a memory in signal communication with the processor, wherein the memory stores instructions executable by the processor to:

receive a first signal from a situationally-aware surgical hub indicative of a surgical state;

display a recommendation on the display based on the first signal;

assign a priority level of the recommendation based on the first signal;

communicate the assigned priority level of the recommendation via the display;

receive a second signal indicative of an anticipated surgical action;

adjust the communicated priority level of the recommendation according to the anticipated surgical action, wherein adjusting the communicated priority level of the recommendation comprises displaying the recommendation with at least one of marking, emphasizing, highlighting, or flashing.

2. The surgical system of claim 1, wherein the memory further stores instructions executable by the processor to select one or more recommendations from a plurality of possible recommendations based on the first signal.

3. The surgical system of claim 1, wherein the anticipated surgical action is based on a position of a surgical device at a surgical site.

4. The surgical system of claim 1, wherein the display is positioned on a surgical device.

5. The surgical system of claim 1, further comprising an imaging system comprising a video monitor positionable in an operating room, wherein the display is positioned on the video monitor.

6. The surgical system of claim 1, wherein the display comprises a screen of an operating room visualization system.

7. The surgical system of claim 1, wherein the memory further stores instructions executable by the processor to

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determine the priority level that is assigned to the recommendation based on the surgical state.

8. The surgical system of claim 1, wherein the memory further stores instructions executable by the processor to compare the determined the priority level to a predetermined threshold, wherein the recommendation is displayed based on the priority level stratifying the predetermined threshold.

9. The surgical system of claim 1, wherein the memory further stores instructions executable by the processor to compare the determined the priority level to a predetermined threshold, wherein the recommendation is displayed based on the priority level stratifying the predetermined threshold.

10. The system of claim 1, wherein the instructions to adjust the communicated priority level of the recommendation according to the anticipated surgical action comprise instructions to elevate the priority level.

11. The system of claim 1, further comprising an imaging system configured to capture a live feed of a surgical site.

12. The system of claim 11, wherein the memory further stores instructions executable by the processor to display the live feed of the surgical site on the display.

13. The system of claim 12, wherein the memory further stores instructions executable by the processor to overlay the recommendation on the live feed.

14. The system of claim 11, wherein the second signal indicative of an anticipated surgical action is based on proximity of a surgical device to the surgical site.

15. The system of claim 14, wherein the second signal indicative of an anticipated surgical action is based on a trajectory of movement of the surgical device.

16. The system of claim 14, further comprising the surgical device.

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17. The system of claim 14, wherein the surgical device is a surgical stapler, and wherein the anticipated surgical action is a tissue transection.

18. The system of claim 1, wherein the second signal indicative of the anticipated surgical action is based on an increased confidence level associated with the surgical state.

19. The system of claim 1, wherein the surgical state corresponds to a step of a surgical procedure.

20. A surgical system, comprising:

a control circuit comprising a processor and a memory in signal communication with the processor, wherein the memory stores instructions executable by the processor to:

receive a first signal from a situationally-aware surgical hub indicative of a surgical state;

display a recommendation on a display based on the first signal;

assign a priority level of the recommendation based on the first signal;

communicate the assigned priority level of the recommendation via the display;

receive a second signal indicative of an anticipated surgical action;

adjust the communicated priority level of the recommendation according to the anticipated surgical action, wherein adjusting the communicated priority level of the recommendation comprises displaying the recommendation with at least one of marking, emphasizing, highlighting, or flashing.

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